

European National Transfer Accounts Manual V.2.0.0

A description of data and methods

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European National Transfer Accounts (ENTA) provide age- and sex-specific data on income, transfers between age groups, and consumption for European countries. The data can be accessed through direct download from Zenodo (`https://doi.org/10.5281/zenodo.20360413`) or the data explorer (`https://ntaeurope.eu/entadata`), where descriptive information for users can also be found. This manual provides a codebook and an overview of the methodology and results. It also serves as a starting point for those with a deeper interest in the methodology, who may wish to consult the code, which is publicly available at `https://github.com/ntaeurope/enta`.

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European National Transfer Accounts are based on data from Eurostat, including the microdata of the European Union Statistics on Income and Living Conditions (EU-SILC), the Household Budget Survey (HBS), and the EUROMOD tax-benefit simulation model. Additionally, ENTA uses the Austrian consumption survey (Konsumerhebung) from 2010 and 2015, provided by Statistics Austria. The responsibility for all conclusions drawn from the data lies entirely with the authors.

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0. Organisation of the ENTA manual

This manual refers to ENTA version V.2.0.0, which includes improved imputation of consumption in EU-SILC, and revisions to income top-coding and the treatment of missing data, ensuring consistent handling across all income types, years, and countries. It also extends Croatia back to 2009, adds the 2024 aggregates, and publishes household saving under the code SH.

The manual describes the data and methods used to generate European National Transfer Accounts (ENTA) for 26 European countries. The age- and sex-specific data cover the income reference years 2008 to 2023, Croatia from 2009 and Bulgaria to 2022; the aggregate data cover 1995 to 2024, Bulgaria again to 2022. ENTA are mainly based on Eurostat data and provide age- and sex-specific information about the economy: on primary income, the redistribution of income through government and private transfers, and the use of disposable income for consumption and saving.

The ENTA data can be used to answer questions such as:

- Is the income of the young declining over time?
- Which countries redistribute the most to older age groups?
- Which age groups provide the most transfers to children?

National Transfer Accounts (NTA), in general, take aggregate data from National Accounts (NA) and estimate their age- and sex-specific distribution using data from a range of sources, including microdata from surveys and administrative records. This manual provides an overview of how this is done for ENTA based on Eurostat data. Furthermore, it serves as a starting point for those with a deeper interest in methodology and data. All code is publicly available, and with access to Eurostat microdata, all results can be reproduced.

How this manual is organised

The manual has three parts, answering three different questions: how the data are made, what each variable means, and how to read a result.

Part I: Description of methodology

Chapter 1 gives an overview of the basic ideas behind the global NTA and their European counterpart, ENTA. It describes the similarities and differences between ENTA and the European System of Accounts (ESA). The data sources are described in Chapters 2 and 3 and in Appendix D.

Chapter 2 describes in detail how the sequence of ENTA is derived from aggregate (non-age-specific) ESA data and related sources. Furthermore, it provides an overview of the aggregate data, including its structure, cross-country differences, and changes over time.

The age- and sex-specific ENTA data are described in Chapter 3. The chapter includes information on data sources and methods for estimating age- and sex-specific income, government transfers, private transfers, and consumption.

ENTA data are provided in two formats. The *ENTA–ESA format* maintains the organisation and terminology of the European System of Accounts and focuses on the household sector. The *NTA format* combines private sectors, includes government-sector information, and uses the concepts and variable definitions of the global NTA project, enabling direct comparison with NTA data from other countries. Chapter 4 describes the relation between the two formats.

All users should be aware of the limitations of the ENTA datasets, which are described in Chapter 5. That chapter closes the methodology part because it depends on everything before it. It covers both the general limitations of each quantity and every place where the ENTA code treats one country differently from the rest, because a source is missing, is reported on a different age grid, or is defined differently there. It follows the same order as Chapter 3 and the codebook, so the limitations of a quantity are found under the name that quantity carries everywhere else in the manual.

Part II: Codebook and project reference

Appendix A describes the units in which ENTA data are provided.

Appendix B consists of a codebook of all aggregate and age-specific ENTA variables of the ENTA data in ESA format, and Appendix C describes the variables of the ENTA data in NTA format.

Appendix D describes the organisation and content of the code and data of the ENTA project. This information is important for those who want to replicate ENTA, or parts of it. The code to generate ENTA data can be accessed through direct download from GitHub (`https://github.com/ntaeurope/enta`), where it is provided as supplementary material.

Part III: Information needed for the interpretation of the results

Some ENTA series contain abrupt level shifts, sign changes or single-year spikes that look like data errors but are not. Appendix E documents them: changes caused by the economy, such as the rise in interest rates from 2022 or the pandemic years; changes caused by national policy and law; and, as signposts into Chapter 5, the breaks that are caused by the way the data are built or reported rather than by any change in the economy. It closes with three questions that cover most cases where a single country looks unusual.

The distinction between this part and Chapter 5 is the distinction between what happened and how the data are made. A user who has noticed something odd in a series should start in Appendix E; a user who wants to know what the data can and cannot support should start in Chapter 5.

Part I

Description of methodology

1. National Transfer Accounts: Analysing the Economy by Age and Sex

1.1 What is ENTA?

European National Transfer Accounts (ENTA) provide age- and sex-specific data on primary income, its redistribution through public and private transfers, and its use for consumption and saving, for 26 European countries over the period 2008–2023. ENTA are built mainly from Eurostat microdata (EU-SILC, HBS, EUROMOD) and the European System of Accounts (ESA), and are the European implementation of the global National Transfer Accounts (NTA) project.

The combination of 26 countries observed over 16 years makes ENTA a unique resource for studying the economic implications of demographic change in Europe. Population ageing is reshaping European economies at different speeds and through different channels; ENTA allow researchers to track, country by country and year by year, how the balance between working-age contributors and older beneficiaries evolves and how public and private transfers between age groups adjust in response.

Three features of the dataset open lines of analysis that single-country or single-year NTA cannot support. First, the time dimension makes it possible to study how economies absorb shocks that hit age groups unevenly — the 2008–09 recession, the COVID-19 pandemic, the 2022 energy crisis — and whether the welfare state cushions those shocks differently for the young and the old. Second, the cross-country dimension reveals the diversity of European welfare systems: the same demographic structure can coexist with very different mixes of public pensions, family benefits, private intra-household transfers, and publicly provided health and education. ENTA make those differences visible in a single, comparable accounting framework. Third, the sex dimension throughout the dataset enables systematic analysis of gender gaps in income, transfers and consumption, and of how those gaps vary with age, across countries, and over time.

The rest of this chapter places ENTA in that wider context. It first introduces the NTA concept in general and the global NTA project ENTA belongs to (Section 1.2); it then describes what is specifically European about ENTA (Section 1.3); and it closes by explaining, in general terms, how NTA integrate age and sex into National Accounts, and how ENTA specifically relates to the European System of Accounts (Section 1.4).

1.2 What are National Transfer Accounts (NTA)?

National Transfer Accounts (NTA) constitute a framework to measure and analyse the economy from a demographic perspective, with a focus on age and sex. The fundamental idea of NTA is the use of microdata to integrate age and sex into National Accounts (NA).

Basic NTA measure primary income by age and sex; the redistribution of primary income between age groups and sexes through public and private transfers; and the age- and sex-specific use of disposable income for consumption and saving. NTA provide the data and tools for understanding the macroeconomic consequences of demographic change, as well as for analysing inequalities between demographic groups.

A major innovation of NTA is the inclusion of private transfers within households in their accounting framework. National Accounts take households as a unit and ignore transactions within households. Estimates of primary income and consumption by age reveal a large gap between consumption and income during childhood and in old age. In the NTA context, this gap is called the *life cycle deficit* (LCD). The LCD of older persons is largely financed by public transfers, which are captured in NA and microdata on income. By contrast, the LCD of children is covered mostly by transfers from their parents, which are usually not captured in economic data. Extending the accounting system by age and sex requires consideration of these transfers within households.

Figure 1.1 illustrates what ENTA data look like, using employment income and total consumption (private plus public) by age for France in 2023. Children and older persons consume much more than they earn, whereas persons of working age earn more than they consume. The gap between the two curves at young and old ages is the life cycle deficit; the surplus at working ages is what finances it.

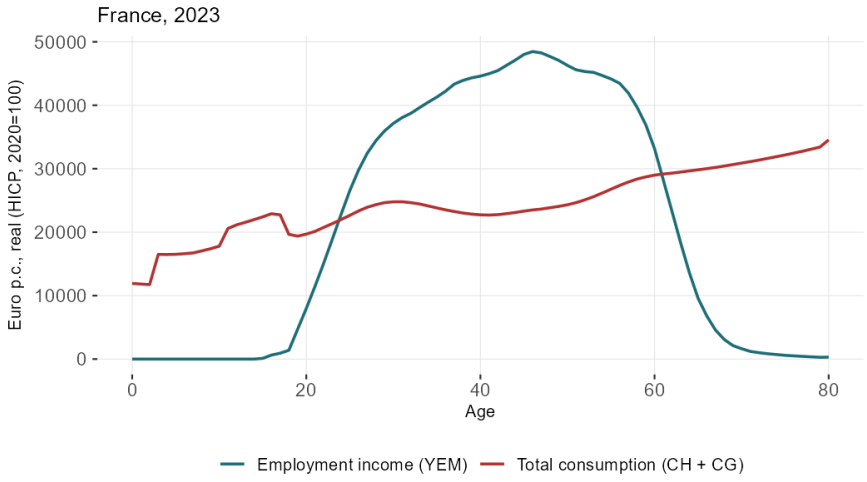


Figure 1.1: Employment income and total consumption by age, France, 2023. Total consumption is the sum of private consumption (CH) and public consumption (CG); euro per capita, real (HICP, 2020 = 100)

1.2.1 The global NTA project

NTA are developed and maintained by a network of research teams in more than 60 countries. More information about the NTA network, the framework, and NTA data can be found on the website of the NTA project at `https://www.ntaccounts.org`. The general NTA methodology is described in detail in a manual published by the United Nations (United Nations, 2013). d'Albis and Moosa (2015) provide a concise description of the concept, and Lee and Mason (2011) describe the methodology and

NTA results for a wide range of countries. A useful reference is also the AGENTA manual (Istenič, Hammer, Šeme, Dolinar, & Sambt, 2019), which describes an earlier version of European NTA data (reference year 2010) that are available on `www.wittgensteincentre.org/ntadata`.

NTA provide information on economic quantities by demographic characteristics and can be used for a range of purposes and research questions. Originally, NTA were designed to better understand the macroeconomic consequences of population ageing. See, for example, Mason, Lee, and members of the NTA network (2022) for a comprehensive overview. In particular, NTA have been used to assess the potential for a demographic dividend (e.g., Dramani & Oga, 2017; Ladusingh & Narayana, 2012). Among the central NTA topics are also the consequences of demographic change for public transfer expenditures and revenues (e.g., Loichinger, Hammer, Prskawetz, Freiberger, & Sambt, 2017).

NTA have evolved over time and are currently used to study heterogeneities by age and other socioeconomic characteristics. Examples include differences between birth cohorts (Mérette & Navaux, 2025), between sexes (Hammer, Spitzer, Vargha, & Istenič, 2020), by education (Abio, Patxot, Souto, & Istenič, 2021), by income quintile (Hammer, Christl, & De Poli, 2023), by parental status (Vanhuysse, Medgyesi, & Gál, 2023), by race (Oosthuizen, 2019), and by religion or culture (Shraberman & Weinreb, 2024).

1.3 European National Transfer Accounts

1.3.1 Using harmonised Eurostat data

Rather than analysing NTA for each country separately, European NTA teams collaborate intensively to take advantage of harmonised data for EU countries provided by Eurostat to generate and analyse NTA for all European countries jointly. The harmonised European data greatly facilitate and improve cross-country comparisons. An important earlier research output was the European NTA data from the EU-funded AGENTA project (see Section 1.2.1), which cover only the year 2010.

The new European NTA (ENTA) data, constructed mainly as part of the SUSTAINWELL and the Generational Satellite Accounts projects, build on this work and provide age- and sex-specific NTA data for 26 European countries over the period 2008–2023 (`https://www.ub.edu/sustainwell-eu-project`).

1.3.2 Differences between ENTA and global NTA

The organisation of ENTA data differs from the usual organisation of NTA data. The NTA manual (United Nations, 2013) organises the quantities in a life cycle account, which measures the age-specific difference between consumption and labour income, and in a set of age reallocation accounts that measure how the life cycle deficit is financed. These include accounts that capture public transfers, private transfers, and asset-based reallocations.

ENTA instead maintain an ESA-like organisation — ESA, the European System of Accounts used across the EU, is introduced in Section 1.4.3 below — with a primary income account, redistribution

accounts, and a use of disposable income account. Keeping a close link to ESA has two main advantages. First, it facilitates quality control of the ENTA data, which is a challenge for 26 countries over 16 years. Since there is a direct link between ENTA and ESA quantities, it is much easier to identify deviations that may be due to coding errors or differences in the data between countries. Second, it is more intuitive, especially for users familiar with NA but not with NTA.

To ensure that ENTA data can be compared with the data of the global NTA project, ENTA are provided in two formats. One keeps a close link to ESA, also in terms of terminology, and a second dataset uses the terminology of the global NTA project. The NTA-format data file also adds more detail than standard NTA: it splits private-sector quantities into the household sector and the corporation sector.

1.4 Integrating age and sex into National Accounts

The central feature of the NTA methodology is the combination of data and concepts from NA with distributional information from microdata. In doing so, NTA capture the advantages of both data sources: the accounting framework and the higher completeness and reliability of macro data in NA, and the distributional detail from microdata.

NTA use the accounting framework and data from NA. The NTA dataset is constructed and organised such that the central accounting identities of NA hold for each individual and age group: disposable income (YD), defined as primary income (Y), less transfers paid (T), plus all transfer benefits received (B), equals the sum of consumption (C) and saving (S).

$$Y - T + B = C + S = YD \tag{1.1}$$

This relation is also referred to as the NTA flow identity. NTA do not only use the concepts from NA, but also the data: the sum of age-specific means of income, public transfers, consumption, and saving over all individuals in all age groups equals exactly the values for the total economy as captured in NA.

However, the integration of demographic characteristics into NA and the focus on redistribution require two major adjustments. First, in NTA, institutional units are defined and aggregated differently than in NA. Second, quantities are measured at basic prices rather than at market prices.

1.4.1 Economic accounting with people

The most fundamental difference between NTA and NA is the definition and aggregation of institutional units. An institutional unit is defined as an *economic entity that is capable, in its own right, of owning assets, incurring liabilities, and engaging in economic activities and in transactions with other entities.*[1] In NA, an institutional unit could be a household, a corporation, or a government department. Institutional units in NA are grouped into sectors: households, corporations, government

[1] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Institutional_unit

units, non-profit institutions serving households (NPISH), and institutional units abroad (grouped into the rest-of-the-world sector, ROW). NA measure the value of economic transactions in which institutional units engage and provide their total value by type of transaction and by sector.

In contrast, in NTA the institutional units are individuals. This enables their aggregation by socioeconomic characteristics such as age and sex. Defining individuals as institutional units has an important consequence: it requires the measurement of transfers between household members. Transfers in NA are defined as *two-party transactions in which one party provides a good, service, or asset to the other but does not receive recompense in return*—a "something for nothing" transaction, or a transaction without a quid pro quo (United Nations, 2011, §3.58). The most important transfers are taxes and social contributions, social benefits, government consumption, and transfers from parents to children.

1.4.2 Measurement at basic prices

NA value transactions at market prices, whereas NTA value them at *basic prices*, i.e., excluding taxes on products and including subsidies on products (D21, D31).[2]

This is necessary for a consistent treatment of redistribution. At market prices, consumption includes taxes less subsidies on products. Disposable income would then have to cover these taxes, although it is meant to be income *after* taxes and benefits, and the taxes would never show up as a transfer to the government. NTA therefore treat taxes on products as a transfer from the private sector to the government in the redistribution accounts, and measure consumption at basic prices, net of these taxes. In ESA, by contrast, they are primary income of the government. Without this reclassification, redistribution through government transfers could not be integrated consistently into ENTA.

1.4.3 ENTA and the European System of Accounts (ESA)

ESA is the system of national and regional accounts used by the members of the European Union. It is EU-specific but consistent with the international guidelines for the compilation of national accounts (United Nations, 2011) and is described in detail in EUROSTAT (2013). The data are publicly available through the Eurostat database. ENTA use annual sector accounts as their central dataset and complement them with additional data, such as data on taxation trends and detailed information on public and private consumption expenditure.

In general, the ENTA aggregates are calculated to be as close as possible to the central quantities in ESA: the measure of total primary income in ENTA corresponds largely to the *balance of primary incomes* in ESA; total income after public and private redistribution in ENTA corresponds to *disposable income*; and the ENTA quantities of *consumption* and *saving* correspond to their equivalents in ESA.

[2] The basic price is the amount receivable by the producer from the purchaser for a unit of a good or service produced as output, minus any tax payable and plus any subsidy receivable by the producer as a consequence of its production or sale. Source: `https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Basic_price`

1.4.4 The sequence of European transfer accounts

NTA inherit accounting identities from NA. Primary income plus all transfers received less transfers paid equals disposable income, which is used for consumption and saving. In ENTA, this accounting identity for the private sector can be written as follows:

1. Primary income (Y)
2. − Government transfer contributions (T)
3. + Government transfer benefits in cash (BS)
4. + Funded pensions, received less paid (BPPEN)
5. + Private sector transfers, received less paid (BPB; transfers within households net to zero in the aggregate but not by age)
6. = Disposable income (YD)
7. − Private consumption of households (CH)
8. +/− Government consumption (CG): received as an in-kind transfer (+), but at the same time constitutes consumption (−)
9. = Saving (SH)

ENTA provide detailed information for each of these components and organise this information in separate accounts. The preparation of ESA data for use in NTA results in a condensed and intuitive organisation of NA data (Table 1.1). It starts with an income account measuring primary income by type; captures public and private redistribution in the transfer accounts; and measures disposable income and its use for consumption and saving.

Table 1.1: The sequence of European National Transfer Accounts

0. Production account (only at aggregate level, not age-specific)		
0	Gross Domestic Product	GDP
0.1	Taxes less subsidies on products	TPS
0.2	Consumption of fixed capital	CFC
0.3	Employment income, net from ROW	YEROW
0.4	Property income, net from ROW	YPROW
1	Primary income	Y
1. Primary income (primary distribution of income account in ESA)		
1	Primary income	Y
1.1	Employment income	YEM
1.1.1	Compensation of employees	YE
1.1.2	Mixed income from self-employment	YM

continued on next page

Table 1.1 (continued)

1.2	Asset income	YA
	2. Government transfer contributions (taxes & social contributions)	
2	Taxes, total	T
2.1	Taxes on employment income	TYEM
2.2	Taxes on assets and asset income	TYA
2.3	Taxes on income, non-employed	TB
2.3.1	Taxes on pay-as-you-go pensions	TBSPEN
2.3.2	Taxes on non-pension social benefits	TBSN
2.3.3	Taxes on benefits, funded pensions	TBPPEN
2.4	Taxes on consumption	TCON
2.5	Other current transfers (paid less received)	TX
2.6	Government to ROW transfers (total economy)	TROW
	3. Government transfer benefits in cash (social benefits)	
3	Social benefits	BS
3.1	Social benefits, pensions	BSPEN
3.2	Social benefits, family	BSFAM
3.3	Social benefits, unemployment	BSU
3.4	Social benefits, other	BSX
	4. Funded pensions, mandatory	
4	Funded pensions (received less paid)	BPPEN
4.1	Contributions	TPSOCfc
4.2	Benefits	BPSOCfc
	5. Private transfers	
5.1	Private transfers between households	BPB
5.1.1	Social contributions, nonfin corp.	TPSOCnfc
5.1.2	Social benefits, nonfin corp.	BPSOCnfc
5.1.3	Private transfers from ROW (received less paid)	BROW
5.1.4	Other private	BPX
5.2	Intra-household transfers received	BPW
5.3	Intra-household transfers paid	TPW
	6. Disposable income	
6	Disposable income	YD
	7. Government consumption	
7	Public consumption	CG
7.1	Public consumption, education	CGE

continued on next page

Table 1.1 (continued)

7.2	Public consumption, health	CGH
7.3	Public consumption, other	CGX
	8. Private consumption	
8	Private consumption	CH
8.1	Private consumption, education	CHE
8.2	Private consumption, health	CHH
8.3	Private consumption, housing	CHR
8.4	Private consumption, other	CHX
	9. Saving	
9	Saving	SH

2. ENTA aggregate data

The sequence of transfer accounts

2.0 Production

In the ENTA context, the production account shows how primary income is derived from GDP. It identifies the economic quantities that are part of GDP but not of primary income, thereby explaining the difference between production (GDP) and income—a gap that varies considerably across European countries and over time.

The production account is not part of age-specific ENTA and is available only at the aggregate level. Neither production nor the income generated from production can be directly assigned to individuals or age groups. Only after distributing income to the owners of the factors of production is it possible to measure the income of individuals and subpopulations. ENTA includes the production account to highlight the difference between the widely used GDP and primary income, which is the starting point of the age- and sex-specific ENTA.

The difference between GDP and primary income, as defined in ENTA, consists of four components: taxes on products (TPS), consumption of fixed capital (CFC), property income received from the rest-of-the-world sector less the property income paid (YPROW), and compensation of employees received from the rest of the world less the compensation paid (YEROW). The rest-of-the-world sector (ROW) refers to all institutional units in other countries.

The first step towards the ENTA measure of primary income is the calculation of GDP at basic prices, i.e. less the taxes on products. Measurement at basic prices corrects for the double counting of taxes on products at the level of the total economy (see also Section 1.4.2). For the economy as a whole, GDP measured at basic prices corresponds to the sum of value added across all sectors.

The second step is to move from gross to net values by subtracting the *consumption of fixed capital* (CFC) from value added. "Gross" in the NA context refers to quantities that include CFC, while "net" refers to quantities without CFC. Specifically, gross value added minus CFC is termed *net value added*. CFC represents part of the production costs and expenditures that must be met to maintain capital stock.[1] For this reason, the ESA 2010 manual also highlights that it is more meaningful to express income in net terms (EUROSTAT, 2013, §8.06).

The third step in deriving an income measure from GDP involves two changes: moving from a production measure to an income measure, and shifting from a domestic focus to a national focus by accounting for primary income flows to and from other countries. The *compensation of employees* and *net property income* received from other countries are added to net value added at basic prices,

[1] See the Eurostat glossary on Consumption of Fixed Capital: `https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Consumption_of_fixed_capital_(CFC)`

while the income paid to other countries is subtracted. Accounting for these income flows yields the *balance of primary incomes* at basic prices at the sector level and *net national income* for the total economy.

ENTA primary income corresponds largely to the ESA quantities *balance of primary income* at the sector level and net national income for the total economy. There is a small difference, as taxes less subsidies on products paid to the rest of the world (ROW) are not part of the balance of primary income in ESA. However, they are part of primary income in ENTA and are captured as transfers to the ROW in the transfer accounts.

At the sector level, there is also a small difference between the ESA and ENTA concepts, as taxes on production are assigned to the primary income of private sectors in ENTA rather than to the government. They are captured as a transfer to the government in the transfer accounts later in the sequence. Consequently, ENTA primary income for the corporation sectors is higher than the balance of primary incomes in ESA, because it refers to income before paying these taxes; the balance of primary income in ESA refers to income after paying the taxes on production. For the government sector, ENTA primary income is lower than the balance of primary incomes in ESA, because it refers to income before receiving taxes on production.

Table 2.1: The production account: from GDP to primary income, 2023

ESA quantity	Germany	Ireland	Luxembourg
0 GDP	50,660	99,247	123,218
0.1 - Taxes less subs. on products (TPS)	-4,387	-5,249	-10,735
0.2 - Consumption of fixed capital (CFC)	-10,300	-24,307	-16,729
0.3 + Employment income, net from ROW (YEROW)	68	-81	-21,539
0.4 + Property income, net from ROW (YPROW)	1,624	-25,142	-17,780
1 Net primary income, basic price (Y)	**37,666**	**44,467**	**56,434**

The difference between GDP and income varies significantly across European countries (Table 2.1). Income flows to other countries are particularly large in Ireland and Luxembourg. As hubs for many multinational companies, a large part of production is reported in these countries; however, part of the generated income is paid as property income to owners in other countries. In Luxembourg, a large part of the compensation of employees is additionally paid to employees not residing in the country. Income paid to the rest of the world amounted to 25% of GDP in Ireland in 2023 and 32% in Luxembourg.

The development of GDP and net primary income differs over time. In general, the capital intensity of European economies is increasing; therefore, the consumption of fixed capital is also increasing. Consequently, net primary income increases less than GDP. This is clearly visible in countries with stagnating GDP, such as Austria. While real GDP per capita was 1% higher in 2023 than in 2008, real net primary income per capita was 2% lower (Figure 2.1).

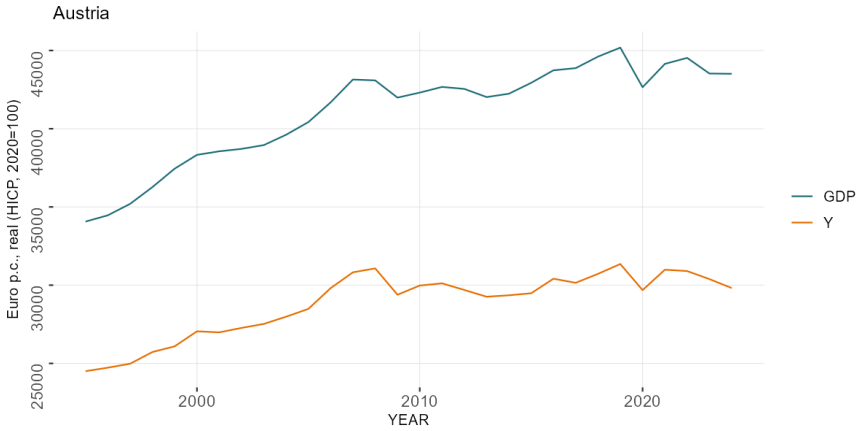


Figure 2.1: GDP and primary income (Y) over time, Austria

2.1 Primary income

The ENTA income account summarises the information from the *allocation of primary income account* in ESA and provides details on the types and amounts of primary income received. *Primary income* refers to income earned directly from production, including wages, salaries, and income from ownership of assets — before any taxes or other transfers are paid. ESA distinguishes between *compensation of employees (YE)*, *mixed income (YM)*, and asset income (YA), which consists of *operating surplus (YO)* and net *property income (YP)*.

The compensation of employees includes wages before the payment of taxes and social contributions, as well as employers' social contributions. Mixed income refers to the income of unincorporated enterprises, often also referred to as income from self-employment, which implicitly consists of an element reflecting the return on labour input and an element reflecting the return on employed capital; hence the term "mixed income." Property income refers to income received by the owner of financial assets, for example in the form of interest or dividends, or by renting out land. The operating surplus of households includes the imputed rent for owner-occupiers and income from renting out dwellings. For the corporate sector, it refers to the surplus from production before paying interest or dividends. For use in ENTA, all income quantities are measured net of the consumption of fixed capital.

There are small differences between ESA and ENTA regarding the allocation of *taxes less subsidies on production* among sectors and their classification. In ENTA, the taxes less subsidies on production (paid) are treated as primary income of the private sectors. Later in the sequence of accounts, they are captured as a transfer to the public sector. By contrast, in the *balance of primary income* in ESA, they are treated as income of the public sector. A second difference is that part of the taxes on production, less the subsidies, is classified in ENTA as part of the compensation of employees. Especially in Austria and Sweden, the taxes on production contain a range of taxes on payroll and employment. In ENTA, these taxes are treated similarly to employers' social contributions and included in the primary income of employees; consequently, they are later treated as a transfer from employees to the government.

Table 2.2 gives an overview of the relative importance of the components of primary income. The *compensation of employees* is, by far, the most important source of income. While the table provides a detailed overview of the types of income of the household sector, only total income is reported for corporations and the government sector. The tables include only values for which age-specific information is available in ENTA, and this is not the case for the income components of the corporate and government sectors.

Table 2.2: Primary income account with values for Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
1. Primary income (Y)	**37,666**	**35,028**	**2,789**	**-152**
1.1 Employment income (YEM)		**28,589**		
1.1.1 Compensation of employees (YE)		26,842		
1.1.2 Mixed income from self-empl. (YM)		1,747		
1.2 Asset income (YA)		**6,440**		
1.2.1 Property income (YP)		5,640		
1.2.2 Operating surplus (YO)		799		

The development of primary income since 2008 differs markedly across EU countries. Real primary income per capita grew strongly in Eastern European countries from a low base (Poland +32% between 2008 and 2023, Lithuania +48%), and also in Denmark (+18%) and Germany (+17%). It stagnated or declined at a high level in Austria, France, Sweden and Finland, and fell in Southern European countries, particularly between 2007 and 2015 (Italy −17%, Spain −11%, Greece −29%). Figure 2.2 illustrates these developments for Denmark, Poland, France, and Italy.

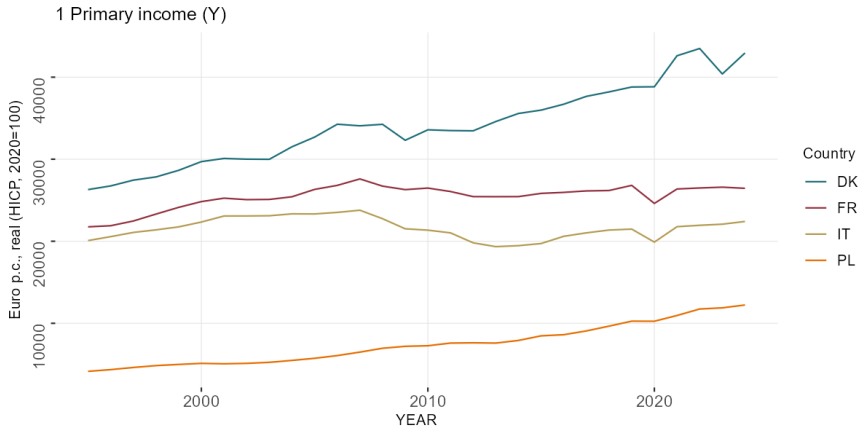


Figure 2.2: Primary income in selected countries and its change over time, prices of 2020

2.2 Government transfer contributions (taxes & social contributions)

Government transfer contributions in ENTA consist of all flows to the government sector in the form of taxes, social contributions, and other transfers. In ESA, these flows are captured in the secondary distribution of income accounts and consist primarily of *current taxes on income and wealth*, *social contributions*, and *other current transfers*. In ENTA, the public transfer account also captures public redistribution through *taxes on products*, with value added tax as the most important component, and other taxes on production. These indirect taxes are not part of the redistribution of income accounts in ESA.

The ENTA aggregates are based not only on ESA data but also on European data on taxation trends (European Commission, 2026) to distinguish taxes and social contributions by source or "economic function" (`https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table`). Based on the categorisation by economic function, the taxes and social contributions used in ENTA are classified into taxes on employment income (TYEM), taxes on assets and asset income (TYA), taxes on the income of non-employed persons (TB), taxes on consumption (TCON), and other transfers (TX).[2] The categorisation by economic function facilitates the allocation of tax payments by age, sex, and other demographic characteristics.

Additionally, EUROMOD is used to decompose the taxes paid by non-employed individuals into taxes on public pay-as-you-go pensions, taxes on non-pension social benefits, and taxes on mandatory funded pensions. Note that this decomposition is based on microdata; the values are estimates. In particular, small positive values may emerge even when the tax rates are effectively zero; for example, for taxes on non-pension social benefits.

Table 2.3: Public transfer contributions account, Germany 2023, Euro per capita

	Total economy	**Households**	**Corp.**	**Gov.**
2. Total taxes (TAX)		**-17,262**	**-1,351**	**18,613**
2.1 Taxes on employment income (TYEM)		-11,369		
2.2 Taxes on assets and asset income (TYA)		-536		
2.3 Taxes on income of non-employed (TB)		-1,415		
2.3.1 Taxes on pay-as-you-go pensions (TBSPEN)		-1,238		
2.3.2 Taxes on non-pension social benefits (TBSN)		-11		
2.3.3 Taxes on funded pensions (TBPPEN)		-166		
2.4 Taxes on consumption (TCON)		-4,319		
2.5 Other current transfers (TX)		377		
2.6 Government to ROW transfers	-338			-338

The tax burden in European countries differs considerably, with large changes over time. In general,

[2] Detailed lists of taxes and their assignment to functions for each country are available here: `https://taxation-customs.ec.europa.eu/document/download/600d56d1-ea6e-416e-9981-1c8b9c603b6b_en?`

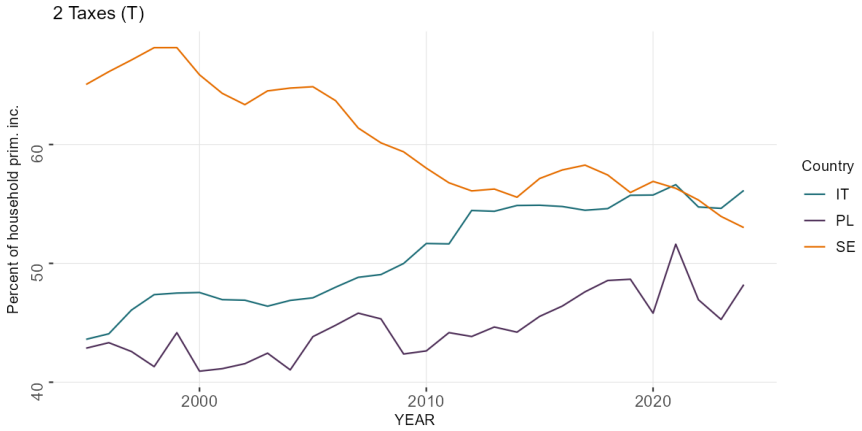


Figure 2.3: Taxes as a percentage of primary income and their change over time, selected countries, household sector

taxes in Eastern European countries are lower than in Central European countries. Among the most remarkable developments is the decline of taxes relative to income in Sweden: they were above 60% until 2007 (65–68% in the late 1990s) and declined to about 56–57% between 2011 and 2019 and to 54% in 2023. By contrast, taxes relative to income in Italy increased from less than 45% in 1995 to about 55% since 2012.

2.3 Government transfer benefits in cash

Government transfer benefits in cash consist of *social benefits* paid by governments to residents, with pensions as the largest component. The ENTA concept differs from the way government expenditure is usually captured in the data: government expenditure typically includes social benefits paid to residents of other countries, whereas ENTA focuses on transfers to residents of the country itself. The ENTA expenditures by function of government are proportionally adjusted to the ESA estimates of social benefits received by the household sector from its own government.

ENTA distinguishes social benefits by function, using ESA together with data on public expenditure classified by the functions of government (COFOG).[3] ENTA aggregates the detailed COFOG categories into fewer components to ensure consistency with the classification in microdata and to have enough observations to estimate age profiles. ENTA distinguishes pensions, including the COFOG categories of old age, disability, and survivors; the category of family & children, which also includes the COFOG category of education; as well as the categories of unemployment, health, and others. Table 2.4 shows that pensions dominate all other categories of social benefits.

Eurostat's COFOG source data does not cover Slovakia before 2010. For 2008 and 2009, the functional shares of social benefits are therefore assumed to be the same as in 2010, the earliest year with a reported breakdown; the resulting category values are then rescaled proportionally so that they sum

[3] See `https://doi.org/10.2908/GOV_10A_EXP`

to the ESA-based total social benefits (BS) for the household sector in the actual year, keeping the imputed years consistent with the rest of the sequence of accounts.

Table 2.4: Government transfer benefits in cash, Germany 2023, Euro per capita

	Total economy	Households	Corp.	Gov.
3. Social benefits, cash (BS)		**7,838**		**-7,838**
3.1 Pensions (BSPEN)		6,165		
3.2 Family & children (BSFAM)		490		
3.3 Unemployment (BSU)		688		
3.4 Other, incl. health (BSX)		495		

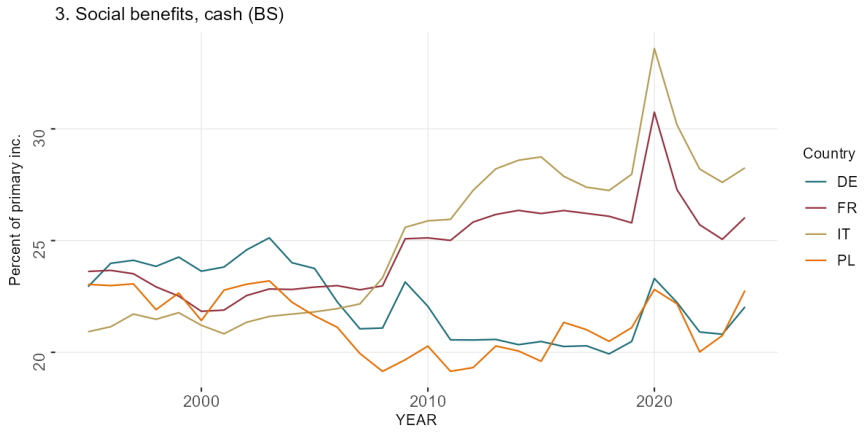


Figure 2.4: Government transfer benefits in cash relative to primary income, selected countries

2.4 Funded pensions

ENTA distinguishes between funded pensions and pay-as-you-go pensions. This requires approximations because funded pensions are treated very differently in ESA and in microdata. In ESA, (mandatory) funded pensions are captured as social contributions and benefits flowing between the household sector and financial corporations. In the microdata, they are included in public pensions together with benefits from pay-as-you-go pension systems.

At the aggregate level, the ENTA contributions to funded pensions consist of the social contributions paid by households to financial corporations. The benefits from funded pensions consist of the social benefits paid by financial corporations to households. The difference corresponds largely to the *adjustment for the change in pension entitlements* in ESA. Treating funded pensions as a separate category facilitates flexible use depending on the exact research question.

ENTA does not use this adjustment (D8) in its accounts. In ESA it is recorded after disposable income and moves the accumulation of pension entitlements from the saving of pension funds to the saving of households. ENTA keeps contributions and benefits gross in disposable income, as ESA

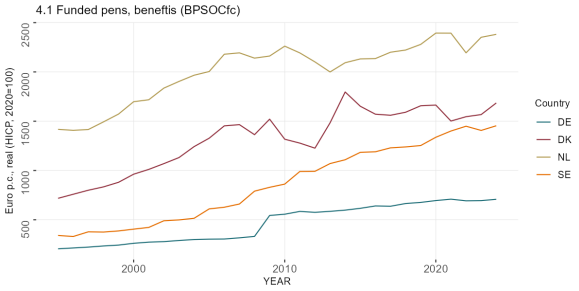


Figure 2.5: Benefits from funded pensions relative to primary income, selected countries

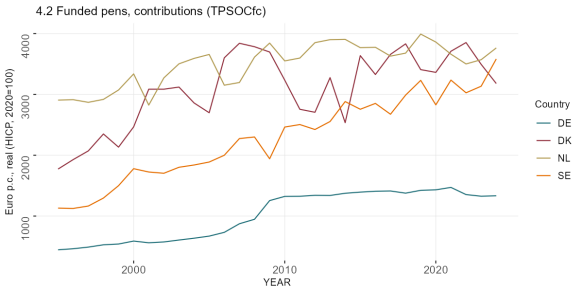


Figure 2.6: Contributions to funded pensions relative to primary income, selected countries

does, and leaves D8 out of disposable income and saving for all sectors. D8 itself is not published; ENTA saving plus D8 equals ESA net saving. In Denmark, the Netherlands and Sweden, household saving in ENTA is therefore noticeably lower, and corporate saving higher, than ESA net saving.

While funded pension schemes exist in many European countries, they are of major importance only in Denmark, the Netherlands, and Sweden. Figures 2.5 and 2.6 show the contributions and benefits relative to the primary income of households.

2.5 Private transfers

Private transfers in ENTA consist of transfers between individuals mediated by the private sector. NTA distinguishes between inter-household transfers, i.e. transfers between households, and intra-household transfers, i.e. transfers within households. Aggregate data are available only for transfers between households.

The most important types of private transfers between households in ESA are the *social contributions* and *social benefits* mediated by corporations, *non-life insurance premiums* and *claims*, as well as *miscellaneous current transfers*. The latter include transfers from households to the NPISH sector and transfers between households, such as remittances and alimony payments. The age-related redistribution through these transfers is small and difficult to measure. ENTA therefore focuses on the net value, measured as the difference between transfers received and paid, and allocates it by a receipt profile, employment income plus pensions, which is zero below age 15 (see Chapter 3).

ESA reports, for each sector, the total of miscellaneous current transfers (D.75) paid and received, but not who pays whom. ENTA needs the flows between households, corporations, government and the rest of the world, and derives them from the sector totals with four proportional rules, applied in this order:

1. Transfers paid by the rest of the world go to households and corporations in proportion to what each receives, at most what the two receive together; any remainder goes to government.
2. Transfers received by the rest of the world come from households and corporations in proportion to what each pays, with the same limit; any remainder comes from government.
3. Government pays households and corporations in proportion to what they still have to receive, at most that amount; any remainder stays within government.
4. Households and corporations pay the remaining receipts of households and government in proportion to their remaining payments; the rest goes to corporations.

Government is not part of the first two rules, since its international transfers are recorded as D.74 and D.76. The rules reproduce the payments of every sector and the receipts of households, government and the rest of the world exactly; the small difference between total D.75 paid and received in the source falls on the receipts of corporations. The rules determine only how household D.75 is split between government (`TX`), other households and corporations (`BPX`) and the rest of the world (`BROW`); the household total and household disposable income do not depend on them.

Table 2.5: Funded pensions and private transfers, Germany 2023, Euro per capita

	Total economy	Households	Corp.	Gov.
4. Funded pensions (received less paid, BPPEN)	**0**	**-752**	**752**	
4.1 Funded pens., contributions (TPSOCfc)		-1,578	1,578	
4.2 Funded pens., benefits (BPSOCfc)		826	-826	
5.1 Private transfers between households (BPB)	**-323**	**-172**	**-151**	
5.1.1 Social contr., private (TPSOCnfc)		-320	320	
5.1.2 Social benefits, private (BPSOCnfc)		211	-211	
5.1.3 Transfer benefits from ROW (rec. less paid, BROW)		-191	-131	
5.1.4 Other private transfer benefits (rec. less paid, BPX)		129	-129	

2.6 Disposable income

Accounting for all types of transfers results in disposable income. It represents the part of income "disposable" for the purpose of consumption and saving. Disposable income in ENTA corresponds to disposable income in ESA, with two differences that follow from the NTA concepts: it is measured

at basic prices, i.e. after taxes less subsidies on products, and it is built from the ENTA definitions of primary income, taxes and social benefits described above. It therefore does not equal Eurostat's B.6n exactly; at market prices the household figure does.

Table 2.6: Disposable income and its use, Germany 2023, Euro per capita

	Total economy	**House-holds**	**Corp.**	**Gov.**
1 Primary income	37,666	35,028	2,789	-152
2 Government transfer contr. + Gov. to ROW	-338	-17,262	-1,351	18,275
3 Government transfer benefits in cash	0	7,838		-7,838
4 Funded pensions		-752	752	
5 Private transfers	-323	-172	-151	
6 Disposable income	**37,005**	**24,680**	**2,039**	**10,285**
7 Public consumption	**10,869**			**10,869**
7.1 Education				1,716
7.2 Health				4,106
7.3 Other				5,047
8 Private consumption	**22,318**	**22,318**		
8.1 Education		178		
8.2 Health		1,176		
8.3 Imputed rent		2,351		
8.4 Other		18,612		
9 Saving	**3,819**	**2,363**	**2,039**	**-583**

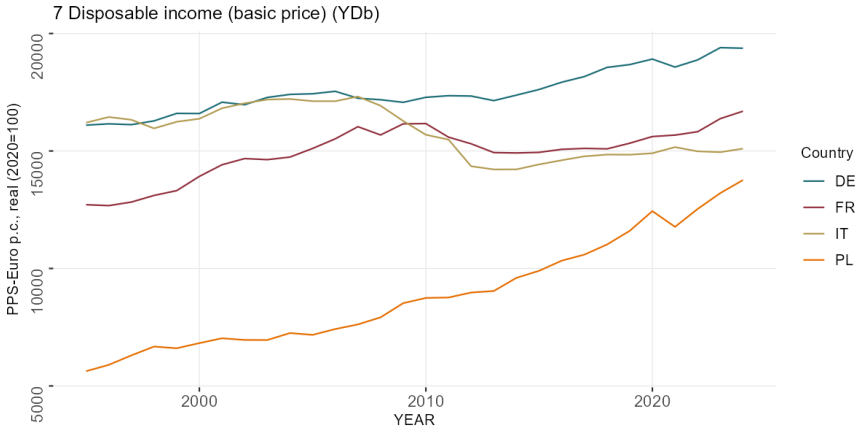


Figure 2.7: Disposable income and its change over time

2.7 Government transfer benefits in kind—government consumption

Government consumption in ENTA comprises the *final consumption* of the government sector as defined in ESA. The distinction by function is based on data on government expenditure by function.[4]

All final consumption of the government sector in ENTA is classified as a transfer from the government to households. ENTA does not directly distinguish between individual consumption (social transfers in kind) and collective consumption. However, the distinction of the categories of education and health identifies the most important components of individual government consumption.

As with social benefits, Slovakia's COFOG-based expenditure by function is not available before 2010. For 2008 and 2009, each category's share of total government consumption (Cb) is assumed to be the same as in 2010 and applied to the actual total for those years. Unlike social benefits, government consumption by function has no independent ESA-based total to rescale against afterwards, so this share-of-total approach is applied directly rather than as a rescaling step.

[4] See `https://doi.org/10.2908/GOV_10A_EXP`

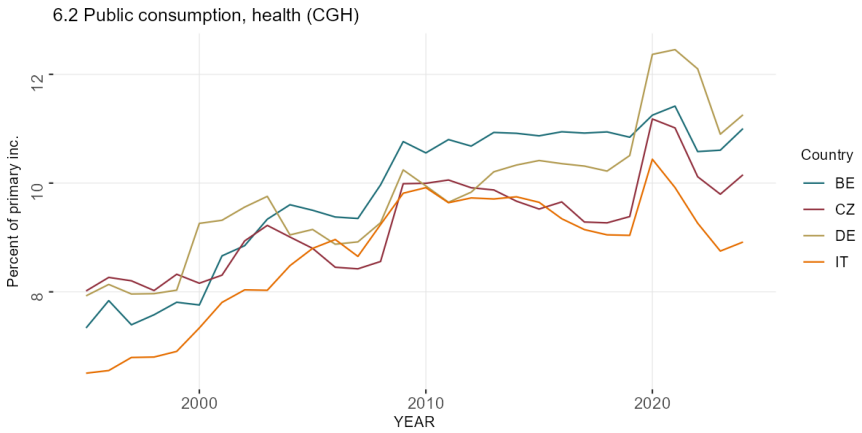


Figure 2.8: Government health consumption relative to primary income, selected countries

2.8 Consumption of households

Final consumption represents expenditure on goods and services that directly satisfy individual needs. Among the most important components are expenditure on housing (including energy), mobility, and food. ENTA consumption corresponds to the final consumption of the household sector in ESA, measured at basic prices.

To distinguish the strongly age-dependent components of education and health, as well as imputed rents, we use data on household final consumption expenditure by purpose.[5]

[5] See `https://doi.org/10.2908/NAMA_10_CO3_P3`

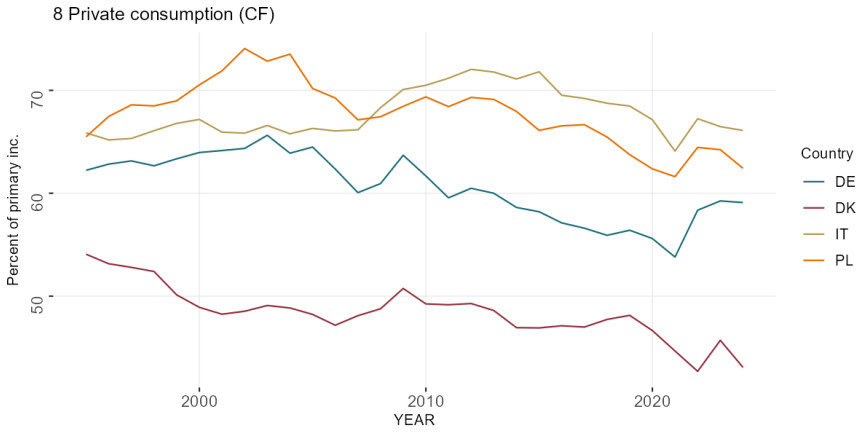


Figure 2.9: Private consumption as a share of primary income, change over time

2.9 Saving

ENTA aggregate data include sector-specific saving, which corresponds to net saving in ESA less the adjustment for the change in pension entitlements (D8; see the section on funded pensions). Household saving rates (saving relative to disposable income) differ considerably across countries. In the 2010s they were highest in France (10–13%), Germany and Austria (about 8–10%), and much lower in Southern and Eastern Europe. They are low or negative in Denmark, the Netherlands and Sweden, partly because the saving of funded pension schemes is not attributed to households in ENTA. An important development over time is the saving rate in Italy, which fell from about 11% in 2005 to 2–5% from 2010 onwards. Most countries show a peak in 2020–2021, when consumption was restricted during the pandemic.

3. ENTA age-specific data

ENTA data consist of age profiles reporting age-specific per capita means for all ages from 0 to 80. The final age group (80) includes the population aged 80 and older. Age profiles are provided separately for men, women, and both sexes combined.

The age profiles derived from EU-SILC and from the consumption imputation are smoothed. Survey-based estimates of age-specific means exhibit considerable variation between single ages, reflecting the high uncertainty arising from the limited number of observations in each age group. ENTA smooths these profiles over age using Friedman's SuperSmoother, implemented via Stata's `supsmooth` command. It improves the age-specific estimates by taking into account the information from neighbouring age groups. The employment profiles from the Labour Force Survey are not smoothed, because the much larger sample makes it unnecessary.

Smoothing of the income, tax and benefit profiles starts at age 18; the values below age 18 are the observed means. The consumption profiles are smoothed over all ages. Negative income is set to zero at the individual level before the profiles are calculated. In the income and tax profiles, ages 15–17 are additionally capped so that they never exceed the value at the next older age (age 18 is the anchor), which prevents spikes just below the smoothed range, and all values below age 15 are set to zero. The benefit profiles apply neither the cap nor the zero below age 15.

The smoother errs most at the ends of the age range. The open top age group (80 and older; 73 and older for Germany 2014–2018) therefore stays in the smoothing but keeps its unsmoothed mean, in all smoothed income, tax and benefit profiles. It averages several hundred persons per sex over some twenty years of age and needs no smoothing, while a smoothed value there can differ markedly from the data (German women aged 73 and over in 2018: 18,070 euro of pensions smoothed against 14,289 euro observed). Unemployment benefits are smoothed differently, because they end abruptly at an age that differs by country, sex and year (see Section 3.3).

3.1 Primary income by age and sex

The age- and sex-specific estimates of primary income in ENTA are based on data from the European Statistics on Income and Living Conditions (EU-SILC). ENTA distinguishes between employment income (YEM), consisting of income from employees (YE) and mixed income (YM), and asset income, which includes property income (YP) and operating surplus (YO).

To avoid age profiles being shaped by outliers, all income components are top-coded. Values that exceed six times the mean of those who receive a certain type of income are replaced with six times the mean.

3.1.1 Employment income

The ENTA estimates of employment income (YEM) are based on several EU-SILC variables. Income of employees is captured in the EU-SILC variables PY010G (wages), PY021G (in-kind components), and PY030 (employers' social contributions) for each individual. A small part is captured in the income of persons below age 16 (HY110G), which is available only at household level and is split evenly among the household's 15-year-olds; if there is no 15-year-old, it is assigned instead to children aged 7–14, and failing that to 16-year-olds. These four EU-SILC variables together largely represent the same concept as the *compensation of employees* in ESA. The EU-SILC equivalent of *mixed income* in ESA is the income of self-employed persons (PY050G).

Although ENTA separate the income of employees (YE) and self-employed persons (YM), we recommend using the age profile of employment income (YEM), which includes both types of income. The reasons are reduced uncertainty due to the larger sample and because smoothing works much better for the combined profile.

Country comparison

The shape of the age profiles of YEM differs considerably across countries. In general, the profiles peak at a much younger age in Eastern European countries, while Western European countries are characterised by a peak between ages 50 and 60. Figure 3.1 shows these patterns for Austria, France, Italy, and Poland.

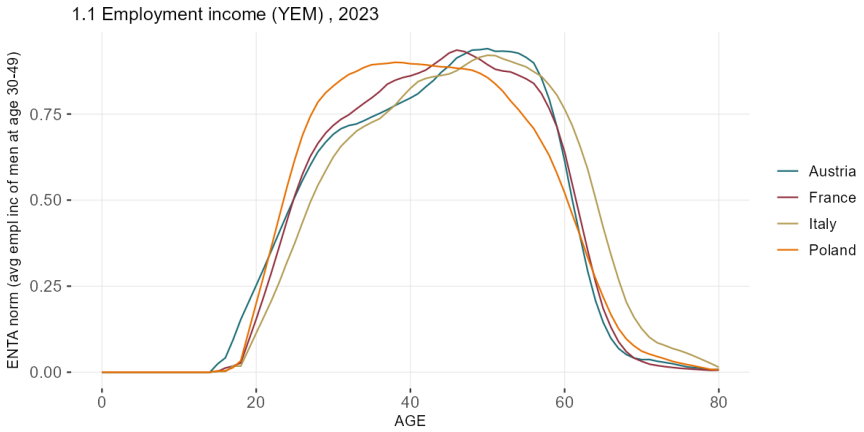
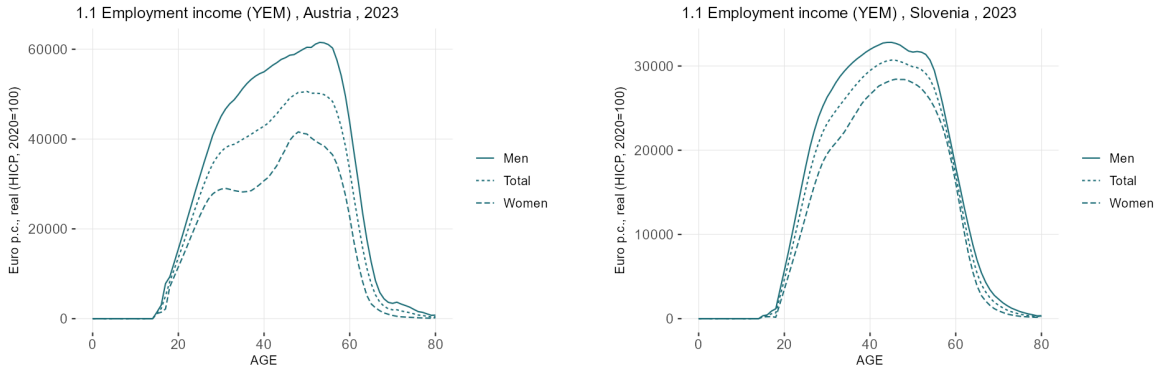


Figure 3.1: Employment income by age in AT, FR, IT, and PL, relative to the average of men aged 30–49

Employment income of men and women differs considerably in all countries, with country-specific patterns. Income and employment of women are high in Baltic countries and Slovenia; however, they are particularly low in German-speaking countries (high part-time rates) and in Southern European countries (low employment rates). Figure 3.2 shows the age profiles by sex for Austria, characterised by large differences, and Slovenia, a country with small differences in employment income between men and women.

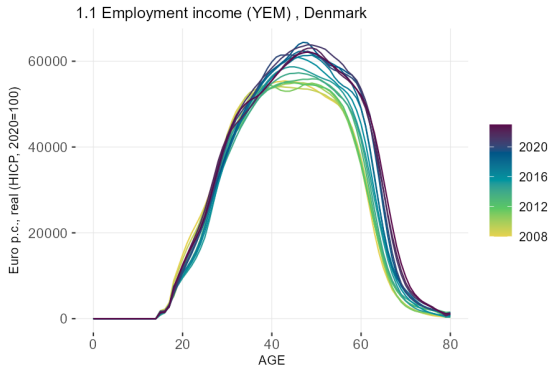


(a) Austria (b) Slovenia

Figure 3.2: Employment income by sex, 2023

Changes over time

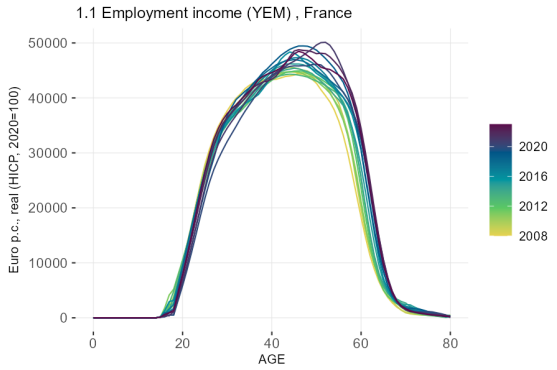
Among the innovations of ENTA is the availability of data from 2008 to 2023, highlighting changes over time. Several of these changes are striking. First, the higher average employment income of the age group 55 to 64 in all countries, which is mostly a result of increasing employment rates. Second, the stagnation of per capita income for the working-age population below age 40 in Western and Northern European countries. Third, the decline of employment in Greece, Italy, and Spain. And fourth, the large increase in Eastern European countries (Figure 3.3).



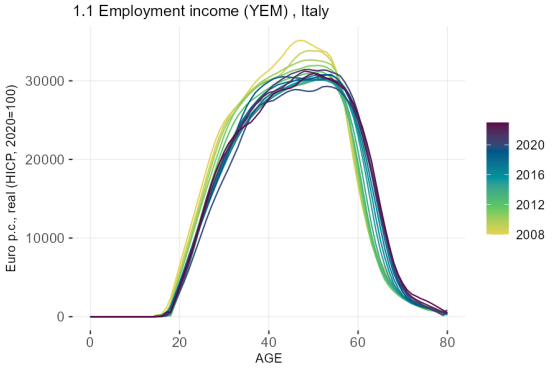
(a) Denmark



(b) Poland



(c) France



(d) Italy

Figure 3.3: Employment income over time, selected countries

3.1.2 Asset income

The age- and sex-specific estimates of ENTA asset income (YA) are based on the following EU-SILC variables: ESA *property income* (received) is distributed according to the combined EU-SILC categories *interest, dividends, and profit from investments in unincorporated businesses* (HY090) and *rental of property or land* (HY040). Property income paid by households is approximated by *interest repayments on the mortgage* (HY100G). ENTA provide the net values, defined as property income received minus property income paid.

In Denmark, property income also includes interest paid before the year 2019. Property income before 2019 is approximated using the information on rental income (HY040) and imputed rent. This delivers a meaningful approximation, but note that for the years until 2019 the age-specific values are not fully comparable to those of other countries.

It is important to note that all the results on asset income based on EU-SILC should be taken with caution. Wealthy households are under-represented in the survey, and total capital income in EU-SILC reflects only a fraction of capital income in ESA.

To allocate the *operating surplus* of households to individuals by age and sex, we use information on ownership status and size of dwelling in terms of the number of rooms. This is an approximation because information on imputed rent (HY030) is not available for all years and countries. Operating surplus is assigned to all adult household members, except for adults who are below 40 years of age and live with their parents. In that case, it is assumed that the parents are the owners of the dwelling and the recipients of the imputed rent.

3.2 Taxes and social contributions

Public transfer contributions in ENTA include all transfers from private sectors (households & NPISH, corporations) to the national government. The largest components are taxes on income, social contributions, and consumption taxes. ENTA does not distinguish between taxes and social contributions; the term "taxes" refers to both. ENTA distinguishes different types of taxes. In particular, taxes on income are decomposed by type of income:

- Taxes and social contributions on employment income (TYEM)
- Taxes on asset income (TYA)
- Taxes paid by non-employed (TB)
- Taxes on government pensions (TBSPEN)
- Taxes on non-pension benefits, such as unemployment and family benefits (TBSN)
- Taxes on funded pensions (TBPPEN)
- Other taxes and transfers (TX)
- Consumption taxes (TCON)

The estimation of age- and sex-specific income taxes by type combines two data sources: the tax estimates from the tax-benefit microsimulation model EUROMOD and income data from EU-SILC.

The EUROMOD data are used to estimate tax rates by age and type of income. EUROMOD microdata include estimates of total taxes paid by each individual (the variable *ils_tax*) as well as a detailed breakdown of social contributions paid. The estimation is restricted to individuals with a meaningful taxable income: those whose income from employment, property, pensions and non-pension benefits is at least the 10th percentile of this tax base in their country and year. We also exclude individuals whose taxes exceed 70% of income in the tax base.

To obtain tax rates by type of income, we regress individual-level total tax payments plus other social contributions (variable *ils_sicot*) on income components. To allow tax rates to vary with income level,

employment income and pension benefits are each split into two regressors depending on whether the individual falls in the lower or upper half of the disposable-income distribution, and the regression also includes the number of children under 15 in the household, interacted with each income half. The estimated coefficients can be interpreted as effective tax rates by income type. These coefficients are then used to allocate each individual's total tax payment across their income sources. This method is robust: since most individuals have one dominant income source, their tax payments are effectively assigned to that source regardless of the precise regression coefficients.

We then add social contributions paid out of employment income and compute age-specific effective tax rates for each income type as the ratio of the allocated tax payments to the corresponding income within age groups. Because a ratio of this kind becomes unreliable where few people in a group have the income, the rates are computed for four broad age groups — 0–19, 20–29, 30–59, and 60 and older — rather than ten-year groups.

The tax rate on property income is computed once per country and year and applied to all ages, since property income is concentrated among few, mostly older, households. All rates are capped at 70%. These rates are not sex-specific; sex variation in the resulting tax estimates comes entirely from applying the same age-specific rate to sex-differentiated income in EU-SILC.

The regression regularly returns negative coefficients for non-pension benefits, and occasionally for pensions or employment income. These do not indicate negative taxes. In most countries, family benefits are received by parents who also benefit from child tax credits or allowances, and tax-free unemployment or sickness benefits replace part of a year's taxed earnings. Recipients of such benefits therefore pay less tax on their other income, and the regression attributes this reduction to the benefit. Negative coefficients are set to a very small positive value before each individual's estimated tax components are scaled to the individual's actual tax payment. As a result, individuals with other income have practically all of their taxes — including the effect of child tax credits — attributed to that income, while individuals living on benefits alone have their full tax payment attributed to the benefits. Alternative specifications (a separate coefficient for family benefits, non-negativity constraints, or minimum rates derived from the taxes of benefit recipients) did not give more stable results across countries. A consequence is that taxes on non-pension benefits (TBSN) are low in countries where family benefits are large and tax-free; in some of them, for example Hungary, Slovakia and Luxembourg, taxes on other non-pension benefits such as unemployment benefits are probably understated.

These age- and income-type-specific tax rates derived from EUROMOD are subsequently applied to EU-SILC income data to impute tax payments. This imputation approach is necessary because year-specific EUROMOD input data are not available for all years since 2008, and EUROMOD output data are published with a considerably longer time lag than EU-SILC. Since EUROMOD is itself based on EU-SILC, the two data sources are in principle compatible, though merging them at the micro level is not straightforward for all countries. The imputation via age-specific tax rates circumvents these practical difficulties while producing reliable age profiles. Note that the estimates of the tax rates reflect the year-specific EUROMOD models, which are available for each year and with little

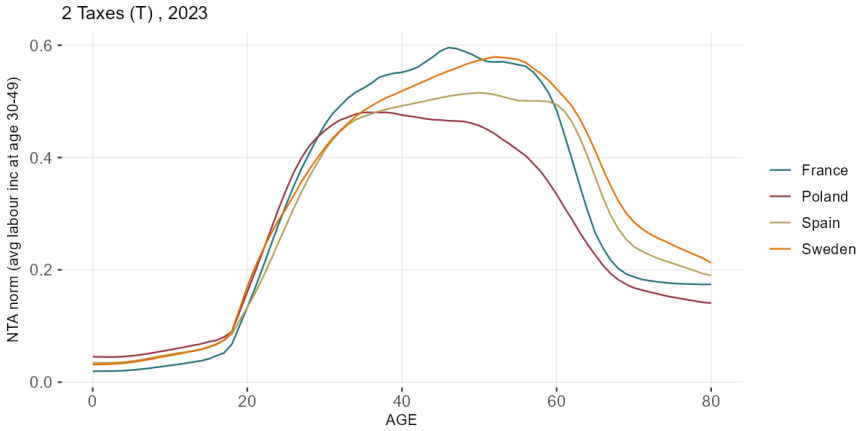


Figure 3.4: Taxes by age in FR, PL, ES, and SE, 2023

time lag — in contrast to year-specific input-microdata.

A sensitivity analysis comparing age profiles constructed directly from EUROMOD output with those based on the imputed EU-SILC data confirms that the results are essentially identical. Note, however, that this imputation approach is designed for age- and sex-specific analysis; analyses along other dimensions (e.g., income deciles or household type) should use EUROMOD data directly, as these capture individual-level variation in tax allowances and benefit taxation more precisely.

Age-specific values for consumption taxes (TCON) are allocated using the age profile of private consumption expenditure. Other current transfers (TX) are allocated by a receipt profile — labour income plus pensions — applied to both components, the household part (TXH) and the part flowing to non-profit institutions serving households (TXNPISH). The reason is that these transfers are already contained in household disposable income before the intrahousehold transfers are estimated, so they reach dependent household members through those transfers rather than directly. Allocating them to children as well would count them twice, and the double count appeared as positive saving below age 15. The profile is therefore zero below age 15, and the total profile is used for both sexes.

Figure 3.4 shows total taxes by age relative to average labour income at ages 30–49. The shape of the profile largely follows the age profile of employment income. At prime working age, the level is highest in France and Sweden and lowest in Poland, which reflects the high taxation of labour income in the former and the low rate in the latter.

3.3 Government transfer benefits in cash

The age- and sex-specific estimates of government transfer benefits in cash are based on EU-SILC. ENTA distinguishes the following categories:

- Pensions (BSPEN)
- Family benefits (BSFAM)

- Unemployment benefits (BSU)
- Other social benefits (BSX), including health

Note that subsidies are captured in the government transfer contributions, which are defined as taxes paid less subsidies received. In principle, EU-SILC distinguishes social benefits by function in a similar way to COFOG in the aggregate data on government expenditure. However, the exact concepts in EU-SILC and the aggregate data are not fully consistent, and for the smaller subcategories, there are too few observations to estimate age profiles by sex. This is why ENTA combines several types of transfers into larger groups to ensure consistency with the aggregates and the reliability of the age-specific estimates.

In ENTA, the social benefits (other than social transfers in kind) in the COFOG functions "old age", "survivors", and "disability" are combined into the category pensions (BSPEN). The age- and sex-specific estimates are based on the EU-SILC variables old-age benefits (PY100G), survivor benefits (PY110G), and disability benefits (PY130G). The ENTA category family and children (BSFAM) includes the COFOG categories "social protection family & children" and "education". Age-specific estimates are based on the EU-SILC variables family-related allowances (HY050G) and education-related allowances (PY140G). Family-related allowances are assigned to the parents of the children. Within couples, they are allocated inversely to the partners' shares of personal gross income (bounded between zero and one), so that the partner with the lower income receives the larger part. Where no child can be attributed, they are shared among all adults of the household, also inversely to their income shares. Where the income of the couple or of the household is not positive, or there is only one adult, the shares are equal. Unemployment (BSU) is based on PY090.

Unemployment benefits end within a year when the unemployed move into retirement or pre-retirement schemes, at an age that differs by country, sex and year (Polish women 57 to 63, Italian women 60 to 67, men mostly 65 to 66). Smoothing across that age would spread benefits into ages at which almost nobody receives them. ENTA therefore determines, for each country, year and sex, the age at which BSU ends: the first age from 57 at which BSU stays at or below 15% of its mean at ages 50–56 for three ages in a row, replaced by the median over the previous, current and following year, or 62 where no such age is found. The profile is smoothed up to four years before that age, the last three years before it are a 1-2-1 moving average of the observed values, and from that age on the observed values are kept. In a split-sample test (fitted on a random half of the households, scored on the other half, 26 countries, 2007–2023) this reduced the error at ages 60–64 by 6.5% against smoothing up to age 61, with no loss at younger ages.

Where benefits in PY090 are in fact early-retirement pay, as with the Danish *efterløn*, pensions take over from them at the same age: their sum changes smoothly, while each part has a step. Keeping the step in unemployment benefits but smoothing it out of pensions would count both before that age and neither after it; before this was corrected, total benefits in Denmark showed a spike at 65 and a dip at 66. From three years before the age at which BSU ends to two years after it, pensions are therefore the smoothed sum of unemployment benefits and pensions, minus the unemployment benefits. The adjustment matters mainly for Denmark, and to a lesser extent for Belgium, Finland

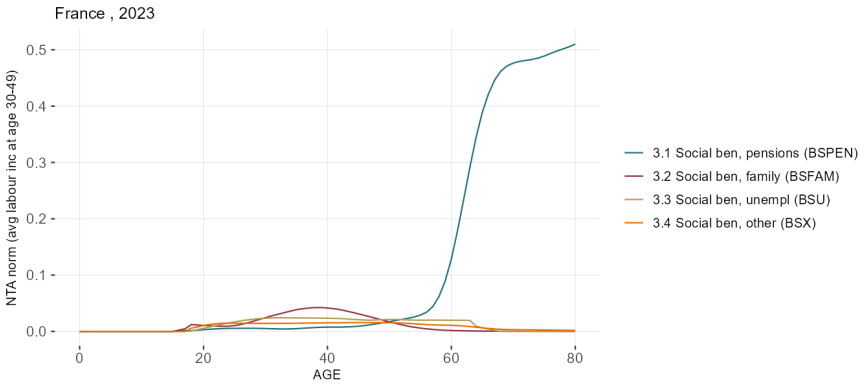


Figure 3.5: Social benefits by type and age, France 2023

and Ireland; elsewhere unemployment benefits are small at these ages.

The category other social benefits (BSX) combines social exclusion (HY060G), housing allowances (HY070G), and sickness benefits (PY120), distributed to all adults in equal shares, because the amounts are small and there are simply too few observations to allow age- and sex-specific estimates.

The age profiles for France in Figure 3.5 show the typical age patterns. In general, government benefits in cash consist mainly of pensions, which are directed to the population aged 60 and older. Of some importance are family benefits, which are concentrated at ages 30–45. Unemployment benefits are rather equally distributed over the whole working age.

The age pattern of government cash transfers differs across countries, with Southern European countries being the most generous to the older population. The differences between countries are also dominated by differences in the size of government pensions. Northern European countries are characterised by moderate government pensions and late retirement. Italy in particular is characterised by large pensions relative to labour income in prime working age. Other types of benefits are of less importance in terms of total size; in Denmark, Sweden, and the Netherlands, somewhat more cash transfers are provided to the young, especially in comparison with Southern Europe.

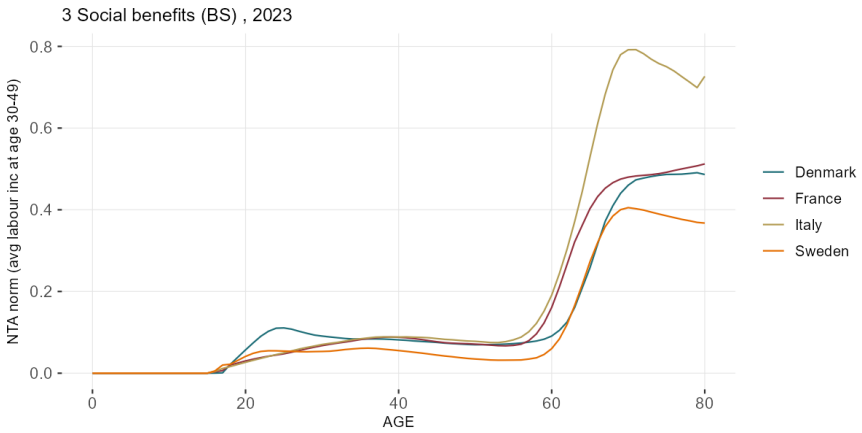


Figure 3.6: Social benefits in cash by age; DK, FR, IT, and SE in 2023

3.4 Funded pensions

Funded pensions in ENTA consist of contributions paid to mandatory funded pension systems and the benefits received from these systems. The aggregate values for contributions and benefits are approximated by social contributions paid by households to financial corporations and the benefits paid by financial corporations to households.

Unfortunately, the microdata make no distinction between contributions to funded and pay-as-you-go systems, nor between benefits from funded and pay-as-you-go systems. However, to allow a flexible adaptation of ENTA to research questions that may differentiate between funded and pay-as-you-go systems, and to maintain consistency with ESA at the sector level, ENTA distinguishes between the two types of pensions. Because the microdata cannot separate the two, ENTA assumes that the shape of the age profiles does not depend on the type of contributions and benefits. To allocate contributions by age and sex, ENTA uses the age profiles of taxes on employment income; benefits are allocated based on the age profiles of pension benefits including mandatory funded pensions (BSPPEN).

3.5 Private transfers

NTA distinguish between private transfers between households and private transfers within households.

3.5.1 Private transfers between households

On average, private transfers between households have little influence on the disposable income of households in EU countries. Private transfers between households comprise all transfers mediated by the household sector (e.g. remittances, alimony payments) and the corporation sector (non-life insurance premiums and claims). Note that social contributions and benefits paid to and received from financial corporations are not included here but under funded pensions.

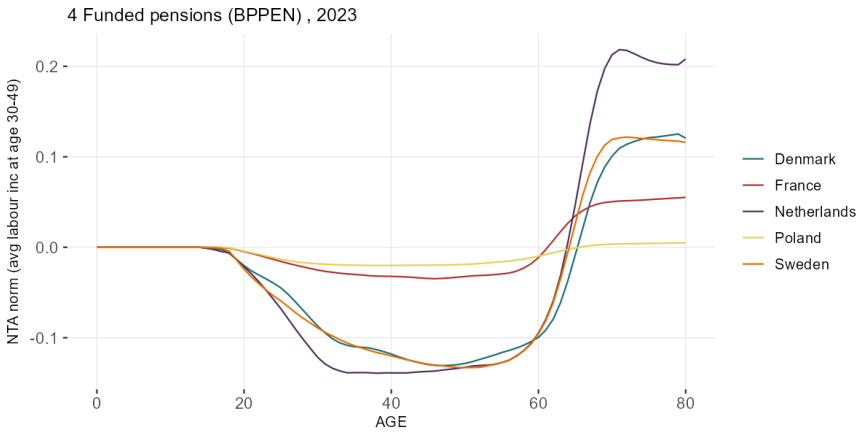


Figure 3.7: Funded pensions in selected countries, 2023

In principle, some private transfers are captured in EU-SILC, with alimony payments and remittances being important components. However, three challenges limit their usefulness for constructing age profiles. First, the number of persons involved in such transfers is small in each age group, resulting in high uncertainty in age-specific estimates. Second, there appear to be data quality issues, with substantial discrepancies between aggregate transfers paid and received. Third, the concept of private transfers in EU-SILC does not map directly onto the corresponding National Accounts aggregate. For these reasons, ENTA relies on assumptions about how private transfers redistribute resources across age groups — see below.

Transfers mediated by non-financial corporations

Social contributions and benefits mediated by non-financial corporations include, for example, the continued payment of wages during illness (Nelisse, 2021). Although some of these transfers may be captured among health-related benefits in EU-SILC, the data do not clearly distinguish between the government and private components (the reported amounts appear too large to reflect only government transfers), and the number of observations is too low to estimate reliable age profiles. ENTA therefore assumes that contributions follow the age profile of taxes on employment income (TYEM), while benefits follow the age profile of employment income (YEM).

Other private transfers

For the remaining private transfers (e.g. remittances, alimony, non-life insurance), ENTA treats the net values (received minus paid) as flows received from outside the household and distributes them by the age profile of employment income plus pensions (YEM + BSPPEN), taken for both sexes together rather than sex by sex, because the household member who receives such a transfer is as often a woman as a man. Crediting these flows to all adults per capita instead would give children a share of a transfer whose receipt they do not carry, on top of an intra-household transfer that already covers their consumption.

3.5.2 Private transfers within households

The estimation of intra-household transfers and private consumption in ENTA goes hand in hand. Estimates of intra-household transfers require microdata that include income and consumption at the individual level. ENTA creates such data by imputing consumption from HBS into EU-SILC. The methodology is described in Section 3.8.4. In the NTA framework, intra-household transfers are derived from the difference between consumption and income at the individual level: if an individual's income falls short of their consumption, NTA assumes that the difference is covered through transfers from other household members.

Estimating intra-household transfers

ENTA assumes that couples share their income: the first type of intra-household transfers in ENTA are transfers between two partners to equalise their income. Since partners are usually of a similar age, this assumption has little effect on the net redistribution between age groups.

After redistributing between partners, based on the estimates of individual income and consumption, each person is classified as having a *surplus* (income exceeds consumption) or a *deficit* (consumption exceeds income). These individual surpluses and deficits are then aggregated at the household level to determine whether total household income is sufficient to cover total household consumption.

Case 1: Household surplus covers the deficit. If the combined surplus of all household members is large enough to cover the combined deficit, the estimation of intra-household transfers is straightforward. A household-specific "internal tax rate" is calculated as the ratio of the total deficit to the total surplus. Each member with a surplus is assumed to transfer this proportion of their surplus to those with a deficit, fully financing their consumption shortfall.

Case 2: Household surplus falls short of the deficit. If the total household surplus is insufficient to cover the total deficit, it is assumed that the available surplus of individual members is used to cover at least part of the deficit. Some members will remain with an uncovered deficit. This remaining deficit is assumed to be covered by the adult income earners in the household, with a contribution proportional to their income.

The situation in which the income of a household falls short of consumption arises frequently in the microdata because the imputed values reflect the high uncertainty in the consumption data. Estimates of consumption for a specific observation are only loosely related to income. These discrepancies cancel out when averaged over whole age groups, so the resulting age profiles remain reliable estimates.

3.6 Disposable income

Disposable income in ENTA reflects all types of primary income and all transfers, including transfers within households. It is calculated based on the age profiles of primary income and transfers:

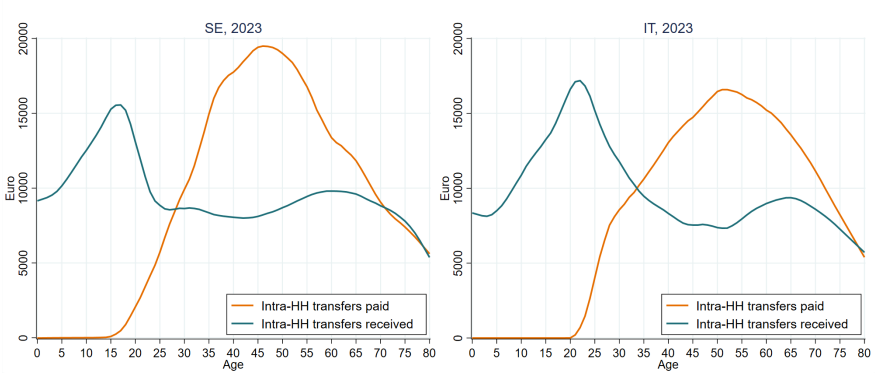


Figure 3.8: Intra-household transfers, paid and received, Sweden and Italy, 2023

$$YD = YEM + YA - T + BS + BPPEN + BPB + BPW - TPW \tag{3.1}$$

where YD is disposable income, YEM is employment income, YA is asset income, T is taxes and social contributions, BS is government transfer benefits in cash, $BPPEN$ is funded pensions (received less paid), BPB is private transfers between households, BPW is private transfers within households received, and TPW is private transfers within households paid.

Disposable income varies considerably by age. Children have little or no primary income; their disposable income consists almost entirely of intra-household transfers received from their parents. During working age, disposable income is dominated by employment income, reduced by taxes and social contributions. In old age, employment income declines and is replaced by public pension benefits and, in some countries, by funded pension benefits and asset income.

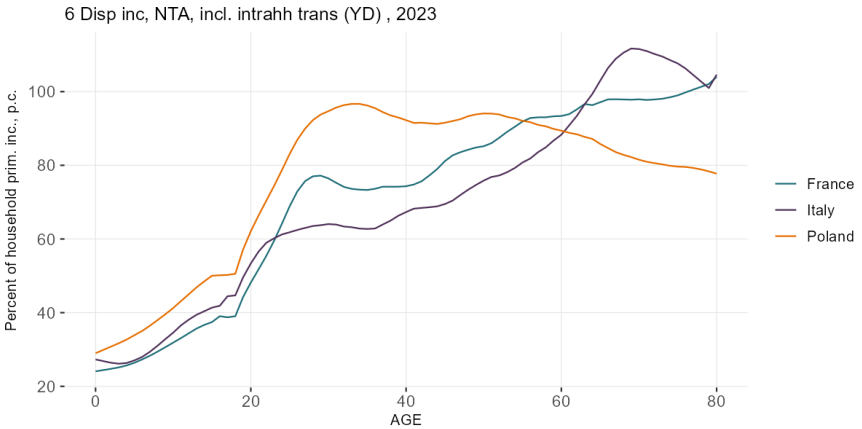


Figure 3.9: Disposable income by age in IT, FR, and PL, in percent of mean primary income of households, 2023

In most countries, NTA disposable income has a first peak in the late twenties or early thirties, before people have children (Figure 3.9). It is lower at ages 35–50, because part of income is transferred

to children. Disposable income rises again in old age in most countries, as transfers to children decline and pensions replace employment income. In Poland, disposable income is particularly high at ages 27–50, at about 90–95% of mean household primary income against 60–75% in France and Italy, owing to high employment income of the young and low fertility, which results in fewer intra-household transfers that need to be paid.

3.6.1 Alternative measures of disposable income

Asset income and private transfers in particular exhibit considerable year-to-year variation, owing to the high uncertainty in their age-specific estimates. Because these components play an important role in disposable income, ENTA provide additional measures that are better suited for the analysis of changes over time.

The first measure ($YD1$) consists of employment income plus social benefits and funded pensions (received less paid), less the taxes on employment income, benefits, and consumption. Estimates of these components are the most reliable and the best suited for comparisons over time. This measure excludes asset income, taxes on asset income, other current transfers, and all private transfers.

$$YD1 = YEM - TYEM + BS + BPPEN - TB - TCON \tag{3.2}$$

Of particular importance in the NTA context are private transfers within households and their effect on disposable income at each age. Because they are based on consumption data, estimates vary considerably over time. However, the estimates do reflect important differences between countries. The second measure ($YD2$) extends $YD1$ by adding private transfers within households:

$$YD2 = YD1 + BPW - TPW \tag{3.3}$$

The third measure ($YD3$) extends $YD2$ by including the operating surplus of households, which consists mainly of imputed rent from owner-occupied housing. This measure excludes property income, which is the income component most difficult to distribute by age and sex:

$$YD3 = YD2 + YO \tag{3.4}$$

3.7 Government consumption

Only expenditure in the functions of education and health is distributed to age groups. For the other consumption components, it is assumed that all residents consume the same amount.

3.7.1 Education

To distribute the consumption expenditure of the government sector to age groups, ENTA uses information on enrolment by age, sex, and education level,[1] as well as information on total expenditure by education level from the data on government expenditure by COFOG. The expenditure data allow a distinction of only three educational groups: pre-primary & primary education, secondary education, and tertiary education. To each enrolled person, ENTA assigns average level-specific spending.

3.7.2 Health

Age profiles of public health care consumption are taken from a single source for all countries: age- and sex-specific per capita public health expenditure approximated from the 2018 Ageing Report of the European Commission and the Ageing Working Group (AWG) (European Commission, 2018). One profile per country is used for all years; in each year it is scaled to the government health expenditure aggregate, so the level follows the aggregate while the shape stays fixed.

Using one source for every country keeps the profiles comparable across countries. Earlier versions of ENTA combined three sources: profiles from the AGENTA project (Istenič et al., 2019) for 15 countries, the 2018 Ageing Report for 10 countries for which AGENTA had no country-specific profile, and national administrative data for Austria. The 2021 Ageing Report is not used, although it is more recent. Its profiles are very similar for most countries, but for some the old-age part is considerably steeper, which most likely reflects a difference between the two editions rather than a change over time (see Section 5.8). The 2018 edition also lies closer to the middle of the period covered by ENTA.

3.8 Private consumption

ENTA includes information on age- and sex-specific expenditure in four categories:

- Education services (CHE)
- Health (CHH)
- Housing (imputed rent of owner-occupiers) (CHR)
- Other consumption (CHX)

The estimates of private consumption in the categories of education, health, and other consumption are based on the Household Budget Survey (HBS), while housing consumption (imputed rent) is based on EU-SILC.

[1] `https://ec.europa.eu/eurostat/databrowser/view/educ_uoe_enra02/default/table` and `https://ec.europa.eu/eurostat/databrowser/view/EDUC_ENRL1TL/default/table`

3.8.1 Household Budget Survey

The Household Budget Survey (HBS) is a consumption expenditure survey carried out periodically in European countries. The HBS contains information on household expenditure classified according to the Classification of Individual Consumption by Purpose (COICOP), together with socio-economic characteristics of households, such as household income and the age and sex of household members.

There are considerable challenges associated with the use of HBS for age- and sex-specific estimates. First, age-specific estimates from the HBS are available only for the survey years 2010 and 2015. The 2020 wave is available but records only whether a household member belongs to the age group 60 and older, severely limiting its usefulness for constructing detailed age profiles. In some countries, HBS data are available for only one or two of these years.

Second, the consumption data in HBS are characterised by high uncertainty for several reasons. Consumption expenditure is not observed over an entire year. Typically, expenditure on frequently purchased items is recorded over a two-week observation period, and this information is complemented by retrospective reporting of spending on larger or less frequent items over the previous year. As a result, the recorded expenditure for any individual household is a noisy measure of its true annual consumption. In addition, expenditure varies considerably across households. Consequently, age- and sex-specific estimates are subject to a high level of uncertainty that is difficult to quantify.

These limitations constrain the level of detail at which consumption can be meaningfully analysed. Reliable estimates can only be obtained for sufficiently large population subgroups and broad consumption categories. Estimates for small subgroups or infrequent consumption items should be interpreted with caution.

3.8.2 Consumption in the categories of education and health

Age- and sex-specific estimates of private consumption expenditure on education and health are based on pooled HBS data for 2010 and 2015. To account for price changes between the two time points, consumption is measured relative to average consumption in each of these years. Data for 2020 are excluded both because of the disruptions caused by the COVID-19 pandemic and due to the limited age information in the 2020 wave. The shape of the age profiles is assumed to be constant over time. Given that household spending on education and health services tends to be small and infrequent, the data would not support the identification of changes in the age pattern over time.

The estimation of age profiles from the individual microdata requires the allocation of household-level spending to individual household members. This is carried out using an iterative method, proceeding as follows:

1. In the initial step, health and education spending is distributed equally among all household members, and preliminary age profiles are estimated from these individual-level values.

2. In each subsequent iteration, the age profile from the previous step is used as a set of weights to redistribute household spending among its members. New age profiles are then re-estimated.

3. This process is repeated 15 times, by which point the age profiles have converged.

3.8.3 Imputed rent

The age profile of the housing consumption in terms of imputed rent (COICOP consumption category 4.2[2]) is allocated to individual household members according to the NTA equivalence scale (described below in Section 3.8.4).

3.8.4 Other consumption

Other consumption comprises all private consumption expenditure not classified as education, health, or imputed rent. Because consumption age profiles are needed also for years without an HBS survey, and because estimates of private intra-household transfers require individual-level consumption data in EU-SILC, other consumption is imputed into EU-SILC using a regression model. The model is estimated as a generalised linear model (gamma family, log link) rather than plain OLS, since OLS produced implausible negative predicted consumption values for some households; the log link guarantees positive predictions by construction.

The model uses HBS from 2010 and 2015 and relates standardised consumption expenditure per household member (from HBS, standardised means that consumption is measured relative to per capita consumption in the specific year) to a set of household characteristics available in both HBS and EU-SILC. These characteristics include the percentile of per capita household income, the number of persons in the household, the number of persons under age 20, the number of persons aged 20–24 and 65–69, and the number of persons in the age groups 70–74, 75–79, and 80+. The age-count variables capture the tendency for per capita consumption to vary systematically with the household's age composition, in particular its decline in households with very old members. In some countries, household income is not available in HBS. In that case, the income percentile is approximated using the percentiles of consumption per capita.

We experimented with a range of other, more sophisticated, imputation methods such as matching and multiple imputation. These methods introduce the high uncertainty of HBS data and the estimates into ENTA, with the result that the age- and sex-specific estimates are hardly useable for comparisons of countries and over time.

The estimated household consumption is allocated to individual household members using the NTA equivalence scale. This scale assumes the following:

- Children aged 0–4 consume 40% of an adult's consumption.
- Adults aged 20 and older consume equal amounts.
- Between ages 4 and 20, the consumption weight increases linearly from 0.4 to 1.0.

For Germany from 2022 onward, the age-16 weight is set to 0.9 rather than following the linear ramp: EU-SILC's age variable for Germany in these years groups ages 16–19 together under the single value

[2] https://unstats.un.org/unsd/classifications/Econ/Download/COICOP_2018_pre_copy_edit_publication.pdf

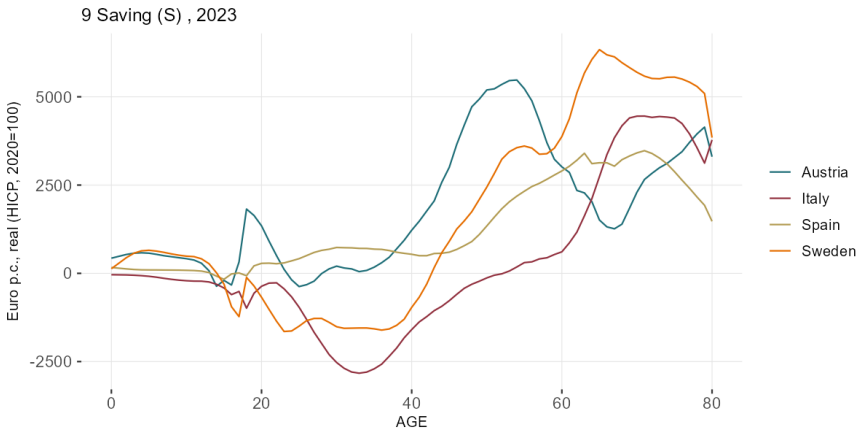


Figure 3.10: Saving by age, 2023, selection of countries

16, so the age-16 weight has to stand in for the whole 16–19 range rather than age 16 alone.

3.9 Saving

Private saving is calculated as a residual, defined as the difference between disposable income and consumption:

$$SH_i = YD_i - CH_i \tag{3.5}$$

where SH_i denotes the saving of individual i, YD_i is disposable income, and CH_i is total private consumption (including education, health, imputed rent, and other consumption).

The residual nature of the saving estimate means that it inherits the measurement errors and assumptions embedded in both the income and consumption estimates. Figure 3.10 shows the estimated age profile of per capita saving for selected countries.

The age profile of saving in ENTA does not follow the simple life-cycle pattern. Saving is close to zero for children. In Italy and Sweden it is negative at ages of about 20–45, in Italy until the mid-fifties, while in Austria it is positive at most ages, with a peak in the fifties. In old age saving is positive in all countries shown in Figure 3.10, at about 3,000 to 6,200 euro per year at age 70, where simple life-cycle models predict pronounced dissaving. Whether the elderly dissave on average depends on the country and on the level of pension income relative to consumption; in many European countries, relatively generous public pension systems result in moderate or even substantial positive saving among the elderly.

4. Translating ESA/ENTA into NTA concepts and terminology

4.1 NTA variables

Although NTA are based on National Accounts, they use a different classification and terminology. ENTA tries to stay as close as possible to the ESA terminology and data, but also provides the data for comparison with results from the global NTA project. ENTA data are therefore provided in two formats: the *ENTA–ESA format*, which maintains a close link to ESA terminology and sector structure, and the *NTA format*, which follows the conventions of the global NTA project. This section describes the main differences between the two formats and how NTA variables are derived from the ENTA–ESA data. Tables 4.1 and 4.2 list all NTA variables and show their correspondence with the ENTA–ESA format.

Providing data in the ESA-related format has two practical advantages. First, the underlying microdata come from household surveys, which map more naturally to the ESA household sector than to the NTA private sector that also includes corporations. Second, the ESA sequence of accounts is more familiar to most users than the NTA life-cycle framework.

4.1.1 Treatment of mixed income

The largest conceptual difference between the two formats concerns income. NTA distinguish between labour income (YL) and asset income (YA). Compensation of employees (YE) is unambiguously labour income. Mixed income (YM), however, contains both a labour and a capital component. In the ENTA–ESA format it is reported as a single category within employment income. In the NTA format, following standard NTA practice, two-thirds is classified as labour income and one-third as asset income of households:

$$YL = YE + \tfrac{2}{3}\, YM, \tag{4.1}$$

$$YAH = YA + \tfrac{1}{3}\, YM. \tag{4.2}$$

Household asset income (YAH) thus includes property income (interest, dividends, rent), the operating surplus (mainly imputed rents from owner-occupied housing), and the capital share of mixed income. Total private-sector asset income combines households and corporations: $YAF = YAH + YAC$.

Asset income of the government (YAG) is often negative, reflecting interest payments on public debt. In ENTA, taxes less subsidies on production are treated as part of government income.

4.1.2 Sectoral detail

The ENTA–ESA format reports quantities for the household sector (including NPISH). The NTA format combines households and corporations into private-sector totals. This applies to asset income (YAC), taxes ($TGOTC$), private transfers ($BPBC$), and saving (SC). The NTA format also includes income and saving of the government sector (YAG, SG), as well as net government transfers to and from the rest of the world ($TGNF$).

4.1.3 Terminology for public and private transfers

NTA use a distinct terminology for transfers. All flows from private sectors to the public sector are *transfer outflows*; social benefits received from the government are *transfer inflows*. Total public transfer outflows (TGO) comprise all taxes plus the difference between non-tax government income and government saving, representing public resources that do not stem from taxes; these are allocated by age according to the profile of total taxes:

$$TGO = TGOT + YAG - SG.$$

Public cash transfer inflows are distinguished by function: pensions ($TGICPEN$), family ($TGICF$), unemployment ($TGICU$), and other ($TGICX$). The "other" category corresponds to other social benefits (BSX) from the ENTA–ESA format, which already includes sickness benefits. Public consumption is treated as an in-kind transfer inflow and split into education (CGE), health (CGH), and other (CGX). Total government inflows are:

$$TGI = TGIC + CG.$$

Private transfers are divided into transfers *between* households (TFB) and *within* households (TFW). Between-household transfers include flows from the corporation sector ($BPBC$):

$$TFB = BPB + BPBC.$$

4.1.4 Disposable income

Disposable income of the private sector (YDF) combines the household and corporation sectors:

$$YDF = YDH + YDC.$$

Household disposable income is:

$$YDH = YL + YAH + TFB - BPBC + TFW - TGOTH + TGIC + TPEN.$$

Corporate disposable income equals corporate saving, since corporations do not consume:

$$YDC = YAC - TGOTC + BPBC - TPEN.$$

Adjusted disposable income of households ($YDHA$) adds government consumption as an in-kind transfer: $YDHA = YDH + CG$.

4.1.5 Saving

Total saving combines private and government saving:

$$S = SF + SG, \qquad SF = SH + SC.$$

Private-sector saving (SF) includes household saving (SH, published under the same code in the ENTA–ESA format) and corporate saving (SC).

4.1.6 Implementation

The Stata programme translates ENTA–ESA data into NTA format. It combines the age profiles estimated from microdata with ESA aggregates and applies the variable mappings described above to generate the final NTA dataset.

Table 4.1: NTA variable definitions

Number	Variable	Abbrev.	ENTA–ESA format
	1. Income		
1	Primary income	Y	
1.1	Labour income	YL	$YE + \frac{2}{3}\ YM$
1.2	Asset income, private	YAF	$YAH + YAC$
1.2.1	Asset income, households	YAH	$YA + \frac{1}{3}\ YM$
1.2.2	Asset income, corporations	YAC	YC
1.3	Asset income, government	YAG	
	2. Transfers to government		
2	Taxes, total	$TGOT$	$T + TC$
2.1	Taxes, paid by households	$TGOTH$	T
2.2	Taxes, paid by corporations	$TGOTC$	TC
2.3	Public transfers, net from ROW	$TGNF$	
	3. Transfers from government (cash)		
3	Social benefits	$TGIC$	BS
3.1	Social benefits, pensions	$TGICPEN$	$BSPEN$
3.2	Social benefits, family	$TGICF$	$BSFAM$
3.3	Social benefits, unemployment	$TGICU$	BSU
3.4	Social benefits, other	$TGICX$	BSX
	4. Funded pensions		
4	Funded pensions (received less paid)	$TPEN$	$BPPEN$
	5. Private transfers		
5	Private transfers	TF	
5.1	Private transfers between households	TFB	$BPB + BPBC$
5.2	Private transfers within households	TFW	$BPW - TPW$
5.2.1	Intra-household transfers received	$TFWI$	BPW
5.2.2	Intra-household transfers paid	$TFWO$	TPW
	6. Disposable income		
6	Disposable income, private sector	YDF	
6.1	Disposable income, households	YDH	YD
6.2	Disposable income, corporations	YDC	

Table 4.2: NTA variable definitions (continued)

Number	Variable	Abbrev.	ENTA–ESA format
	7. Government consumption		
7	Public consumption	CG	
7.1	Public cons., education	CGE	
7.2	Public cons., health	CGH	
7.3	Public cons., other	CGX	
7.4	Adjusted disp. income, households	$YDHA$	$YDH + CG$
	8. Private consumption		
8	Private consumption	CF	CH
8.1	Private consumption, education	CFE	CHE
8.2	Private consumption, health	CFH	CHH
8.3	Private consumption, housing	CFR	CHR
8.4	Private consumption, other	CFX	CHX
	9. Saving		
9	Saving, total	S	$SF + SG$
9.1	Saving, private	SF	$SH + SC$
9.1.1	Saving, households	SH	SH (same code in ESA format)
9.1.2	Saving, corporations	SC	SC
9.2	Saving, government	SG	
	NTA aggregates		
10.1	Total government transfers, net	TG	$TGI - TGO$
10.2	Total government inflows	TGI	$TGIC + CG$
10.3	Total government outflows	TGO	$TGOT + YAG - SG$
10.3.1	Government outflows, asset-based	$TGOABR$	$YAG - SG$
10.4	Total consumption	C	$CF + CG$

5. Limitations

ENTA combines a wide range of European datasets to provide consistent and detailed age- and sex-specific information about the economy in European countries and its change over time. However, the data relies on survey data, assumptions, and approximations of certain quantities. This chapter provides an overview of the limitations, to avoid misinterpretation of results regarding differences between countries and changes over time.

It also documents every place where the ENTA code treats one country, or a small group of countries, differently from the rest. Each such rule exists because a source is missing, is reported on a different age grid, or is defined differently for that country. None of them are adjustments made to improve the appearance of a result. The list covers every conditional in the code that names a country; where a rule has no justification recorded in the project, this chapter says so rather than supplying one.

Section 5.2 holds what applies to every profile. Sections 5.3 to 5.10 follow the order of Chapter 3 and of the codebook, and each one gives the general limitations of a quantity first and the country-specific ones after. Patterns that are in the data because they happened — economic events, policy reforms — are not limitations and are in Appendix E instead.

5.1 Simple rules for analysing ENTA data

The age-specific ENTA data has limitations that are not necessarily visible. The following rules should be followed when working with ENTA data, to avoid misinterpretation of patterns, differences across countries, or changes over time:

1. **Do not use age-specific saving estimates for analysis.** Age-specific saving is calculated as a residual (disposable income minus consumption) and inherits all measurement errors and biases from both components. It cannot be used for meaningful analysis; the aggregate saving of each sector, which comes from the national accounts, is not affected.

2. **Do not compare asset income across countries or over time.** Asset income is not well captured in EU-SILC: wealthy households are under-represented, and the highly skewed distribution produces noisy age-specific estimates.

3. **Do not use public consumption to analyse changes over time.** Age profiles of public health expenditure are based on a single reference year. Education profiles are based on enrolment data, which change over time, but per-student expenditure by level does not vary by age within each level.

4. **Use private consumption and intra-household transfers for cross-country comparisons, not for changes over time.** Consumption estimates are based on Household Budget

Survey data from 2010 and 2015 only. Large cross-country differences are reflected in the data, but changes over time in the shape of consumption age profiles are not captured.

5. **Employment income, taxes, and social benefits in cash are most reliable.** These quantities are well captured in the survey data and can be compared across countries and over time. Consistent changes over several years point to significant trends. Even so, estimates are subject to the usual uncertainty of survey data.

Despite these limitations, the organisation in a consistent accounting system has considerable value. ENTA data are complete for each country and year and can be used to illustrate the generational economy: age-specific income, its redistribution between age groups through public and private transfers, and its use for consumption and saving. However, care should be taken with comparisons across countries and over time, in particular for those variables identified above.

5.2 What applies to every profile

The limitations in this section are not specific to one quantity. They come from the coverage of the data, the age grid it is reported on, the harmonisation of the microdata, or the way profiles are built, and they therefore reach every profile in the dataset.

5.2.1 Scaling to the aggregates

Almost every age profile in ENTA is rescaled so that the sum over ages, weighted by population, equals the corresponding national accounts aggregate. A treatment applied to survey microdata therefore changes the *shape* of a profile across ages, not its level: the level comes from the aggregate. The exceptions are the treatments that act on the aggregates themselves — the administrative sources behind benefits and government consumption (Sections 5.5 and 5.8) and the ESA sector accounts (Sections 5.4 and 5.10) — which do change levels.

5.2.2 Coverage

Age profiles are produced for 26 countries: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and Sweden. The same list appears in every do-file that loops over countries.

- **Malta** has aggregates but no age profiles. It is absent from every profile country list in the code. The reason is not recorded there.
- **Croatia** starts in 2009, not 2008. Croatian EU-SILC begins with the 2009 income reference year. EUROMOD, which supplies the estimated tax rates, begins two years later in 2011; for 2009 and 2010 the Croatian tax rates and the pension share of taxes on benefits are carried back from 2011 (Section 5.4.2).
- **Bulgaria** ends in 2022; the 2023 ESA data were not available when the aggregates were built.

- The published aggregate file is wider than the profile set. It carries 33 reporting units — the 26 above plus Malta, Norway, Iceland, the United Kingdom, Albania, Turkey and Ukraine — because it is built directly from the Eurostat sector accounts. Some of these units are dropped again when the file with the unit conversion factors is built.

Profiles run from 2008 to 2023, aggregates from 1995 to 2024.

5.2.3 Germany: a coarser age grid, and a survey break

German EU-SILC does not report single years of age throughout. From 2014 to 2018 the top age category is 73 and older; from 2022 the data are delivered in five-year age groups above 20, with 16 to 19 collapsed into a single category. ENTA maps German ages onto the grid the data actually support before any collapse and restores single-year ages afterwards, so that the merged profile is defined on the same age axis as every other country. Within a group each single year carries the group value. All EU-SILC-based data for the income reference years 2022 and 2023 are therefore mean values in five-year age groups, and for 2014 to 2018 the age group from 73 to 80 holds the mean of the age group 73 and older.

Two consequences follow. Smoothing is switched off for Germany after 2021 — there is nothing to smooth on a five-year grid — so the German income, tax, benefit and consumption profiles are smoothed only up to 2021. And in the consumption equivalence scale, age 16 stands for the whole 16–19 range from 2022 on, so its weight is raised from the value the scale would give a 16-year-old (0.85) to 0.9, closer to the average of the four ages it now represents.

Germany also changed its EU-SILC survey methodology in 2020, the survey for income reference year 2019; data before and after this break are not fully comparable.[1] The visible effect on the sex split of pensions and asset income is described in Appendix E.

5.2.4 Harmonisation of EU-SILC over time

Some EU-SILC variables changed definition or coding during the period covered. ENTA harmonises them so that a profile does not shift only because the survey changed. Years below are income reference years, one year before the survey year.

Activity status and adults. The self-defined activity status is collected in three successive variables: `PL030` until 2007 (in some countries also 2008), `PL031` from 2008 to 2019 and `PL032` from 2020. For persons without a personal interview, the basic activity status `RB210` is available until 2019 and `RB211` from 2020; `RB211` uses the codes of `PL032`, not those of `RB210`. ENTA maps all five variables onto one classification (education, part-time and full-time employment, unemployment, domestic work, retired or disabled, other inactive). Persons in the personal file for whom none of them is available — mainly in Sweden and the Netherlands, where only a selected respondent answers the personal questionnaire — are classified as employed if they have employment or self-employment income and as unemployed if they receive unemployment benefits.

[1] `https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Einkommen-Konsum-Lebensbedingungen/Lebensbedingungen-Armutsgefaehrdung/Methoden/meth-auswirkungen-corona-EU-SILC.html?nn=210056`

Asset income, housing costs and some household-level benefits are assigned to the adults of a household. Adults are all persons aged 35 and over and, at ages 18 to 34, persons in employment, unemployed or in domestic work, and persons in education who do not live with a parent. Because the definition uses the harmonised classification, it is the same in every year.

Personal identifiers. Couples within a household are identified with a within-household person number, which ENTA derives from the order of the personal identifiers in the household. Couples matter for the distribution of family benefits and of income within couples.

5.2.5 Bulgaria 2010

In the Bulgarian survey for 2010 the income components do not add up to the published household total. Total household income (`HY010`) is continuous with the neighbouring years (18.9, 18.9 and 17.9 billion euro in 2009, 2010 and 2011), but the sum of its components is 1.6 billion lower in 2010 and the difference affects 78% of households. Sickness benefits are almost absent (0.01 billion against 0.17 and 0.16 billion in the neighbouring years) and old-age pensions are lower (2.6 billion against 2.9 and 3.2 billion). ENTA builds income from the components. Because the profiles are scaled to the national accounts, the level is not affected, but the age and sex pattern of the missing parts is lost for Bulgaria 2010.

5.2.6 Smoothing and the open top age group

The income, tax, benefit and consumption profiles derived from EU-SILC are smoothed over age with Friedman's SuperSmoother; the employment profiles from the Labour Force Survey are not. Smoothing of income, tax and benefit profiles starts at age 18, and ages 15–17 of the income and tax profiles are capped at the age-18 anchor (see Chapter 3). The smoother errs most at the ends of the age range, so the open top age group (80 and older; 73 and older for Germany 2014–2018) stays in the smoothing but keeps its unsmoothed mean, in all smoothed income, tax and benefit profiles. Unemployment benefits are smoothed differently, because they end abruptly at an age that differs by country, sex and year.

Several country treatments described elsewhere in this chapter reappear at the profile stage, because that is where they bite:

- **Germany**: smoothing of income, taxes, benefits and consumption stops at 2021.
- **France**: property income is left out of the strongly smoothed group. The stronger smoothing setting fails outright on the French property-income series — the smoother returns missing values — so France uses the default setting for these two variables, as every country does for all other variables.
- **Croatia**: tax profiles are smoothed from 2009, the first Croatian EU-SILC year.

5.2.7 Population

Romania's open-ended top age category (100 and over) is implausibly large relative to the United Nations figures up to 2001 — a factor of 35 to 44, against 2 to 8 from 2002 on, which is the normal range of cross-source disagreement at that age. The UN figure is used for Romania at age 100 through 2001. This is not a missing-data case and is therefore not caught by the general fallback that handles missing population values.

5.2.8 Non-profit institutions inside the household sector

Net other current transfers (`TX`) are negative for households in most countries and years, which is counter-intuitive for a variable that represents transfers paid. The household sector in ENTA is the combined sector S14 and S15, so non-profit institutions serving households are included with households. Those institutions are net recipients of government funding, and their receipts outweigh what households themselves pay in the combined figure. The split between the two components is not observed directly in the source data and is estimated in ENTA under a proportionality assumption, so the size of the non-profit part carries more uncertainty than the combined total.

The size of the non-profit component varies considerably: measured against the consumption of the combined sector, it is 6.3% in Luxembourg, 5.4% in Norway, 4.1% in France, 3.7% in Austria and 2.8% in Germany, but only about 1.3% in Romania.

Other current transfers, and the consumption of the household sector, both include a non-profit component that has no household age pattern. This matters most for the countries at the top of that list.

5.2.9 Cyprus: repairs to the aggregate source

Other subsidies on production paid (D.39) are not reported for Cyprus and are set to zero. Non-life insurance claims paid (D.72) by financial corporations are missing for 2012 and 2013 and are set to the whole-economy figure for those two years.

Miscellaneous current transfers paid (D.75) by financial corporations, general government and the rest of the world, and non-life insurance claims paid by the rest of the world, are also missing for 2012 and 2013. Left at zero, D.75 received would exceed D.75 paid by 714 million euro in 2012 and 427 million euro in 2013, and the allocation of transfers between sectors would book that difference as household payments, understating household disposable income by the same amount. The total of other current transfers (D.7) is reported for all three sectors, so the missing values are derived from it: for financial corporations and government as D.7 less its other components, for the rest of the world as the amount that balances D.75 paid against D.75 received. After this, net saving computed as disposable income minus consumption equals the published B.8n for every Cypriot sector.

5.2.10 Import repairs in the microdata

Three further country-specific rules exist in the import code. They repair the delivered files rather than the data they contain, and none of them changes a published quantity.

- **Ireland** and **Austria**: some records point to a partner identifier that does not exist in that year's register. These rows are dropped. In Austria a second, smaller problem occurs where the partner pointer disagrees with the reciprocal one; those duplicates are dropped as well. The treatment was confirmed as specific to these two countries.
- **Poland**: the personal-data file codes one variable (`PB120`) with asterisks in place of a number, which is stripped on import.
- **Italy**: the raw personal-data file for two waves has to be renamed by hand before import. This is an operational note in the code, not a data treatment.

5.3 Primary income

5.3.1 Asset income

Wealthy households are systematically under-represented in EU-SILC, and in most countries total asset income in the survey captures only a fraction of what is recorded in ESA. The age-specific estimates are subject to high uncertainty and vary considerably over time, due to the low number of observations and the highly right-skewed distribution. Consequently, asset income shows the basic age pattern but should not be used to compare countries or analyse changes over time.

5.3.2 Employment income: four countries on five-year age groups

Four countries' public-use Labour Force Survey files carry only five-year age groups: the Netherlands, Czechia, Ireland and Slovakia. This is visible directly in the prepared files — each has 21 distinct age values (16 for Slovakia), against 60 to 119 for every other country. Employment rates in full-time equivalents are therefore merged into the compilation on five-year groups for these four countries and broadcast to the single years inside each group, while the true single-year age is retained for everything else.

The practical consequence is that the employment profile is a step function for these four countries and a smooth single-year profile elsewhere. Everything derived from employment inherits that. Comparisons of profile curvature across these countries and the rest are affected; comparisons of totals are not.

5.3.3 Denmark: property income before 2019

Danish EU-SILC reports interest and dividend income (`HY090G`) net of interest paid, so the variable takes negative values and cannot be used as gross property income received. In contrast to other countries, this resulted in a considerably higher share of households with negative property income,

inconsistent with the aggregate data. For Denmark before 2019, property income received is therefore built from rental income (HY040G) alone, with imputed rent added. Total primary income for Denmark keeps the standard definition.

5.3.4 Private pensions before 2010

Income from individual private pension plans (PY080G) is part of Eurostat's total household income (HY010, HY020) only from 2010 on; before, it is recorded but not included in the total. ENTA adds private pensions to property income received in every year, so the concept is the same throughout. As a consequence, household cash income in ENTA exceeds HY020 before 2010 by the private pensions. The difference is largest in Sweden (1.7–1.8% of household disposable income in 2007–2009), Spain (1.3–1.5%) and Finland (1.3%), and 1% or less elsewhere.

5.3.5 Mortgage interest not reported

Property income paid (YPp) would be zero at every age for Hungary 2018, France 2020 and Germany 2007 and 2008, making asset income too high in those years. EU-SILC does not deliver usable data on interest repayments on mortgages (HY100G) for those country-years. France and Germany report the variable as not collected or missing; Hungary reports a value of zero with a flag indicating "no income", which is indistinguishable from real data by any missing-value check. The surrounding years show several hundred paying households in each case, so the zero is a reporting failure rather than a real absence.

The affected profiles are replaced by the average of the neighbouring years. Germany 2007 is unusable in the same way and is deliberately left alone, since no earlier year is available, so German 2008 is built from 2009 only.

5.3.6 Ireland: private pensions in 2018

In Ireland, the age and sex profile of property income received (YPr) would shift towards older men in 2018 and back in 2019. The Irish EU-SILC for income year 2018 reports income from individual private pension plans (PY080G) for 609 persons, against 124 to 171 in 2016–2020. The amount per recipient is in line with the other years, so the weighted total rises to 3.1 billion euro, against about 0.5 billion in the neighbouring years. The cause of the jump is not known. Private pensions are part of property income received in ENTA, and 72–79% of them go to men. The total of YPr comes from the national accounts, so only its age and sex pattern is affected.

The private-pension part of the Irish 2018 profile is replaced by the average of 2017 and 2019. Rental and investment income are kept as reported.

5.4 Taxes and social contributions

5.4.1 General

Consumption taxes are allocated using the age profile of total private consumption, without accounting for age-specific differences in the consumption basket and the corresponding tax rates. Tax rates by type of income are estimated from the EUROMOD tax-benefit model, which may not fully capture all country- and year-specific tax rules.

5.4.2 EUROMOD: country-specific income mappings

EUROMOD output uses a common variable naming convention, but the mapping from national income concepts onto it is not uniform. Eleven countries need an explicit statement of which national variables belong to which ENTA aggregate:

Table 5.1: Country-specific EUROMOD income mappings

Country	Added or redefined
AT	military and civil service pay, income of those under 16, tips
CY	military service allowance
DE	health insurance contributions of the self-employed (an expenditure item)
EL	health insurance contributions of the self-employed
HR	compensation for carers and foster parents; student income
IE	occupational pensions (public and private); a national medical-expenses item
IT	non-taxable employment income, additional jobs, several self-employment and pension components
NL	a benefit component outside the standard benefit aggregate
PL	private transfers other than maintenance payments
SK	lump-sum employment income; company shares and wages
SI	basic employment and self-employment income, additional jobs, student income

These are definitional, not corrections: without them the components would not add up to EUROMOD's own totals for those countries.

Croatia. The tax-attribution regressions start in 2011, two years after Croatian EU-SILC. For the 2009 and 2010 income reference years the estimated tax rates by age group, and the share of taxes on benefits falling on pensions, are carried back from 2011. What is carried back is the *age pattern* only: as with every other age profile, the level of the Croatian tax profile in 2009 and 2010 is set by rescaling it to that year's own national accounts aggregate, and the pension share is applied to that year's own reported total of taxes on benefits. Before 2009 there is no Croatian microdata at all, so the aggregate total of taxes on benefits is reported without its pension/non-pension split.

Luxembourg and the Netherlands. The estimated effective tax rate on employment income at ages 60 and over is the only place where the ceiling on estimated tax rates binds. Very few people

over 60 in these two countries have employment income in the microdata, and those who do sit high in the income distribution, so the ratio of tax to income in that cell is not a reliable estimate of a tax rate. The rate is capped there; six of the eight countries examined never come close to the cap.

5.4.3 Very small microsimulation tax rates

In some countries the age-specific tax rates on benefits produced by EUROMOD are very small — a few tenths of a percent — even though the corresponding aggregate is substantial. A tax rate for an age group is computed as tax divided by benefit within that group, and where few people in a group receive the benefit the ratio is unreliable. Rates below a threshold are therefore set to zero.

Where the rate is unusable at the ages that hold most of the benefits — above age 60 for pensions, between 20 and 60 for other benefits — the age profile of the benefit itself is used as the shape instead, and the level is set from the aggregate. This is equivalent to applying a single average tax rate across all ages. It affects Czechia, Slovakia, Germany and Ireland most.

5.4.4 Czechia and Slovakia: taxes on benefits

The effective tax rate on social benefits implied by the national accounts is far higher than the rate produced by the EUROMOD microsimulation: about 10% against 0.2% for Czechia, and 19% against 0.5% for Slovakia. No other country shows a discrepancy of this size; the next largest is a factor of seven, and most countries agree within a factor of 1.5.

Explanation, not confirmed. In both countries the state pays health insurance contributions on behalf of pensioners, and pensions are largely exempt from income tax. If those state-paid contributions are classified as labour taxes of the non-employed, they appear as tax on benefits in the aggregate while the household itself pays almost nothing — which is what the microsimulation reports. The two sources would then be measuring the accounting flow and the net household burden respectively, and neither would be wrong. This interpretation fits the magnitudes and the timing but has not been verified against a documented source.

Taxes on benefits for Czechia and Slovakia therefore carry more uncertainty than for other countries, and the level should be understood as the national accounts concept rather than an out-of-pocket household payment.

5.4.5 Greece: taxes on assets and on production, 2014

Taxes on assets (`TA`) paid by Greek households fall by more than half between 2013 and 2014, from 4,894 to 2,153 million euro, and stay at the lower level. In the same year taxes less subsidies on production (`TPN`) rise from 149 to 2,334 million euro. Taxes on asset income and assets (`TYA`) fall with `TA`.

The shift is in the national accounts themselves and is recorded identically on both sides of the transaction: households' other current taxes (D.59) paid fall from 4,894 to 2,153 while their other taxes on production (D.29) paid rise from 549 to 2,724; government's receipts of the two items move

in the same way (4,919 to 2,207 and 3,830 to 6,011). About 2.7 billion euro of household taxes are therefore recorded under a different transaction from 2014 onwards. The underlying tax change is not documented in the data.

`TA`, `TYA` and `TPN` for Greek households are not comparable across 2013 and 2014, and neither are the age profiles, since taxes on assets and taxes on production are allocated to ages with different shapes. Total taxes are hardly affected.

5.4.6 The benefit-tax decomposition in the early years

Taxes on benefits are reported as a total and as a split into taxes on public pensions, private pensions and non-pension benefits. The two come from different places. The total is taken from the DG TAXUD tax revenue data and is available from 1995. The split is that total multiplied by a pension share estimated from EUROMOD, which does not reach back nearly as far. Its nominal coverage begins in 2005 or 2006 in eighteen countries and in 2007 in the rest, but in the first year or two EUROMOD simulates no taxes at all on the non-employed, so the share it yields carries no information. The published split therefore begins in **2007** in every country, and in 2009 in Croatia, the first year of Croatian EU-SILC, with the 2009 and 2010 values carried back from 2011 (Section 5.4.2).

Before that first year the split is not measured, and ENTA reports the total on its own rather than a decomposition of it. It is published as missing and not as zero, because a zero would state something the data never said — that none of the benefits taxed in, say, Austria in 1995 were pensions. The total is unaffected in every year.

5.4.7 Denmark: social contributions before 2005

The breakdown of social contributions into employers' actual, employees', self-employed and imputed contributions is not reported for Denmark before 2005, although the total is. The total is split using the 2005 shares, separately for each sector and for the paying and the receiving side. This affects the aggregates only; Danish age profiles start in 2008.

5.5 Government transfer benefits in cash

Benefits in cash are among the better-measured ENTA quantities: the age and sex pattern comes from EU-SILC, where benefit receipt is reported directly, and the level from the national accounts. The limitations below concern the *split* of the total into categories — pensions, family, unemployment and the rest — which does not come from the survey at all. It comes from government expenditure by function (COFOG), and it is the part of the benefit data that most often breaks between years.

5.5.1 Government expenditure by function (COFOG)

- **Slovakia, 2008 and 2009**: Eurostat's COFOG series does not cover Slovakia before 2010. Two different repairs follow. For the cash benefit categories the 2010 values are carried back as a placeholder and then rescaled against the true ESA benefit total, so the placeholder affects only

the split between categories. For public consumption by function there is no such aggregate to rescale against, so each category is assumed to hold the same share of total public consumption as in 2010 and is scaled by Slovakia's actual 2008 and 2009 consumption.

- **Bulgaria**: several categories — education and health benefits, post-secondary education spending, and public spending on social services — are not reported and are set to zero rather than left missing.
- **Germany, Poland** (reference year 2012) and **Portugal** (reference year 1999): the split of social benefits between cash and in kind is missing for part of the period. The share observed in the reference year is applied to the years where only the combined figure exists.
- **Ireland from 2022**: spending shifts between primary and secondary education, which is a reclassification rather than a real change. The combined primary-plus-secondary total is kept and split in the proportions observed in 2021.

The same COFOG data distribute government consumption by function; see Section 5.8.

5.5.2 Ireland: break in the benefit breakdown, 2011

Public pensions (`BSPEN`) fall by about a quarter between 2010 and 2011, from 16,313 to 12,017 million euro, while total social benefits in cash are almost unchanged (23,606 and 23,640). Family and unemployment benefits rise by about the same amount.

In the Irish COFOG data for 2011, spending on old age, disability and survivors falls by 4,341 million euro (old age 10,134 to 8,879, disability 3,993 to 2,210, survivors 2,387 to 1,085), and spending on family and unemployment rises by 4,289 million euro (family 2,438 to 4,966, unemployment 3,826 to 5,589). Near-offsetting changes of this size in a single year point to a reclassification between functions rather than a change in the benefits paid; the reason is not documented in the data.

The split of Irish benefits into pensions and other benefits — and therefore the age profiles of these components, which differ strongly by age — is not comparable across 2010 and 2011. Total benefits are not affected.

5.5.3 Early years of the benefit breakdown

The COFOG source does not reach back to the start of the aggregate period in every country. Where it does not, the four benefit categories — pensions (`BSPEN`), family (`BSFAM`), unemployment (`BSU`) and the residual (`BSX`) — are all reported as missing together:

- **through 2000**: Belgium, Greece, Finland, Italy, Latvia, Poland, Romania and Sweden
- **through 1999**: Germany and Lithuania
- **through 1998**: Portugal
- **through 2007**: Slovakia

- **2002 alone**: Bulgaria, an interior gap between reported years on either side

Total benefits in cash (BS) come from ESA and are reported in all of these years; it is only the split into categories that is absent. The four categories are therefore published as missing and not as zero, because a zero would assert what the source never said — that a country paid social benefits of which none were pensions. Read them as unknown, not as nil.

5.5.4 Denmark: early-retirement pay at ages 60–65

In Danish EU-SILC, unemployment benefits (BSU, from PY090) at ages 60 to 65 are largely early-retirement pay (*efterløn*), which is administered by the unemployment insurance funds. They are therefore much larger at these ages than in other countries — 7,000 to 12,000 euro per person at ages 63–65 in 2008–2017 — and they end at 65–66, where pensions take over. Two consequences of the smoothing rules (Section 3.3) follow.

Around 65. The hand-over from early-retirement pay to pensions is handled explicitly, so total benefits (BS) pass smoothly through the age at which BSU ends. The split between BSU and pensions at ages 63–68 is, however, partly modelled rather than observed, and should not be read year by year.

Around 62, in 2016–2018. The age from which early-retirement pay is received rises over the period, and from about 2016 the rise in BSU is a step at 62 rather than a gradual increase from 60. The profile is smoothed up to four years before the age at which BSU ends — 62 in Denmark — and follows the observed values after that, so a step exactly at the edge of the smoothed range is flattened: in 2016, BSU at 62 is 2,655 euro in ENTA against 5,682 in the survey, and BS is about 7–8% below the mean of its neighbours at 62 before rising at 63. The effect is confined to one or two ages and a few years, and it is of the same size as the ordinary irregularities in benefit profiles at ages 55–60 elsewhere; it was therefore left in place. For Denmark in these years, read benefits at ages 61–63 together rather than age by age.

5.6 Funded pensions

The microdata cannot distinguish between contributions to or benefits from funded versus pay-as-you-go pension systems. ENTA assumes the same age profile shape for both. This limitation is most relevant for countries where funded pensions are substantial, in particular Denmark, the Netherlands, and Sweden.

Hungary's funded-pension series changes level permanently in 2011 for a reason that is not a limitation of the data: the mandatory private pillar was nationalised. See Appendix E.

5.7 Private transfers

5.7.1 Transfers between households

The concepts of private transfers in aggregate data (including social contributions and benefits to and from corporations, non-life insurance premiums and claims) differ from those in microdata (mainly remittances and alimony payments). Furthermore, in microdata they suffer from data quality issues. ENTA distributes their net value with a receipt profile (employment income plus pensions) that carries no information on who actually pays or receives them. These transfers are comparatively small and included for completeness. An analysis focusing on private transfers between households should use other data sources.

5.7.2 Transfers within households

Intra-household transfers are not observed but imputed as the difference between individual consumption and income. They depend on assumptions about how consumption expenditure is distributed within the household, which are the same for all countries and years. Differences between countries and years reflect mainly differences in economic household structure, such as the employment of youth and young adults and how long they live with their parents.

The identification of couples, which governs how family benefits and income within couples are distributed, is described in Section 5.2.4.

5.7.3 Germany and Ireland: the split of other current transfers

Other current transfers (`TX`) are reported separately for households (`TXH`) and for non-profit institutions serving households (`TXNPISH`). The source data carry this split from 1995 for 24 of the 26 countries, but only from 2012 for Germany and from 2010 for Ireland. In the earlier years the whole of `TX` is carried in `TXH`, and `TXNPISH` is reported as missing rather than as zero, since the split is unknown for those years rather than known to be nil. The total `TX` is unaffected and stays comparable over time; the two components are not. German `TXH` moves from −14,374 million euro in 2011 to +6,762 million in 2012, and Irish `TXH` from −678 million in 2009 to +1 million in 2010. In both cases the jump reflects only the point at which the NPISH part begins to be separated out, not any change in the economy. For these two countries, `TXH` and `TXNPISH` should not be compared across those years; use `TX` instead.

5.8 Government consumption

5.8.1 Public health consumption

Age profiles of public health expenditure come from one source for all countries, the 2018 Ageing Report, and refer to a single year. The same shape is applied to every year and scaled to that year's aggregate, so the profiles do not capture changes in the age pattern of health spending over time;

only the level changes.

For some countries, the shape depends on which edition of the Ageing Report is used. The 2021 edition gives very similar profiles for 20 of the 26 countries. For Spain, Slovakia and Ireland, however, its ratio of per capita spending at ages 65 and over to spending at ages 20–64 is 28–42% higher than in the 2018 edition, and for Cyprus, France and Finland it differs by 11–13%. A change of this size within three years is not plausible, so the difference most likely reflects revisions between the editions. For these six countries, the steepness of the old-age health profile is uncertain.

5.8.2 Education enrolment

Enrolment by single year of age is used to distribute public education spending over ages. Eurostat delivers it in two files — ISCED 1997 until 2012, ISCED 2011 from 2013 — with country-specific age banding. Countries report different mixtures of single years, five-year bands and open-ended groups, and a band label sometimes means something other than what it says.

ENTA distributes every reported figure over single years of age with a reference age profile: students per person by country, level, sex and single year of age, averaged over 2015–2017. Where those years are themselves affected by one of the problems below, the nearest unaffected years are used instead (for example 2018–2020 for French tertiary and 2022–2024 for Romania); where a country has no unaffected year for a level, the EU average profile is used, and where that is zero as well, population alone. Each reported group is spread over its ages in proportion to that profile times population, so the group total is preserved.

Groups that hold more ages than their label. These are found in the data — the reported parts add up to the total, so the missing ages are inside the last group — except where noted.

- **ISCED 1997, 35 and over inside 35–39**: Italian and Romanian tertiary, German upper secondary. **30 and over inside 30–34**: French and Polish tertiary, Ireland 2008–2009, Greece 2008–2011.
- **Croatia, tertiary**: the reported 40-plus group is small (400–750 students) next to a 35–39 group as large as 30–34. The two together are treated as 35 and over, which makes the age pattern continuous into 2013.
- **Romania, lower and upper secondary and post-secondary**: single age 21 holds everyone aged 21 and over until 2012, single age 24 holds everyone aged 24 and over in 2013.
- **Greece, tertiary 2010–2013**: single age 29 holds everyone aged 29 and over.
- **France, upper secondary 2020**: single age 30 holds everyone aged 30 and over.
- **France 2010 and 2011**: secondary, post-secondary and tertiary enrolment at age 29 is a copy of the 25–29 band and is set to missing, so that the band is used instead.

Groups the source has already spread. In the ISCED 2011 file some countries report equal values across consecutive single ages or groups, which is an even split rather than a measurement.

Such segments are detected from the data (France, Poland, Greece, Ireland, Hungary, Slovenia from 2013, Romania 2022, Latvia) and redistributed with the reference profile. The opposite case — a whole five-year group reported at its first age and zero at the other four — occurs in Romania between 2013 and 2021 and in Italy in 2013 and is spread over the group. Slovenian ISCED 1997 assigns band totals for ages 20–29 to single years, so the band is used there instead of the single years.

Students of unknown age. Students whose age is not reported are otherwise left out, which spreads them like the students of known age. Where they are demonstrably adults they are distributed over the older ages with the reference profile: ages 25 and above for Polish tertiary 2008–2012 and Romanian secondary 2015–2021, ages 24 and above for Romania 2014, and ages 20 and above for Belgian secondary and tertiary from 2014.

Breaks between years. Where a country-year reports implausibly few students at older ages, or no age information at all, the enrolment rate is replaced by the rate of the nearest years without such a break and the level is rebuilt from population. The cases are found by rule: no age information (for example Czech post-secondary), students aged 55 and over below a tenth of the neighbouring years (Romania, Latvia, Italy, Ireland, Greece), a short explicit list (Ireland 2018, Belgium 2015, Dutch tertiary, Greek tertiary 2014, Bulgaria 2013–2014), and isolated one-year spikes such as Hungary 2013. Trends and permanent level shifts are left alone.

Missing years and levels. A second group of repairs replaces a country-year's enrolment *rate* with the rate of another year and rebuilds the level from population:

Table 5.2: Enrolment repaired from another year

Country and level	Years repaired	Rate taken from
EL, all levels	2009	mean of 2008 and 2010
DK, pre-primary/primary	2008, 2009 (age 5 only)	2011
EL, pre-primary/primary	2018 and 2020 onwards	2019
HU, pre-primary/primary	2013	2014
PL, pre-primary/primary and secondary	2020 onwards	2018
IE, pre-primary/primary and secondary	2011 (ages 12–13)	2012
LU, tertiary	up to 2010	2011

Greece 2009 is the largest case: Eurostat reports no Greek enrolment at all for that year. Luxembourg is the clearest: before 2011 no tertiary enrolment is reported, which would otherwise leave Luxembourg with no students and hence no share of tertiary education spending. Irish post-secondary enrolment is *not* repaired; students move from post-secondary to upper secondary in 2017 and 2018, and the two levels are combined before public education spending is distributed, so the reclassification cancels.

Men and women add up to the total at every level and age: the sexes are reconciled in the source data first, and the distributed totals are split with each sex's share of students.

Finally, twelve countries — Belgium, Bulgaria, Croatia, Czechia, Denmark, Estonia, France, Ireland, Latvia, Poland, Slovakia and Slovenia — are assumed to have full enrolment between ages 6 and 16, overriding a computed rate below one.

5.9 Private consumption

5.9.1 Limited information on changes over time

Age-specific consumption profiles from the Household Budget Survey (HBS) are available only for 2010 and 2015. The 2020 HBS data contain very limited information by age and are likely influenced by year-specific effects of the COVID-19 pandemic. All consumption estimates are therefore based on pooled data from these two survey years.

The imputation of consumption in EU-SILC captures changes due to age-specific variation in household structure and income. However, other factors that could influence age-specific consumption are not reflected. The shape of the age profiles for education and health consumption is assumed to be constant over time; other consumption is imputed using a regression model based on the pooled HBS data. Age-specific ENTA consumption data are not suitable for analysing changes over time beyond changes in the overall level of consumption.

5.9.2 High uncertainty

HBS data are characterised by high measurement noise. HBS records expenditure over short observation periods (typically two weeks for frequent purchases). The recorded expenditure for any individual household is a noisy measure of annual consumption. Reliable estimates can be obtained only for broad consumption categories and sufficiently large population subgroups.

5.9.3 Role of assumptions

HBS records consumption at the household level. Allocating expenditure to individuals requires assumptions. ENTA uses the NTA equivalence scale, which assigns children aged 0–4 a weight of 0.4 relative to adults, increasing linearly to 1.0 at age 20. Actual consumption patterns within households may differ from this assumption.

5.9.4 Survey coverage by country and wave

The HBS is a three-wave survey (2010, 2015, 2020) and not every country has all three waves in the Eurostat delivery ENTA uses. The code therefore keeps four separate country lists:

Austria's 2010 and 2015 consumption data are not taken from the Eurostat HBS at all. They come from the Austrian national household budget survey, prepared separately because the Austrian file uses different variable names and a different layout, and appended to the Eurostat 2020 wave.

Two further treatments:

Table 5.3: HBS wave availability as coded in `Micro_HBS.do`

Waves available	Countries
2010, 2015, 2020	BE BG CY CZ DE DK EE EL ES FI FR HR HU IE IT LT LU LV RO SI SK
2010, 2015 only	PL PT SE
2015, 2020 only	NL
2020 only	AT

- **Italy (all waves), Luxembourg 2010, Czechia 2010 and 2020**: the delivered file has no household income variable. Cash disposable income, which is used to align HBS households with EU-SILC households, is proxied by total consumption expenditure for these country-waves.
- **Portugal**: the HBS age classification stops at a 75-and-over group; there is no 80-plus category. When consumption profiles are merged into EU-SILC, Portuguese 80-plus individuals are matched to the 75 group and the true age group is restored afterwards.

5.9.5 Sweden: no private education consumption

Private consumption of education (`CHE`) is zero for Sweden in every year, both in the age profiles and in the aggregates, while every other country reports a positive amount — typically 2 to 4 per cent of other private consumption.

The Household Budget Survey does not provide education consumption for Sweden, so no age profile can be estimated. Rather than publish a profile borrowed from a different consumption category, ENTA reports Swedish education consumption as zero and includes the amount in other private consumption (`CHX`) instead, on both the profile and the aggregate side. The amount involved is small: about 540 million euro a year, or roughly 0.35 per cent of `CHX`.

Total household consumption is unaffected, and so is the non-durable aggregate — the treatment only moves the amount between two components. But Sweden must be excluded from any comparison of private education spending across countries, since its zero is a convention and not a measurement.

5.10 Saving

As noted in the simple rules, saving is calculated as the residual of disposable income minus consumption. It inherits all measurement errors from both components, making its age distribution an approximation only.

5.10.1 Net saving in the aggregates

ENTA saving is the published ESA balancing item B.8n less the adjustment for the change in pension entitlements (D.8), for every country, sector and year, so that ENTA saving plus D.8 equals ESA net saving. D.8 is not published in the ENTA data; see the codebook entry on D8 in Appendix B. ENTA

also computes saving as its own disposable income minus consumption and keeps it as a control: the two agree to within rounding for every sector of every country. In Denmark, the Netherlands and Sweden, where funded pensions are large, household saving in ENTA is therefore noticeably lower, and corporate saving higher, than in ESA. At the level of the total economy, Ireland's published B.8n differs from the sum of its sectors by up to about 3 billion euro, because Ireland's total operating surplus and mixed income in the sector accounts does not equal the sum of the sectors' values; ENTA reports the published figure.

Part II

Codebook and project reference

A. Units of measurement

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ENTA output is provided in a range of units to facilitate comparison across time and countries. The standard unit is Euro per capita at nominal value, but ENTA also provides real values, purchasing power adjusted values, and quantities expressed relative to income and GDP.

Every ENTA dataset (aggregate and age-specific) stores a single value, in Euro (per capita for age-specific data; as a nominal aggregate total for aggregate data). All other units are calculated from this single value on demand, using a small set of conversion factors saved separately in *ENTA_units.dta* (also exported as *ENTA_units.csv*) — one row per country and year. This replaces an earlier design where every unit was pre-calculated and stored as its own variable; keeping only the base value and the conversion factors keeps the ENTA datasets much smaller, and a unit is calculated only when a user actually selects it. Section 11 documents every variable in *ENTA_units.dta*.

For age-specific data, the base value and the aggregate total coincide with the stored value itself (already per capita). For aggregate data, the stored value is a total (not per capita), so the per-capita conversion (multiplying by 1,000,000 and dividing by POPtot) is itself part of every unit calculation except "Billion Euro" and "Percent of GDP"/"Percent of primary income" (which use the total directly).

Exception: indicators that are rates rather than monetary values — currently only employment in full-time equivalents (EMPLFT) — are never unit-converted; the same value is shown regardless of the unit selected.

1 Euro per capita, nominal

Both aggregate and age-specific data are provided in Euro per capita at nominal value. For countries outside the Euro area, values in national currency are converted into Euro using year-specific exchange rates provided by Eurostat.

2 Billion Euro, nominal

Aggregate ENTA quantities are provided in billion Euro. This unit is useful for assessing the macroeconomic scale of intergenerational transfers and other aggregate flows. It is not available at the age-specific level. Calculated as the nominal aggregate total divided by 1,000 (the total itself is stored in million Euro).

3 Euro per capita, real (HICP, 2020=100)

Both aggregate and age-specific quantities are provided in constant 2020 Euro. Real ENTA values are calculated as follows:

1. Nominal Euro values are converted into national currency using the year-specific exchange rate (EXR_NAC_EUR).
2. The values in national currency are deflated using the Harmonised Index of Consumer Prices (HICP), with 2020 as the base year.
3. Deflated values are converted back into Euro using the 2020 exchange rate (EXR_2020).

This procedure ensures that the deflation is based on the consumer price index of the country in question, while the final unit remains the Euro. For Euro area countries, steps 1 and 3 use an exchange rate of one, since EXR_NAC_EUR and EXR_2020 both equal one throughout.

4 PPS-Euro per capita

ENTA quantities expressed in purchasing power standard (PPS) Euro. PPS values reflect the purchasing power of one Euro in the EU-27 at current prices, without adjustment for inflation. This unit enables cross-country comparisons that account for differences in price levels. The conversion multiplies the nominal per-capita value by PPP_EUR, a purchasing-power-parity factor derived from Eurostat data.

5 PPS-Euro per capita, real (2020=100)

ENTA quantities expressed using the 2020 purchasing power of one Euro in the EU-27. This unit combines the cross-country comparability of purchasing power adjustment with the intertemporal

comparability of inflation adjustment, enabling meaningful comparisons across both countries and time. The conversion first applies the real Euro calculation described above (Euro per capita, real) and then multiplies by PPP_EUR_2020, the 2020-anchored purchasing power factor.

6 Percent of GDP

ENTA quantities expressed as a percentage of Gross Domestic Product. This is the most common way of expressing the relative importance of economic flows such as tax revenues, social benefits, or total government expenditure, and is available only at the aggregate level (calculated as the aggregate total divided by GDP, the total-economy GDP aggregate, times 100). In most cases, however, measurement as a percentage of primary income provides a more meaningful comparison, as discussed below.

7 Percent of primary income

ENTA quantities expressed as a percentage of primary income (net national income at basic prices, total economy). This normalisation facilitates cross-country comparisons of the relative size of economic flows by accounting for differences in overall income levels.

Expressing economic quantities relative to primary income is generally preferable to expressing them relative to GDP. This applies, for example, to tax rates, saving rates, and social benefit ratios, because these flows are financed out of the income of the national population rather than out of production. In countries where a large share of production accrues to foreign owners—such as Ireland—GDP substantially overstates the income available to residents. This measure has the additional advantage of being available at both the aggregate and the age-specific level: the per-capita primary income used as the denominator (Y divided by total population, POPtot) is the same regardless of whether the numerator is an age-specific or an aggregate value.

8 Percent of household primary income

ENTA quantities expressed as a percentage of household primary income (YH divided by total population, POPtot). This measure is particularly relevant when analysing the household sector and comparing it across countries, for instance with respect to tax burdens, the importance of social benefits relative to household income, or saving rates.

9 NTA norm (average labour income at ages 30–49)

The standard unit for comparing age-specific NTA values across countries is the average labour income of persons aged 30–49 (NORMnta). This normalisation controls for differences in income levels across countries. Simple per capita averages over the whole population are influenced by population structure (e.g. the share of pensioners) and retirement age. Normalisation relative to the

income of the prime working-age population avoids these distortions and is therefore more informative in many NTA applications.

10 ENTA norm (average labour income of men at ages 30–49)

The ENTA norm expresses age-specific values relative to the average employment income of *men* aged 30–49 (NORMenta). Like the NTA norm, the ENTA norm controls for differences in income levels across countries. In addition, because it uses male income as the reference, it preserves differences in employment levels between men and women and in the relative income of women. This makes it particularly suitable for analyses of gender differences, where using a norm based on income of both sexes combined would obscure part of the variation of interest.

11 The ENTA_units dataset

ENTA_units.dta (and the identical *ENTA_units.csv*) holds the conversion factors described above, one row per country (CNT2) and year (YEAR). It is built by *ENTA_units.do* and combines four independent sources: age-specific income microdata (for the NTA/ENTA norms), the ESA sector accounts (for primary income and GDP), Eurostat exchange rate and purchasing-power-parity data, and the Harmonised Index of Consumer Prices. Table A.1 lists every variable.

Table A.1: Variables in ENTA_units.dta

Variable	Description
CNT2	Country code (ISO 3166-1 alpha-2)
YEAR	Reference year
NORMnta	NTA norm: average labour income (YL), population-weighted, ages 30–49
NORMenta	ENTA norm: average male employment income (YEM), ages 30–49
Y	Primary income, total economy — ESA aggregate, million EUR (not per capita)
YH	Primary income, household sector — ESA aggregate, million EUR (not per capita)
GDP	GDP, total economy — ESA aggregate, million EUR (not per capita)
POPtot	Total population, all ages
EXR_NAC_EUR	Exchange rate: national currency per Euro, year-specific
EXR_2020	Exchange rate: national currency per Euro, fixed at the country's 2020 value
HICP	Harmonised Index of Consumer Prices, 2020=100
PPP_EUR	Purchasing power of one Euro (PPS), year-specific
PPP_EUR_2020	Purchasing power of one Euro (PPS), fixed at the country's 2020 value
PPP_EU27_2020	Eurostat PPP conversion factor, national currency per PPS unit, EU27=2020 basis (intermediate input used to derive PPP_EUR; not itself used in any unit calculation)

B. ENTA Codebook (ESA concept)

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0 Production Account — Total Economy

The production account is part of the aggregate data only; it is not age-specific. It shows how primary income is derived from GDP. The age-specific analysis of national accounts in ENTA starts with primary income.

0.0 GDP — Gross domestic product

Gross domestic product at market prices, as defined in ESA 2010.

0.1 TPS — Taxes less subsidies on products

Taxes less subsidies on products include, in particular, value added tax (VAT). In national accounts, taxes less subsidies on products appear twice: once in the final use of the private sector and once in the final use of the public sector. ENTA corrects for this double counting to ensure a consistent measurement of redistribution: all quantities in ENTA are measured at basic prices, i.e. excluding taxes less subsidies on products (see Section 1.4.2).

0.2 CFC — Consumption of fixed capital

CFC captures the investment necessary to maintain productive capital due to wear and tear, obsolescence, and normal accidental damage.[1] ENTA measures income and saving net of CFC.

0.3 YEROW — Employment income received from rest of the world

YEROW refers to the compensation of employees received from abroad, less the compensation paid to non-residents. These flows of income are part of primary income. They are particularly important in Luxembourg, where a large part of employment income is paid to residents of Belgium, Germany, or France.

0.4 YPROW — Property income received from rest of the world

Property income received from other countries less property income paid to other countries. This is a particularly important component for Ireland, which records part of the production of multinational companies; however, the income derived from these activities is paid to owners abroad.

0.5 Y — Primary income

Primary income is derived from GDP as follows:

$$Y = GDP - TPS - CFC + YEROW + YPROW.$$

[1] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Consumption_of_fixed_capital_(CFC)

1 Y — Primary Income

Primary income in ENTA captures the income that residents receive for contributing to production, either through their work or the use of their assets. It includes wages before paying taxes and social contributions, income from self-employment, as well as asset income such as dividends, interest, and rents.

ENTA primary income of each sector corresponds largely to the *balance of primary income* in ESA, measured at basic prices. The primary income of all sectors combined represents *national income* in ESA. The age profile is calculated as the sum of the subcomponents:

$$Y = YEM + YA.$$

1.1 POP — Population

Age- and sex-specific population count, the denominator underlying every per-capita age-profile value in ENTA. Sourced from Eurostat population data (with World Population Prospects used to fill specific country-year gaps where Eurostat is incomplete; see *Population (Suppl_population.do)* in Appendix D). Unlike the other indicators, population is a count, not a monetary value, so unit conversions such as real, PPS, or percent of income/GDP do not apply to it.

1.2 YEM — Income from employment

YEM includes the income of employees (YE) and income from self-employment (YM):

$$YEM = YE + YM.$$

We recommend analysing the two components together because the age-specific estimates are more reliable. In particular, the smoothing of the age profiles works much better for the combined profile.

1.2.1 YE — Income of employees

The income of employees in ENTA includes gross wages, employers' social contributions, all other taxes that employers pay on payroll or employment, and in-kind compensation, all before paying any taxes.

In ESA, YE corresponds to the *compensation of employees* plus taxes on production paid on payroll or employment. The age- and sex-specific estimates are based on the EU-SILC variables *employee cash or near cash income* (PY010), *employers' social contributions* (PY030), *non-cash employee income* (PY021), and *income received by people aged under 16* (HY110).

1.2.2 YM — Mixed income / self-employment income

YM captures the income of self-employed persons. Implicitly, it includes both the return to labour and the return to capital of self-employed persons. In ESA, it is therefore captured as *mixed income*. Age

profiles are based on the EU-SILC variable *cash benefits or losses from self-employment* (PY050G).

1.3 YA — Asset income

Asset income of households consists of all types of income derived from assets, such as interest, dividends, and rent; it also includes imputed rents, which capture the value of housing services enjoyed by owner-occupiers.

In ESA, the components of household asset income are captured as *property income* (YP) and *operating surplus* (YO):

$$YA = YO + YP.$$

1.3.1 YO — Operating surplus

The operating surplus of households consists mainly of the value of housing for owner-occupiers (imputed rent). ENTA maintains the distinction between property income and operating surplus: while the distribution of property income is characterised by a concentration among a few households at the top of the distribution, operating surplus plays an important role for all owner-occupiers, and age-specific data is more reliable because of the large number of observations.

The age- and sex-specific estimates of the operating surplus of households are based on the age profile of imputed rent (HY030) in EU-SILC, assigned to the person responsible for accommodation.

1.3.2 YP — Property income

The age- and sex-specific estimates of property income of households are based on the EU-SILC variables *interest, dividends, profit from capital investments in unincorporated businesses* (HY090) and *income from rental of property or land* (HY040).

1.4 Employment status (EMPLSTAT)

Employment status based on the Labour Force Survey. A person is counted as employed if they worked for pay in the reference week (variable WKSTAT) or were absent due to holidays, special working time arrangements, sick leave, or job-related training (variable ABSREAS).

The four Labour Force Survey based indicators — EMPLSTAT, EMPLFT, WHRUSUT and WHRACTT — are published for the full age range 0–80 in every country and year. The Labour Force Survey covers persons aged 15 and over, so these indicators are zero at every age below 15.

1.5 Employment status, EU-SILC based (EMPLSTATsilc)

The second indicator on employment status is based on EU-SILC. A person is counted as employed if their self-reported economic status is employed or self-employed (variable PL030/Pl031/PL032 in EU-SILC).

1.6 Usual working hours (WHRUSUT)

Usual working hours in the main and second job. Based on the variables HWUSUAL and HWUSU2J in LFS.

1.7 Usual working hours, EU-SILC (WHRUSUsilc)

Usual working hours in the main job and beyond. Based on the variables PL060 and PL100 in EU-SILC.

1.8 Actual working hours (WHRACTT)

Actual working hours for the main and second jobs during the reference week. Based on the variables HWACTUAL and HWACTU2J in the LFS.

1.9 Employment, fulltime equivalents (EMPLFT)

Employment in full-time equivalents is based on the usual working hours (HWUSUAL and HWUSU2J in the LFS). If the number of working hours is 35 or more, employment is counted as one full-time equivalent. Fewer working hours are counted proportionally; e.g., an employment of 17.5 hours is counted as 0.5 full-time equivalents.

1.10 Employment, fulltime equivalents, EU-SILC (EMPLFTsilc)

Based on the number of usual working hours in EU-SILC.

1.11 Employment, fulltime in income reference year, EU-SILC (EMPLFTRYsilc) [INACTIVE - not currently computed]

Intended to capture employment in the income reference year, based on the variables PL073, PL074, PL075, PL076 (number of months in full-time and part-time employment). This variable is not currently computed: the underlying calculation is commented out in the code as unused (EMPLFT is built from usual working hours and employment status instead, not from this variable), kept only for reference.

2 T — Taxes and Contributions

"Taxes" in ENTA include all transfers paid by nationals to the government, less the subsidies received. In particular, they also include social contributions. ENTA distinguishes taxes by economic function, i.e. whether they are paid on employment income (TYEM), assets or asset income (TYA), social benefits (TB), or consumption (TCON). They also include a component of *other current transfers* (TX), which includes, for example, fees, other charges, fines, and penalties.

$$T = TYEM + TYA + TB + TCON + TX.$$

The aggregate value for total taxes is based on ESA data; the aggregates for the subcategories are mainly based on data on tax revenues by economic function.[2] The age- and sex-specific values for total taxes are calculated as the sum of the subcomponents.

2.1 TYEM — Taxes and social contributions on employment income

TYEM includes wage taxes, income taxes paid by the self-employed, and social contributions paid by employees, employers, and self-employed persons.

The ENTA aggregates by type of tax are based on the dataset on taxes by economic function. TYEM corresponds largely to *taxes on income from employment* paid by employees and employers, plus *taxes on the income of the self-employed*.

The age- and sex-specific estimates of TYEM combine data from the tax-benefit model EUROMOD and EU-SILC. EUROMOD is used to estimate age-specific tax rates by type of income, including income from employment. These rates are then applied to the income variables in EU-SILC.

2.2 TYA — Taxes on asset income and assets

Taxes on asset income and assets of households. The aggregate values are based on *taxes on capital — income of households* in the data on tax revenues by economic function. Additionally, the aggregate includes *other taxes* in the ESA category *taxes on income and wealth*.

The age-specific estimates use tax rates by type of income calculated from EUROMOD data, which are applied to impute age-specific estimates in EU-SILC. In general, they follow the age profile of property income, based on the corresponding EU-SILC variable (HY090).

2.3 TB — Taxes and social contributions on income of non-employed

TB captures taxes and social contributions paid on transfer benefits, such as pay-as-you-go pensions, benefits from funded pensions, or other social transfers. The aggregate of TB comes from the data on tax revenues by economic function. The subcategories of taxes paid on pensions and taxes paid on non-pension benefits are identified using EUROMOD microdata. The further distinction between taxes paid on pay-as-you-go and funded pensions uses the share of these two types of pensions, assuming the same rate of taxation on both:

$$TB = TBSPEN + TBSN + TBPPEN.$$

2.3.1 TBSPEN — Taxes on pay-as-you-go pensions

Taxes and social contributions paid on pension benefits received from government pay-as-you-go pension systems.

[2] `https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table`

2.3.2 TBSN — Taxes on non-pension social benefits

Taxes and social contributions paid on non-pension social benefits, such as unemployment benefits or family allowances.

2.3.3 TBPPEN — Taxes on funded pension benefits

Taxes and social contributions paid on benefits received from funded pension systems.

2.4 TCON — Consumption taxes

TCON captures primarily the taxes paid when purchasing a specific good or service, with value added tax as the most important component. The corresponding quantity in ESA is *taxes less subsidies on products*. The age profiles are based on estimates of consumption imputed in EU-SILC using information from the Household Budget Survey.

2.5 TX — Other current transfers

TX refers, in particular, to transfers from government to NPISH, as well as penalties or fines. The corresponding quantity in ESA is *other current transfers*. Both components, TXH and TXNPISH, are allocated with a receipt profile: labour income plus pensions (YEM + BSPPEN), the same profile used for BPX and BROW. These transfers are part of household disposable income before the intrahousehold transfers are estimated, so they are already held by the household members who earn or receive them and reach dependent members through the intrahousehold transfers. Allocating them to children a second time would count them twice, which showed up as positive saving below age 15. The profile is consequently zero below age 15, and the total profile is used for both sexes rather than a sex-specific one. Each component is scaled to its own aggregate, so the shape decides only how a total is spread across ages, never its level.

TX is reported separately by recipient sector: TXH for the household sector and TXNPISH for non-profit institutions serving households,

$$TX = TXH + TXNPISH.$$

The split follows the sector detail of the source data. It is available from 1995 for 24 of the 26 countries, but only from 2012 for Germany and from 2010 for Ireland; in the earlier years the whole of TX is carried in TXH and TXNPISH is missing. The total TX is unaffected, but TXH and TXNPISH are not comparable over time across those two breaks — see the country-specific limitations.

3 BS — Social Benefits

BS consists of the social benefits (in cash) paid by the government to residents. In ESA, these transfers are captured as *social benefits other than social transfers in kind*. The most important transfers in the ENTA category of social benefits are government pensions (BSPEN), followed by support for

families (BSFAM) and unemployment benefits (BSU). All other benefits (BSX, including health) are of minor importance.

$$BS = BSPEN + BSFAM + BSU + BSX.$$

The distinction of social benefits by function is based on data on general government expenditure by function (COFOG), which is consistent with ESA 2010.[3]

3.1 BSPEN — Pensions

The ENTA category of pensions includes all types of pension-like benefits, including regular old-age pensions, survivor pensions, and disability pensions.

The aggregate values are derived from expenditure on social benefits in the COFOG categories of old age, survivors, and disability. There is no clear-cut distinction between these functions in the microdata, and although EU-SILC contains a variable for each, it is not possible to distinguish them in the microdata in the same way as in the aggregates. Consequently, the age- and sex-specific data is based on the sum of the EU-SILC variables PY100G (old age), PY110G (survivors), and PY130G (disability).

3.2 BSFAM — Family and children

BSFAM includes cash benefits for families, children, and their education. The aggregate values are based on social benefits, not in-kind, in the COFOG categories family and children (GF10.04) and education (GF09). The combination enables better harmonisation of microdata and aggregate data. The age- and sex-specific data are based on the EU-SILC variables HY050G (family/children allowances) and PY140G (education-related allowances).

3.3 BSU — Unemployment

BSU mainly captures unemployment benefits. In the data on government expenditure by COFOG, these transfers are captured in the category of social protection and unemployment. The age- and sex-specific estimates are based on the EU-SILC variable PY090G.

3.4 BSX — Other social benefits

Captures social benefits not classified elsewhere. At the microdata level, BSX is based on the EU-SILC variables for social exclusion (HY060G), housing allowances (HY070G), and health-related cash benefits (sickness benefits, PY120G). At the aggregate level, the corresponding COFOG health category is excluded from the BSX aggregate before it is used to rescale the age profile.

At the aggregate level BSX is computed as a residual: the COFOG total for social benefits in cash less the categories that are named separately (old age, survivors, family, unemployment, sickness and disability, health, education). The residual form is deliberate, because it guarantees that the

[3] https://doi.org/10.2908/GOV_10A_EXP

benefit categories sum to the COFOG total and that nothing is silently dropped. It does mean that BSX is slightly wider than its name suggests. It contains the four *other social protection* categories — housing (GF1006), social exclusion not elsewhere classified (GF1007), research and development in social protection (GF1008) and social protection not elsewhere classified (GF1009) — and in addition any cash benefit that a country reports under a function other than health, education or social protection.

In practice that additional part is almost entirely *general economic, commercial and labour affairs* (GF0401). It is a standing item rather than an episodic one, and it is confined to two countries: about 4.6 to 5.5 billion euro a year in Germany and 1.0 to 1.1 billion in Belgium, stable across the whole period including 2020. It raises BSX by roughly 28 per cent for Germany and 42 per cent for Belgium. France reports no such benefits and is unaffected, and elsewhere the amounts are negligible. Users looking at BSX for Germany or Belgium should read it as other social protection plus labour-affairs benefits, not as other social protection alone.

The social protection total itself is internally consistent: COFOG GF10 equals the sum of GF1001 to GF1009 in every country-year, to within rounding. The wider content of BSX comes only from comparing against the all-function total.

Movements in BSX are almost always movements in social exclusion (GF1007). Of the forty largest year-on-year changes across all countries, thirty-six are driven by that category and four by housing (GF1006, all of them Germany); the labour-affairs component never drives a change. Two examples: Latvia in 2022, where GF1007 rises from 43.5 to 258.6 and falls back to 44.9 the following year, and Estonia in 2022, where it rises from 28.7 to 77.2 and stays up.

4 BPPEN — Funded Pensions

BPPEN captures the net transfer from funded pension systems, defined as benefits received less contributions paid:

$$BPPEN = BPSOCfc - TPSOCfc.$$

ENTA approximates contributions with social contributions paid by households to financial corporations and benefits with social benefits paid by financial corporations to households. As in ESA disposable income, both flows enter household disposable income gross. The difference between contributions and benefits is close to the ESA *adjustment for the change in pension entitlements* (D8), which ENTA does not use in its accounts; see D8 below.

In the microdata, it is not possible to distinguish between pay-as-you-go pensions and funded pensions. Both social contributions and benefits are therefore based on the same age profiles as pay-as-you-go pensions: contributions follow the age profile of taxes on employment income (TYEM), and benefits follow the age profile of total pension benefits (BSPPEN).

4.1 TPSOCfc — Funded pensions, contributions

Social contributions paid by households to financial corporations, which operate the funded pension schemes. The age profile follows that of taxes on employment income (TYEM).

4.2 BPSOCfc — Funded pensions, benefits

Social benefits paid by financial corporations to households from funded pension schemes. The age profile follows that of total pension benefits including funded pensions (BSPPEN).

4.3 D8 — Adjustment for the change in pension entitlements (memo item)

In ESA, contributions to and benefits from employment-related pension schemes are recorded as transfers between households and the pension schemes, which are mostly financial corporations. Disposable income (B.6n) includes these flows gross. The adjustment for the change in pension entitlements (D.8) is recorded after disposable income, in the use of disposable income account: it moves the accumulation of pension entitlements from the saving of the pension schemes to the saving of households,

$$\text{ESA net saving (B.8n)} = \text{B.6n} + \text{D.8} - \text{P.3}.$$

D8 is positive for households, negative for financial corporations, and nets to approximately zero for the total economy, the remainder being the rest-of-world leg. It is close to the net contributions to funded pensions, $TPSOCfc - BPSOCfc = -BPPEN$.

ENTA does not use D8 in any of its concepts. It is not part of primary income, disposable income, GDP or saving, for any sector, and it has no age profile. It is not part of the published data either; it is kept in the project's internal aggregate file, where it bridges ENTA saving to ESA saving:

$$SH + D8 = \text{ESA net saving (B.8n)}.$$

The same holds for corporations, with SC and a negative D8. As a consequence, household disposable income at market prices equals ESA disposable income B.6n (the published YDb is at basic prices, i.e. after consumption taxes), and household saving in ENTA is lower than ESA saving by D8 in countries with large funded pension schemes, in particular Denmark, the Netherlands and Sweden; corporate saving is higher by the same amount.

5 BP — Private Transfers

Private transfers include all transfers between private sectors as well as transfers between private sectors and the rest of the world (ROW). Among them are social benefits and contributions that are financed and/or organised by corporations themselves, such as continued remuneration during sick leave. Total private transfers are defined as:

$$BP = BPB + BPW - TPW.$$

5.1 BPB — Private transfers between households

BPB captures inter-household private transfers, including transfers between the household sector and non-financial corporations (BPSOCnfc, TPSOCnfc), other private transfers between households (BPX), as well as transfers received from the rest of the world less transfers paid (BROW):

$$BPB = BPSOCnfc - TPSOCnfc + BPX + BROW.$$

5.1.1 TPSOCnfc — Social contributions, non-financial corporations

Social contributions paid to employers, including imputed social contributions for social benefit schemes that are not financed by explicit contributions. An important example is the continued remuneration of employees in the event of sickness. The age profile follows that of taxes on employment income (TYEM).

5.1.2 BPSOCnfc — Social benefits, non-financial corporations

Social benefits paid by employers, such as the continued remuneration of employees in the case of sickness. The age profile follows that of employment income (YEM).

5.1.3 BPX — Other private transfers

Includes transfers in the national accounts category *other current transfers*. These are flows received from outside the household, so the age profile follows a receipt shape: employment income plus pensions ($YEM + BSPPEN$), taken for both sexes together rather than sex by sex, because the household member who receives such a transfer is as often a woman as a man.

5.1.4 BROW — Transfers from rest of the world

Includes social contributions, social benefits, and other private transfers received from the ROW less those paid to the ROW. This encompasses, in particular, benefits paid to non-resident workers (such as cross-border commuters) and remittances. As for BPX, the age profile follows employment income plus pensions ($YEM + BSPPEN$) for both sexes together.

5.2 BPW — Private transfers within households, received

Intra-household transfers received, representing resources received by household members whose consumption exceeds their individual income (typically children and non-working adults). These are imputed as the difference between each household member's consumption and their individual income net of saving and inter-household transfers.

5.3 TPW — Private transfers within households, paid

Resources provided by household members whose individual income exceeds their consumption (typically prime-age earners).

6 YD — Disposable Income

ENTA provides several measures of disposable income that differ in the income components included. Each successive measure adds components, from the narrowest definition based on employment income and transfers (YD1) to the broadest measure that includes all income and transfer flows (YD).

6.1 YD — Disposable income (NTA)

The broadest measure of disposable income, including all primary income, all public and private transfers, and intra-household transfers:

$$YD = YEM + YA - T + BS + BPPEN + BPB + BPW - TPW.$$

This is the income that is available for consumption and saving. The NTA accounting identity ensures:

$$YD = CH + SH.$$

The aggregate value in ENTA corresponds largely to the disposable income of the household sector in ESA. Age-specific values are the sum of all income and transfer age profiles.

6.2 YD1 — Disposable income from employment and transfers

A narrow measure of disposable income based on employment income and public transfers only:

$$YD1 = YEM - TYEM + BS + BPPEN - TB - TCON.$$

It excludes all asset income (property income and operating surplus), taxes on assets, other current transfers, private transfers between households, and intra-household transfers. YD1 captures the income available to individuals from their own work and the social transfer system. Estimates of these components are the most reliable and best suited for comparisons over time.

6.3 YD2 — Disposable income including intra-household transfers

Extends YD1 by adding private transfers within households:

$$YD2 = YD1 + BPW - TPW.$$

This measure captures the resources available from employment, public redistribution, and family support, but still excludes asset income and private transfers between households.

6.4 YD3 — Disposable income including operating surplus

Extends YD2 by including the operating surplus of households:

$$YD3 = YD2 + YO.$$

This measure includes imputed rent from owner-occupied housing but still excludes property income (dividends, interest, rent), taxes on assets and asset income, other current transfers, and private transfers between households.

6.5 EHI — Equivalised household income

Equivalised household income by age is based on total disposable household income (HY020) in EU-SILC and the OECD modified equivalence scale. EHI is not derived from the ENTA accounting framework but provided as a complementary measure of household welfare from the microdata.

7 CG — Government Consumption

Government consumption measures public consumption expenditure, distinguishing education, health, and other. In NTA, government consumption is categorised as an in-kind transfer. The aggregate values are based on general government expenditure by function (COFOG).[4]

$$CG = CGE + CGH + CGX.$$

7.1 CGE — Government consumption, education

Education expenditure of the government, covering all levels from pre-primary to tertiary education. Aggregates are based on COFOG, category education. The age profiles are based on government expenditure by education level (ISCED) and enrolment by education level, age, and sex.[5] Enrolment data from two Eurostat datasets are combined: *educ_uoe_enra02* (from 2013) and *educ_enrl1tl* (until 2012). Enrolment rates by ISCED level are multiplied by per-student expenditure at each level to obtain age-specific education consumption.

7.2 CGH — Government consumption, health

Government expenditure on health care services. Age profiles are based on the health expenditure profiles of the 2018 Ageing Report of the European Commission and the Ageing Working Group, the same source for all countries. One profile per country is used for all years and scaled to each year's aggregate.

7.3 CGX — Government consumption, other

All remaining government consumption not classified under education or health. In the ENTA framework, all expenditure of the government is treated as transfers. Expenditure that cannot be assigned to a particular age group — such as public administration or defence — is assumed to benefit all individuals equally, with everyone receiving the same per capita amount.

[4] https://ec.europa.eu/eurostat/databrowser/view/gov_10a_exp/default/table

[5] https://ec.europa.eu/eurostat/databrowser/view/educ_uoe_enra02/default/table and https://ec.europa.eu/eurostat/databrowser/view/EDUC_ENRL1TL/default/table

8 CH — Household Consumption

Household consumption expenditure at basic prices, with housing, food, and transport as the largest components:

$$CH = CHE + CHH + CHR + CHX.$$

In ESA, total household consumption is captured as *final consumption expenditure*. For the distinction by purpose, ENTA uses data on household final consumption expenditure by purpose (COICOP 1999).[6]

Age- and sex-specific estimates are based on the Household Budget Survey, which provides information on household structure and consumption expenditure. ENTA uses different methods and assumptions to distribute expenditure within households. Note that household consumption in ENTA is measured at basic prices, i.e. excluding taxes less subsidies on products. The variable name in the code is `Cb`, which is renamed to `CH` in the long-format output.

8.1 CHE — Household consumption, education

Household expenditure on educational services, including tuition fees, textbooks, and other education-related spending. Note: for Sweden, education consumption data are not available from the HBS; CHE is set to zero and the corresponding amount is included in CHX.

8.2 CHH — Household consumption, health

Household out-of-pocket expenditure on health, including co-payments, pharmaceuticals, and medical services not covered by public or private insurance.

8.3 CHR — Household consumption, owner-occupied housing

Imputed rental value of owner-occupied housing, representing the housing services consumed by owner-occupiers.

8.4 CHX — Household consumption, other

All remaining household consumption expenditure not classified under education, health, or owner-occupied housing, including food, clothing, transport, communication, and recreation.

9 SH — Saving, Households

Saving of the household sector, defined as disposable income less consumption expenditure:

$$SH = YD - CH.$$

[6] `https://doi.org/10.2908/NAMA_10_CO3_P3`

It represents the change in household net worth due to current transactions. ENTA saving excludes the adjustment for the change in pension entitlements; adding the memo item D8 gives ESA net saving (B.8n).

Household saving is published under the code `SH`. Earlier releases of the ENTA–ESA file used `S`; in the ENTA–NTA file `S` denotes total saving, so a script keyed on the old ESA code needs updating.

C Corporation Sector

The corporation sector captures income, taxes, transfers, and saving of non-financial and financial corporations. All corporate sector variables use the age profile of property income from the household sector, scaled to the corresponding corporate aggregate from ESA. The suffix C distinguishes corporate variables from their household counterparts.

C.1 YC — Primary income, corporations

Primary income of corporations corresponds to their income after compensating the factors of production (compensation of employees, paying net property income such as interest and dividends) but before paying taxes. The ENTA aggregate values for YC are calculated from the ESA quantities as follows: *net value added* less *compensation of employees* less net *property income* paid. Age profiles are based on property income in EU-SILC.

C.2 TC — Taxes, corporations

Taxes paid by corporations mainly include taxes on production and taxes on corporate income. The age profiles of TC are based on those for property income of households, using information on property income in EU-SILC.

C.4 BPPC — Funded pensions, corporations

Funded pensions from the perspective of financial corporations mirror the flows of the household sector: contributions to funded pension systems constitute an inflow for financial corporations, while benefits represent an outflow:

$$BPPC = -BPSOCfc + TPSOCfc.$$

C.5 BPBC — Private transfers, corporations

Private transfers of corporations include flows to and from the household sector, as well as some flows to and from the rest of the world. The age profile is based on property income of households.

C.9 SC — Saving, corporations

Saving of corporations largely represents retained profits, i.e. the profits that are not paid out to owners but are kept in the company for investment. Total corporate saving consists of saving excluding funded pensions (SCX) and saving from funded pensions (SCP):

$$SC = SCX + SCP.$$

As for households, corporate saving excludes the adjustment for the change in pension entitlements: SC plus the (negative) corporate D8 gives ESA net saving of corporations.

C. Codebook ENTA (NTA concept)

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0 Introduction and General Variables

ENTA (European National Transfer Accounts) is provided in two versions. The first version focuses on the household sector and maintains a close link to the European System of Accounts (ESA). The second version ensures comparability with global NTA terminology and concepts. The ENTA data organised according to NTA terminology can be used for comparisons with global NTA data. However, there are some differences that users need to consider.

0.1 Differences between ENTA in NTA format and global NTA

First, the organisation of the data according to the ESA/NA sequence of accounts is maintained. Rather than showing the life cycle deficit (difference between consumption and income) and how it is covered through transfers and asset-based reallocations, ENTA shows how income is generated, redistributed, and used for consumption and saving.

Second, ENTA provides a more detailed distinction by sector than standard NTA data: it separates the household sector from the corporation sector. Global NTA generally distinguishes between the private sector and the government sector. Private sector quantities are denoted with the suffix F (for "family"), while government sector quantities carry the suffix G — for example, private asset income YAF and government asset income YAG. ENTA further decomposes the private sector into households (H) and corporations (C). For example, YAH refers to asset income of households, YAC to that of corporations, and:

$$YAF = YAH + YAC.$$

Third, in NTA labour income includes a share of self-employment income (mixed income). The standard NTA assumption is that two-thirds of mixed income represents the return to labour and

one-third represents the return to capital. In ENTA in ESA format, employment income includes the full mixed income, reflecting the ESA treatment. The NTA format applies the two-thirds/one-third split:

$$YL = YE + \tfrac{2}{3}\, YM, \qquad YAH = YA + \tfrac{1}{3}\, YM.$$

Fourth, the two formats show taxes on benefits differently. In the ESA format, social benefits are reported gross and taxes on benefits (TB) are a separate tax component. In the NTA format, social benefits ($TGIC$) also equal the ESA gross benefits (BS), and taxes on benefits are included in total taxes ($TGOT$) without being shown separately.

0.2 Overview

Table C.1 shows the NTA variables and their relation with ENTA in ESA format.

Table C.1: ENTA variable definitions in NTA format

Number	Variable	Abbrev.	ENTA-ESA format
0	Population	POP	*POP*
	1. Income		
1	Primary income	Y	
1.1	Labour income	YL	$YE + \frac{2}{3}\,YM$
1.2	Asset income, private	YAF	$YAH + YAC$
1.2.1	Asset income, households	YAH	$YA + \frac{1}{3}\,YM$
1.2.2	Asset income, corporations	YAC	*YC*
1.3	Asset income, government	YAG	
	2. Transfers to government (taxes and contributions)		
2	Taxes, total	TGOT	$T + TC$
2.1	Taxes, paid by households	TGOTH	*T*
2.2	Taxes, paid by corporations	TGOTC	*TC*
2.3	Public transfers, net from ROW	TGNF	
	3. Transfers from government (cash)		
3	Social benefits	TGIC	*BS*
3.1	Social benefits, pensions	TGICPEN	*BSPEN*
3.2	Social benefits, family	TGICF	*BSFAM*
3.3	Social benefits, unemployment	TGICU	*BSU*
3.4	Social benefits, other	TGICX	*BSX*
	4. Funded pensions		
4	Funded pensions (received less paid)	TPEN	*BPPEN*
	5. Private transfers		
5	Private transfers	TF	
5.1	Private transfers between households	TFB	$BPB + BPBC$
5.2	Private transfers within households	TFW	$BPW - TPW$
5.2.1	Intra-household transfers received	TFWI	*BPW*
5.2.2	Intra-household transfers paid	TFWO	*TPW*
	6. Disposable income		
6	Disposable income, private sector	YDF	
6.1	Disposable income, households	YDH	*YD*
6.2	Disposable income, corporations	YDC	

Table C.2: ENTA variable definitions in NTA format (continued)

Number	Variable	Abbrev.	ENTA-ESA format
	7. Government consumption		
7	Public consumption	CG	
7.1	Public cons., education	CGE	
7.2	Public cons., health	CGH	
7.3	Public cons., other	CGX	
7.4	Adjusted disp. income, households	YDHA	$YDH + CG$
	8. Private consumption		
8	Private consumption	CF	CH
8.1	Private consumption, education	CFE	CHE
8.2	Private consumption, health	CFH	CHH
8.3	Private consumption, housing	CFR	CHR
8.4	Private consumption, other	CFX	CHX
	9. Saving		
9	Saving, total	S	$SF + SG$
9.1	Saving, private	SF	$SH + SC$
9.1.1	Saving, households	SH	SH (same code in ESA format)
9.1.2	Saving, corporations	SC	SC
9.2	Saving, government	SG	
	NTA aggregates		
10.1	Net government transfers	TG	$TGI - TGO$
10.2	Total government inflows	TGI	$TGIC + CG$
10.3	Total government outflows	TGO	$TGOT + YAG - SG$
10.3.1	Government outflows, asset-based	TGOABR	$YAG - SG$
10.4	Total consumption	C	$CF + CG$

0.3 Basic variables

AGE — Age group

Single-year age groups for ages 0 to 79. The age group 80 includes all persons aged 80 and older.

Country-specific notes:

- **Germany:** In the years 2014–2018, the oldest available age group is 73. For EU-SILC–based variables, averages for the 73+ group are applied to all older age groups up to age 80. From 2022 onward, age is provided only in five-year age groups.

YEAR — Reference year

Calendar year to which the observation refers.

CNT2 — Country code

Two-letter country code following the ISO 3166-1 alpha-2 standard (e.g. DE for Germany, FR for France).

1 Y — Primary Income

Primary income in NTA is classified into labour income (YL) and asset income (YAF, YAG):

$$Y = YL + YAF + YAG.$$

1.1 YL — Labour income

Labour income represents remuneration for work. Its largest component is gross wages, which include employers' social contributions and other taxes on payroll and employment paid by the employer. Labour income also includes the labour share of self-employment income (mixed income), which is assumed to be two-thirds of total self-employment income following the standard NTA convention:

$$YL = YE + \tfrac{2}{3}\, YM.$$

1.2 YAF — Asset income, private sector

Asset income of the private sector, calculated as the sum of household and corporate-sector asset income:

$$YAF = YAH + YAC.$$

1.2.1 YAH — Asset income, households

Asset income of the household sector, including property income (interest, dividends, rent on land), imputed rents on owner-occupied housing (operating surplus), and one-third of self-employment income (the capital share of mixed income):

$$YAH = YA + \tfrac{1}{3}\, YM,$$

where YA is total household asset income in ENTA–ESA format (property income plus operating surplus) and YM is mixed income.

1.2.2 YAC — Asset income, corporations

Primary income of corporations, corresponding to the operating surplus of corporations less net property income paid to other sectors, before the payment of corporate taxes. Corresponds to YC in the ENTA–ESA format.

1.3 YAG — Asset income, government

Asset income of the government sector. This quantity is negative in most countries, largely reflecting interest payments on public debt.

2 TGOT — Public Transfer Outflows (Taxes and Contributions)

Government transfer outflows (taxes) comprise all transfers paid by the private sector to the government, net of subsidies received. The most important components are income taxes (wage and other income taxes), social contributions (employees' and employers' shares), and taxes on consumption (VAT and excise duties):

$$TGOT = TGOTH + TGOTC.$$

2.1 TGOTH — Taxes paid by households

All taxes and social contributions paid by the household sector to the government, including taxes on employment income, asset income, benefits, and consumption, as well as other current transfers. Corresponds to T in the ENTA–ESA format.

2.2 TGOTC — Taxes paid by corporations

Taxes paid by the corporation sector, mainly including taxes on production and taxes on corporate income. Corresponds to TC in the ENTA–ESA format.

2.3 TGNF — Public transfers, net from rest of the world

Net transfers between the domestic government sector and the rest of the world (ROW), including contributions to and receipts from international organisations and foreign governments. This component affects only the government sector and is included for accounting consistency with total government revenue and expenditure.

3 TGIC — Public Transfer Inflows in Cash

Public transfer inflows in cash consist of social benefits paid in cash by the government to individuals. Pensions are the largest component. The categories correspond to social benefits in the ENTA–ESA format, with the abbreviation prefix "TGIC" (transfers, government, inflows, cash) replacing "BS":

$$TGIC = TGICPEN + TGICF + TGICU + TGICX.$$

3.1 TGICPEN — Public transfer inflows in cash, pensions

Includes all pension-like cash benefits: regular old-age pensions, early-retirement pensions, survivor pensions, and disability pensions. Corresponds to *BSPEN* in the ENTA–ESA format.

3.2 TGICF — Public transfer inflows in cash, family

Cash benefits directed at families and children, including child allowances, parental leave benefits, and education-related cash transfers. Corresponds to *BSFAM* in the ENTA–ESA format.

3.3 TGICU — Public transfer inflows in cash, unemployment

Cash benefits related to unemployment, including unemployment insurance benefits and unemployment assistance. Corresponds to *BSU* in the ENTA–ESA format.

3.4 TGICX — Public transfer inflows in cash, other

All remaining social benefits in cash, including benefits in the categories of health (sickness benefits) and other social benefits (housing, social exclusion). These categories are individually too small for reliable age-specific estimation based on survey data and are therefore grouped together. Corresponds to *BSX* in the ENTA–ESA format.

4 TPEN — Funded Pensions

Funded pension transfers (*TPEN*) capture benefits received from funded (i.e. capital-reserve–based) pension systems, net of household contributions to those systems. Corresponds to *BPPEN* in the ENTA–ESA format.

For the private sector as a whole, funded pension transfers net to zero across age groups, because benefits are paid by financial corporations and received by households — both belonging to the private sector. They do, however, generate a reallocation of resources across age groups: younger workers pay contributions, while retirees receive benefits.

Note: There are several ways to treat funded pensions, depending on the purpose of the analysis:

- as *private transfers* (redistribution within the private sector),
- as *public transfers* (since participation is often mandatory and regulated by government),
- as *saving/dissaving* of households (an asset-based perspective, where contributions represent saving and benefits represent dissaving).

5 TF — Private Transfers

Private transfers include all forms of redistribution that do not involve the government sector:

$$TF = TFB + TFW.$$

5.1 TFB — Private transfers between households

Inter-household transfers, including social contributions and benefits organised by non-financial corporations (such as continued remuneration during sick leave), remittances, regular cash support between households, and transfers to and from the rest of the world. In the NTA format, TFB combines the household component (BPB) and the corporate component ($BPBC$) from the ENTA–ESA format:

$$TFB = BPB + BPBC.$$

There is no reliable microdata to distribute most of these transfers by age. ENTA distributes their net value with a receipt profile (employment income plus pensions) that carries no information on who actually pays or receives them. These transfers are comparatively small and included for accounting completeness.

5.2 TFW — Private transfers within households

Net intra-household transfers, representing the reallocation of resources among members of the same household:

$$TFW = TFWI - TFWO.$$

These are imputed as the difference between each household member's consumption and their individual income net of saving and inter-household transfers. Intra-household transfers are the mechanism through which children and non-working household members receive resources for consumption.

5.2.1 TFWI — Intra-household transfers, received

Resources received by household members whose consumption exceeds their individual income (typically children and non-working adults). Corresponds to BPW in the ENTA–ESA format.

5.2.2 TFWO — Intra-household transfers, paid

Resources provided by household members whose individual income exceeds their consumption (typically prime-age earners). Corresponds to TPW in the ENTA–ESA format.

6 YDF — Disposable Income, Private Sector

Disposable income of the private sector is calculated as primary income less taxes plus social benefits, funded pensions, and private transfers:

$$YDF = YL + YAF + TFB + TFW - TGOT + TGIC.$$

The NTA accounting identity ensures that private disposable income equals private consumption plus private saving:

$$YDF = CF + SF.$$

YDF refers to the combined disposable income of the household and corporation sectors. It can be decomposed as:

$$YDF = YDH + YDC.$$

6.1 YDH — Disposable income, household sector

Disposable income of the household sector, calculated as:

$$YDH = YL + YAH + TFB - BPBC + TFW - TGOTH + TGIC + TPEN.$$

This corresponds to the NTA version of disposable income (YD) in the ENTA–ESA format. It includes all household primary income, less taxes paid by households, plus social benefits in cash, funded pensions (received less paid), and all private transfers (between and within households).

6.2 YDC — Disposable income, corporations

Disposable income of the corporation sector, calculated as corporate asset income less corporate taxes, plus the corporate component of private transfers, less funded pension transfers:

$$YDC = YAC - TGOTC + BPBC - TPEN.$$

Since corporations do not consume, corporate disposable income equals corporate saving: $YDC = SC$.

7 CG — Government Consumption

Government consumption represents the value of goods and services provided by the government to individuals or the community. In NTA, it is treated as an in-kind transfer from the government to individuals and is allocated by age:

$$CG = CGE + CGH + CGX.$$

7.1 CGE — Government consumption, education

Government expenditure on education services, including pre-primary, primary, secondary, and tertiary education. Allocated by age using enrolment data by education level, age, and sex, combined with per-student spending by education level.

7.2 CGH — Government consumption, health

Government expenditure on health care services, including hospital care, outpatient care, and long-term care. Allocated by age using health expenditure profiles from the 2018 Ageing Report of the European Commission and the Ageing Working Group.

7.3 CGX — Government consumption, other

All remaining government consumption not classified under education or health, including public order, defence, general administration, and environmental protection. Allocated equally per capita across all ages.

7.4 YDHA — Adjusted disposable income, household sector

Disposable income of the household sector including government consumption allocated to individuals (i.e. in-kind transfers):

$$YDHA = YDH + CG.$$

In ESA, *adjusted disposable income* refers to disposable income plus individual (but not collective) government consumption. The ENTA definition includes all government consumption, including the collective component, which is allocated equally per capita.

8 CF — Private Consumption

Final consumption expenditure of households at basic prices, with housing (including energy), mobility, and food and non-alcoholic beverages as major components:

$$CF = CFE + CFH + CFR + CFX.$$

Aggregate values are based on ESA final consumption expenditure at basic prices. The breakdown by purpose follows the Classification of Individual Consumption According to Purpose (COICOP 1999).[1] Age- and sex-specific estimates are mainly based on data from the Household Budget Survey (HBS).

8.1 CFE — Private consumption, education

Household expenditure on education, including tuition fees, textbooks, and other education-related spending. Corresponds to *CHE* in the ENTA–ESA format.

8.2 CFH — Private consumption, health

Household out-of-pocket expenditure on health, including co-payments, pharmaceuticals, and medical services not covered by public or private insurance. Corresponds to *CHH* in the ENTA–ESA format.

8.3 CFR — Private consumption, imputed rent

Imputed rental value of owner-occupied housing, representing the housing services consumed by owner-occupiers. Corresponds to *CHR* in the ENTA–ESA format.

8.4 CFX — Private consumption, other

All remaining household consumption expenditure not classified under education, health, or imputed rent, including food, clothing, transport, communication, and recreation. Corresponds to *CHX* in the ENTA–ESA format.

9 S — Saving

Saving is defined as the residual of disposable income less consumption. It represents the change in net worth due to current transactions:

$$S = SF + SG.$$

9.1 SF — Private saving

Saving of the private sector (households and corporations combined):

$$SF = SH + SC.$$

9.1.1 SH — Saving, households

Saving of the household sector, equal to household disposable income less household consumption expenditure:

$$SH = YDH - CF.$$

[1] https://doi.org/10.2908/NAMA_10_CO3_P3

Published under the same code, SH, in the ENTA–ESA format.

9.1.2 SC — Saving, corporations

Saving of the corporation sector, comprising retained earnings (profits not distributed as dividends or other property income). Since corporations do not consume, corporate saving equals corporate disposable income: $SC = YDC$.

9.2 SG — Saving, government

Saving of the government sector, equal to government revenue less government current expenditure (i.e. the government budget balance on current account).

10 NTA Aggregates

NTA is based on a flow identity that holds for each individual and each age group. For the total economy:

$$\underbrace{Y}_{\text{Income}} + \underbrace{TGI}_{\text{Gov. inflows}} - \underbrace{TGO}_{\text{Gov. outflows}} + \underbrace{TFB + TFW}_{\text{Private transfers}} = \underbrace{C}_{\text{Consumption}} + \underbrace{S}_{\text{Saving}}.$$

These aggregate variables are constructed in the code and included in the standard ENTA output, as indicators 10.1–10.4, plus 10.3.1 for the asset-based component of TGO (TGOABR). They can be used to verify the accounting identity.

10.1 TGO — Total government transfer outflows

Total resources transferred from individuals to the government, calculated as:

$$TGO = TGOT + YAG - SG.$$

This expression reflects that government expenditure allocated to individuals is financed by tax revenue ($TGOT$), government asset income (YAG), and government dissaving ($-SG$).

10.2 TGOABR — Government outflows, asset-based

The part of total government transfer outflows financed by government asset income net of government saving, rather than by tax revenue:

$$TGOABR = YAG - SG.$$

It is the non-tax component of TGO: $TGO = TGOT + TGOABR$.

10.3 TGI — Total government transfer inflows

Total resources received by individuals from the government, comprising cash transfers and in-kind transfers (government consumption):

$$TGI = TGIC + CG.$$

10.4 TG — Net government transfers

Net public transfers received by an age group, i.e. total government transfer inflows less total government transfer outflows:

$$TG = TGI - TGO.$$

A positive value indicates an age group is, on net, a beneficiary of public redistribution; a negative value indicates it is, on net, a contributor.

10.5 C — Total consumption

Total consumption of an age group, combining private and government consumption:

$$C = CF + CG.$$

D. Organisation of ENTA code and data

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0 Overview

ENTA combines a range of different datasets. This chapter gives an overview of the data, the way they are combined into ENTA, and the organisation of programming code and data. It should enable users to investigate details about the calculations and replicate results.

Figure D.1 provides a schematic overview. The input data folder contains the code to prepare all source data. The aggregates folder calculates ENTA aggregate values from ESA sector accounts and supplementary data. The age profiles folder generates age- and sex-specific profiles from EU-SILC, HBS, and other microdata. The two compilation files in the root directory combine aggregates and age profiles into the final ENTA datasets.

1 The ENTA Root Directory

All code and publicly available data of the ENTA project are included in the ENTA folder, which is the root of the project. The root directory includes four Stata do-files:

- *ENTA_setup_TU.do* and *ENTA_setup.do* (identical apart from the hardcoded directory paths — TU points to local drives, VID to the VID network shares; use whichever matches the machine ENTA is run on)
- *ENTA_compilation1_esa.do*
- *ENTA_compilation2_nta.do*

and six folders with different functions in data preparation:

- *ENTA_input_data*
- *ENTA_aggregates*
- *ENTA_ageprofiles*
- *ENTA_data_other*

```
ENTA/
|-- ENTA_setup_TU.do / ENTA_setup.do
|-- ENTA_compilation1_esa.do
|-- ENTA_compilation2_nta.do
|-- ENTA_input_data/
|   |-- Aggr_*.do, Micro_*.do, Suppl_*.do
|   |-- Original_input_data/
|   `-- ENTA_input_datasets/
|-- ENTA_aggregates/
|   |-- ENTA_aggregates_compilation.do
|   |-- ENTA_units.do
|   `-- ENTA_aggregates.dta
|-- ENTA_ageprofiles/
|   |-- ENTA_income.do, ENTA_taxes.do, ...
|   `-- ageprofiles/
|-- ENTA_data_other/
|   `-- Health/
|-- ENTA_output_data/
|   |-- ENTA_priv_wide.dta
|   |-- ENTA_esa.dta / .csv / .zip
|   |-- ENTA_nta.dta / .csv / .zip
|   |-- ENTA_nta_complete_wide.dta
|   |-- ENTA_aggregates.dta / .csv
|   `-- ENTA_units.dta / .csv
`-- ENTA_diagnostic/
    |-- ENTA_diagnostic_scan.do
    `-- scan_triage.csv
(the CSV files in ENTA_output_data/ are read by the dataexplorer and
 published externally)
```

Figure D.1: ENTA folder structure

- *ENTA_output_data*
- *ENTA_diagnostic*

The first five follow the flow of the calculation, from source data to final output. The sixth, *ENTA_diagnostic*, is not part of that flow: it checks the finished output and is run after it.

1.1 ENTA_setup_TU.do / ENTA_setup.do

The first step for running the ENTA code is specifying the ENTA root as the working directory in *ENTA_setup_TU.do* (or *ENTA_setup.do*, depending on the machine) and running the code. The directories containing the Eurostat microdata (EU-SILC, HBS, EUROMOD, LFS) can also be specified in this file. The do-file additionally defines several Stata programs that are used throughout the project, including:

- *adjustaggr*: adjusts microdata so that the population-weighted sum matches the ESA aggregate
- *adjust*: adjusts an already-collapsed age profile to the corresponding ESA aggregate
- *adjustgender*: rescales male/female profiles so they are consistent with the total profile
- *adjustsec*: adjusts a profile to an aggregate from a specified sector (e.g. asset income to the corporation sector)
- *control* / *controlcorp*: display the aggregate value implied by a given per-capita age profile, for checking against the ESA aggregate

The program *gensex* (generates sex-specific variables — total, male, female — from a single expression) is defined separately, in *ENTA_compilation1_esa.do*. Age-profile smoothing is not a shared program; each age-profile do-file applies Stata's *supsmooth* directly, per year and variable.

1.2 ENTA_compilation1_esa.do

The file *ENTA_compilation1_esa.do* is the heart of the ENTA project and combines aggregate and age-specific data into European National Transfer Accounts, in ESA terminology. Running the ENTA compilation do-files is the final step of generating ENTA. It uses only the data provided in *ENTA_input_data/ENTA_input_datasets* and the age profiles in *ENTA_ageprofiles/ageprofiles*, and works without downloading any additional data.

The compilation proceeds in the following steps:

1. Build the population base for each country and merge in the age profiles for income, taxes, benefits, and private consumption/intra-household transfers
2. Derive quantities not directly available from the merged profiles (e.g. taxes on asset income (TYP) from the property-income profile, or social contributions and benefits in the non-financial corporate sector (BPSOCnfc/TPSOCnfc) from the employment-income profile)

3. Adjust each age profile to the corresponding ESA aggregate (via the *adjust*/*adjustaggr* programs from the setup file, which merge against *ENTA_aggregates.dta* internally)

4. Calculate derived and composite variables (e.g. intra-household transfers, disposable income, saving) and check flow identities

5. Calculate the corporation sector variables using the age profile of property income as an allocation key, then the government sector variables (including public health and education consumption)

6. Save the combined output in wide format (*ENTA_priv_wide.dta*)

7. Reshape to long format, generate indicator labels, and save the final output (*ENTA_esa.dta* and *ENTA_esa.csv*, plus a zipped copy of the CSV for the Zenodo upload)

Values are saved at nominal per-capita value only. Calculating them in other units (real, PPS, percent of income, NTA/ENTA norms) is done by the ENTA dataexplorer on demand rather than pre-computed here — see Appendix A (Units) for the conversion factors and formulas involved.

1.3 ENTA_compilation2_nta.do

The Stata do-file *ENTA_compilation2_nta.do* transforms the data generated in *ENTA_compilation1_esa.do* into the format and concepts used by the global NTA project. It reads the wide-format output *ENTA_priv_wide.dta* and:

- Splits mixed income into a labour share (two-thirds) and a capital share (one-third)
- Combines household and corporation sectors into private-sector totals
- Renames variables to NTA conventions (e.g. $BS \rightarrow TGIC$, $CH \rightarrow CF$)
- Calculates NTA aggregates (TGO, TGI, C) and verifies the flow identity
- Saves the output in wide format (*ENTA_nta_complete_wide.dta*) and long format (*ENTA_nta.dta* and *ENTA_nta.csv*, plus a zipped copy of the CSV)

1.4 ENTA_input_data (folder)

The folder *ENTA_input_data* includes all code used to prepare the original input datasets for use in ENTA. The code for each dataset is provided in a separate do-file, grouped into three types of data:

- The prefix *Aggr_* refers to code that prepares aggregate data from ESA 2010.
- The prefix *Micro_* refers to code for preparing microdata from EU-SILC, EUROMOD, HBS, and LFS. These files can only be run with access to the respective datasets (`https://ec.europa.eu/eurostat/web/microdata`).
- The prefix *Suppl_* is used for supplementary data, including population by age, government expenditure, household consumption expenditure, data on tax structure, enrolment in education,

and prices. These files use publicly accessible data; the code for downloading them is included in the respective files.

1.4.1 Original_input_data (subfolder)

This subfolder contains the original datasets used for generating ENTA. In particular, all data directly downloaded from Eurostat are saved here. These data are relevant only for those who build ENTA from scratch. ENTA can be compiled without direct access to these data.

1.4.2 ENTA_input_datasets (subfolder)

The ENTA input data (output of the *Aggr_*, *Suppl_*, and *Micro_* do-files) are saved in the folder *ENTA_input_datasets*, provided the source of the data is publicly available. These data are publicly provided as ENTA supplementary material. The microdata need to be stored in a local folder whose path is specified in *ENTA_setup.do*, in compliance with the Eurostat rules for accessing microdata.

1.5 ENTA_aggregates (folder)

The *ENTA_aggregates* folder includes the code to calculate the ENTA aggregate data, which are stored in *ENTA_aggregates.dta*. It uses the datasets from the folder *ENTA_input_data/ENTA_input_datasets*. The folder contains three do-files: *ENTA_aggregates_compilation.do*, *ENTA_units.do* and *ENTA_generate_manual_ta* (described in Section 3).

1.6 ENTA_ageprofiles (folder)

The folder *ENTA_ageprofiles* includes all the code used to generate age profiles. Most of the age profiles are based on EU-SILC microdata. The folder contains separate do-files for income, taxes, benefits, consumption, employment (described in Section 4); public consumption is calculated within *ENTA_compilation1_esa.do*.

1.6.1 ageprofiles (subfolder)

Contains the output of the do-files in *ENTA_ageprofiles*: country-specific Stata files with age- and sex-specific profiles of income (*inc_[CNT2].dta*), taxes (*tax_[CNT2].dta*), benefits (*ben_[CNT2].dta*), consumption expenditure and intra-household transfers (*ctpriv_[CNT2].dta*), and employment (*lfs_[CNT2].dta*). Public consumption is not among them: it is built inside *ENTA_compilation1_esa.do* and never saved per country. The folder also holds *cfe_cfh_1015_[CNT2].dta*, an intermediate written and read again by *ENTA_consumption_intrahousehold_transfers.do* — the HBS-based age and sex pattern of education and health consumption — not a profile in its own right.

1.7 ENTA_data_other (folder)

A single subfolder, *Health*, holding the supplementary data on age-specific public health and long-term-care expenditure that no Eurostat table provides: the health expenditure rates from the 2018

Ageing Report (*HealthExpRates.csv*), and two sources no longer used but kept for reference, the AGENTA profiles and the Austrian source workbooks with their code, *Austria/health_AT.do*. The code beside them — *Combine_public_health.do* — builds *CGH_AWG.dta* in *ENTA_input_datasets* and is therefore part of the input-data step, described in Section 2 with the other input files rather than here.

1.8 ENTA_output_data (folder)

This folder contains the final ENTA datasets in wide and long format. In the wide format, each ENTA indicator is stored as a separate variable, with sex-specific values identified by the suffixes m and f. Each observation corresponds to a country, year, and age group. In the long format, each observation corresponds to a single indicator for a given country, year, age group, and sex. Both formats contain the full set of ENTA variables for all countries and years.

The folder contains:

- *ENTA_priv_wide.dta*: ENTA in ESA format, wide (intermediate file)
- *ENTA_esa.dta*: ENTA in ESA format, long, Euro per capita nominal (other units are calculated on demand by the ENTA dataexplorer, not stored here — see Appendix A). The identical data is also saved as *ENTA_esa.csv*, which is used by both the ENTA dataexplorer and external publication (e.g. Zenodo), and as *ENTA_esa.zip*, built from that CSV.
- *ENTA_nta_complete_wide.dta*: ENTA in NTA format, wide (intermediate file).
- *ENTA_nta.dta*: ENTA in NTA format, long, Euro per capita nominal (other units calculated on demand as above), with the identical *ENTA_nta.csv* and *ENTA_nta.zip*.
- *ENTA_aggregates.dta*: ENTA aggregate data in long format, nominal aggregate total (other units calculated on demand, as above). The identical data is also saved as *ENTA_aggregates.csv*.
- *ENTA_units.dta* and *ENTA_units.csv*: the unit conversion factors (Appendix A).

1.9 ENTA_diagnostic (folder)

The finished output is checked rather than assumed correct. *ENTA_diagnostic_scan.do* reads only finished files — *ENTA_priv_wide.dta*, *ENTA_nta_complete_wide.dta* and *ENTA_aggregates.dta* — and rebuilds nothing, so it can also be pointed at an earlier vintage by changing three paths at the top. It is run after every change to the pipeline. The checks are graded: the first tier tests invariants, where a hit is a defect — every profile must reconcile with its aggregate to within 0.01 per cent, the accounting identities must hold, and quantities that are floored at zero in the pipeline must not be negative. The remaining tiers look for year-on-year level breaks, changes in the shape of a profile, inconsistencies between the male, female and total profiles, and the same checks again on the NTA output.

Level and shape breaks are judgements, not defects: most of them are real events in the data. *scan_triage.csv* is the ledger of those already examined, one line per flag with the reason it was

accepted. The scan reads it and reports only what is not already in it, so a run that prints nothing new means the output is in its known state. *scan_findings.md* and *BSU_smoothing_review.txt* record what earlier passes established.

A second scan, *dataexplorer_scan.R*, runs against the published long CSV files — the form in which users actually receive the data — and so catches anything introduced between the wide Stata files and the publication step.

2 ENTA_input_data (folder)

The input data in this folder are essential for generating ENTA. To build ENTA from scratch, the code in this folder needs to be run first — with one exception. *Micro_impute_cons_into_SILC.do* reads *ENTA_units.dta*, so it runs after *ENTA_units.do* and before the consumption age profile, not with the rest of this folder. Most of the data are downloaded from Eurostat; to re-download them, simply uncomment the respective lines at the beginning of the code. However, the output of the code consists of datasets that contain condensed information and are directly provided with ENTA in the folder *ENTA_input_data/ENTA_input_datasets*.

Input data to calculate aggregates

2.1 Tax structure aggregates (Suppl_tax_structure.do)

The ESA sector accounts provide data on taxes and social contributions. Additionally, ENTA uses information on *tax revenues by economic function* (i.e. the type of income on which they are levied) for the decomposition of taxes by type of income. This information is taken from `https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/explore/all/DATA_ON_TAX`.

Suppl_tax_structure.do imports these data to identify taxes and social contributions on employment (TYEE for payments by employees, TYER for employers' social contributions), on self-employment (TYM), and taxes paid on income that is not from employment or property, such as taxes on pensions and non-pension social benefits (TB). The output is saved as *taxstructure_wide.dta* in *ENTA_input_datasets*.

2.2 ESA 2010 sector accounts (Aggr_ESA2010.do)

The most important dataset for ENTA aggregate data is the *non-financial transactions—annual data* of the ESA sector accounts (`https://doi.org/10.2908/NASA_10_NF_TR`). These data contain sector-specific information on total primary income, total public transfers, private transfers between households, consumption, and saving. The file saves the data as *esa_aggregates.dta* in *ENTA_input_datasets*.

ENTA requires estimates of which sectors transfers originate from and to which sectors they are paid. This allocation is performed in the file *Aggr_ESA2010_allocate_transfers_to_sectors.do*, which is called within the *Aggr_ESA2010.do* code. Because of its length, it is kept in a separate file.

2.3 EUROMOD microdata on taxes (Micro_EUROMOD.do)

The EUROMOD tax-benefit simulation model adds information on taxes to EU-SILC microdata, based on the income and socioeconomic characteristics of households.[1] The output data provide comparable micro-level information on taxes paid and benefits received for all European countries.

Micro_EUROMOD.do uses EUROMOD output data to estimate taxes by type of income, distinguishing between taxes on employment income (TYEM), taxes on property income (TYP), taxes on non-pension social benefits (TBSN), and taxes on pensions (TBSPEN).

The generation of aggregate data in *ENTA_aggregates_compilation.do* requires information about income taxes paid on pensions and non-pension social benefits. This information is based on EUROMOD microdata. Within *Micro_EUROMOD.do*, the section "4. Decomposing income taxes paid by the non-employed" creates the dataset *inctax_nonempl_decomp.dta*. It includes the share of taxes on benefits paid by pensioners (TBSPENsh).

The age-specific estimates of income taxes are also based on EUROMOD data, albeit not directly. First, EUROMOD microdata are used to estimate age-specific tax rates by type of income. Second, these tax rates are used to impute tax payments in EU-SILC (in *ENTA_taxes.do*). The age-specific tax rates are saved in the file *taxr.dta* as TYEMr, TBSPENr, TBSNr, and TYPr.

2.4 Government expenditure by function (Suppl_gov_expenditure.do)

The data on *general government expenditure by function (COFOG)* (`https://doi.org/10.2908/GOV_10A_EXP`) provide aggregate data on government transfers, including both social benefits in cash and consumption by the government sector. *Suppl_gov_expenditure.do* prepares data on government expenditure by function and type.

The file provides aggregate consumption spending on education (CGE), health (CGH), and other (CGX). It also includes social benefits (not in kind) for education (BSE), health (BSH), and social protection in the categories of old age (BSOA), disability (BSD), survivors (BSS), family and children (BSF), unemployment (BSU), and other (BSX). For ENTA, the pension-type benefits BSOA, BSD, and BSS are combined into BSPEN, BSF and BSE into BSFAM, and BSH into BSX. The data are saved as *gov_expenditure.dta* in *ENTA_input_datasets*.

2.5 Household consumption aggregates (Suppl_consumption_hh_coicop.do)

Consumption expenditure by COICOP (`https://doi.org/10.2908/NAMA_10_CO3_P3`) is used to decompose aggregate household-sector consumption in ESA into the following components: education, health, imputed rent, and other consumption. *Suppl_consumption_hh_coicop.do* creates variables for expenditure on health (CHH), education (CHE), housing/imputed rent (CHR), and other consumption (CHX). The output is saved as *consumption_hh_coicop.dta* in *ENTA_input_datasets*. Imputed rent enters as a level (COICOP 04.2); education and health enter as shares of total COICOP consumption applied to ESA household consumption. Where the COICOP data for 2023 or 2024 are

[1] `https://euromod-web.jrc.ec.europa.eu/resources/documentation`

not yet published, each missing value is imputed from the previous year using the growth rate of ESA household consumption at basic prices, so the file requires *esa_ aggregates.dta* and must be run after *Aggr_ ESA2010.do*.

Input data to calculate age profiles

2.6 Population (Suppl_ population.do)

Population data by age and sex are required at several steps in the generation of ENTA. The ENTA age-specific values need to correspond to the aggregate values when summed over the whole population at all ages.

ENTA uses Eurostat and UN population data by age. The Eurostat source is *population on 1 January by age and sex* (`https://doi.org/10.2908/DEMO_PJAN`). For some country-year combinations, age-specific information is missing at older ages; for those specific years, ENTA replaces the values with data from the World Population Prospects 2024.[2] Only the affected years are replaced this way — years for which Eurostat data are complete are kept even if other years for the same country are missing.

Output files in *ENTA_ input_ data/ENTA_ input_ datasets* include:

- *population100.dta*: 100 age groups in long format (sex as a variable)
- *population100w.dta*: 100 age groups in wide format (sex identified with suffix m and f)
- *population80.dta*: 80 age groups in long format
- *population80w.dta*: 80 age groups in wide format
- *population_ agegr15.dta*: total population in 15-year age groups
- *population0.dta*: total population, sex-specific, long format
- *population0t.dta*: total population
- *population_ agegr3.dta*: total population in three age groups (0–19, 20–64, 65 and older)

2.7 Population from UN (Suppl_ population_ UN.do)

Imports the age-specific population data from the World Population Prospects for use in ENTA. The directories need to be specified at the beginning of the do-file. Because of the size of the original files, they are not included in *ENTA_ input_ data/Original_ input_ data*. Data source: `https://population.un.org/wpp/downloads?folder=Standard%20Projections&group=Population`.

[2] `https://population.un.org/wpp/downloads?folder=Standard%20Projections&group=Population`

2.8 EU-SILC microdata (Micro_SILC.do)

The most important data source for ENTA age profiles is EU-SILC.[3] EU-SILC is an income survey carried out every year in all EU countries. It provides detailed and comparable microdata on income in European countries and also contains information on the socioeconomic characteristics of individuals and households. These data serve as the basis for the estimates of primary income and public benefits in ENTA.

The file *Micro_SILC.do* combines the original EU-SILC data provided by Eurostat into a single file for all years for each country. It prepares the ENTA variables on income from employment (YE, YM, YEM), asset income (YP, YO), social benefits (BSP, BSFAM, BSU, BSPPEN, BSX), imputed rent (CHR), and inter-household private transfers.

The directories with the original data and the working directory need to be specified in *ENTA_setup.do*. The output file for country CNT2 (two-letter country identifier) is called *silc_[CNT2].dta* and contains microdata with the central EU-SILC variables, particularly income and benefits.

2.9 Family characteristics in EU-SILC (Micro_SILC_famben.do)

The code is part of *Micro_SILC.do*. It calculates partnership and family characteristics by merging information from partners and children. These characteristics are used, in particular, for the allocation of family benefits to the correct household members.

2.10 HBS microdata (Micro_HBS.do)

The HBS microdata from Eurostat consist of consumption expenditure surveys carried out in European countries (`https://ec.europa.eu/eurostat/web/microdata/household-budget-survey`). The HBS data contain information on expenditure according to the Classification of Individual Consumption by Purpose (COICOP), together with socioeconomic characteristics of households, such as household income and age structure. ENTA uses HBS data from 2010 and 2015. For Austria, ENTA uses the national consumption survey (Konsumerhebung) from 2010 and 2015, because the Eurostat HBS data for Austria are available from 2020 only.

Micro_HBS.do combines the original Eurostat microdata files into a single file for each country. It also creates the necessary variables for income, household structure, and consumption by category (education, health, and other). The data are saved as *HBS_[CNT2].dta* in the HBS working directory.

2.11 HBS microdata for Austria (Micro_HBS_Austria.do)

Austria is the only country for which the Eurostat HBS data cover 2020 only. ENTA uses the Austrian Konsumerhebung from 2010 and 2015 and adapts the two surveys to HBS format.

[3] `https://ec.europa.eu/eurostat/web/microdata/european-union-statistics-on-income-and-living-conditions`

2.12 Imputing consumption (Micro_impute_cons_into_SILC.do)

NTA requires both income and consumption data at the individual level to estimate private transfers within households. Since no single survey contains both, ENTA combines EU-SILC (which provides income data) and HBS (which provides consumption data) by imputing consumption into EU-SILC.

The do-file *Micro_impute_cons_into_SILC.do* links EU-SILC and HBS based on household characteristics available in both surveys. These include:

- Percentile of household disposable income per household member
- Household size
- Number of household members below age 20
- Number of members aged 20–24 and 65–69, and in the older age groups 70–74, 75–79, and 80+

A generalised linear model (gamma family, log link) is estimated on the pooled HBS data for 2010 and 2015 and relates standardised consumption expenditure per household member to these characteristics. It is then used to predict other consumption for every EU-SILC household. The log link keeps the predicted values positive; multiple imputation and matching were tested and rejected (see Section 3.8.4).

The predictions are saved as *CHX1015_[CNT2].dta* in the HBS working directory (*hbswd* in the setup file). This file contains an EU-SILC household identifier and an estimate of other consumption expenditure (excluding health, imputed rent, and education).

2.13 LFS microdata (Micro_LFS.do)

The Labour Force Survey provides age-specific information on employment. Employment data are not part of the ENTA accounting framework, but the information on employment provides important supplementary information for understanding age-specific economic data.

Micro_LFS.do combines LFS microdata from 1995 to 2023. The LFS data are used to estimate sex- and age-specific employment rates in full-time equivalents.

2.14 Enrolment in education (Suppl_enrollment_educ.do)

The age profiles for public education expenditure are based on the aggregates of education expenditure by level and enrolment rates by age. For data up to 2012, ENTA uses *Students by ISCED level, age and sex* (`https://doi.org/10.2908/EDUC_ENRL1TL`); from 2013 onwards, it uses *Pupils and students enrolled by education level, sex and age* (`https://doi.org/10.2908/EDUC_UOE_ENRA02`). The output is saved as *enrollment.dta* in *ENTA_input_datasets*.

2.15 Public health expenditure (Combine_public_health.do)

Unlike the other files in this section, *Combine_public_health.do* lives in *ENTA_data_other/Health/*, not *ENTA_input_data/*.

Age- and sex-specific health expenditure is taken for all countries from the 2018 Ageing Report, approximated from the profiles published there (*HealthExpRates.csv*). The file weights the male and female rates with population, groups ages 80 and over, and keeps the 2018 edition. The 2021 edition, also in the source file, is not used (see the Health subsection of "Public consumption"). The resulting profile is held to this single reference year and applied to every year, where the compilation scales it to the changing aggregate. The AGENTA profiles and the Austrian national data used in earlier versions are still in the file, commented out, and *Austria/health_AT.do* is no longer needed. The output is saved as *CGH_AWG.dta* in *ENTA_input_datasets*.

Input data for ENTA in different units

2.16 Purchasing power and exchange rates (Suppl_ppp_exchange_rates.do)

Exchange rates and purchasing power parities are needed for converting national currency (NAC) into Euro and for providing final data in meaningful and comparable units. *Suppl_ppp_exchange_rates.do* downloads data on purchasing power and exchange rates from Eurostat (`https://doi.org/10.2908/PRC_PPP_IND`) and adapts them for use in ENTA. It generates the file *ppp_exchange_rates.dta* in *ENTA_input_datasets*, which includes:

- EXR_NAC_EUR: exchange rate of the national currency relative to the Euro (NAC per Euro)
- EXR_2020: exchange rate in 2020
- PPP_EUR: purchasing power parities in Euro (purchasing power of one Euro)
- PPP_EUR_2020: purchasing power parity in 2020
- PPP_EU27_2020: Eurostat PPP conversion factor, national currency per PPS unit, EU27=2020 basis (intermediate input for PPP_EUR; not itself used in any unit calculation)

2.17 Consumer prices (Suppl_prices.do)

ENTA results are provided in different units, including real values adjusted by consumer prices.

Data on consumer price inflation are taken from the *Harmonised Index of Consumer Prices—annual data* (HICP, `https://doi.org/10.2908/PRC_HICP_AIND`) and extended with World Bank data for years prior to 1996 (`https://data.worldbank.org/indicator/FP.CPI.TOTL`). The price data are saved as *hicp.dta* in *ENTA_input_datasets*.

Other input data programmes

2.18 Country names (PROG_gen_country_name_from_CNT2.do, Prog_gen_CNT2_

Two utility programmes that convert between the two-letter country code (CNT2) and the full country name. *PROG_gen_country_name_from_CNT2.do* generates the name from the code and is called wherever output is prepared for publication: in both compilation do-files, in *ENTA_aggregates_compilation.do* and in *Aggr_ESA2010.do*. *Prog_gen_CNT2_from_country_name.do* runs the other way, and is needed where a source arrives with names rather than codes — at present only the World Bank price series read by *Suppl_prices.do*.

3 ENTA Aggregate Data (ENTA_aggregates)

The folder *ENTA_aggregates* includes the code for generating the aggregates for ENTA variables and the Stata file with the aggregates at nominal value. The folder contains:

- *ENTA_aggregates_compilation.do*: calculates all ENTA aggregate quantities (nominal totals, million Euro) and saves them for different purposes (the dataexplorer and long-format output)
- *ENTA_units.do*: generates the conversion factors required to express ENTA data in different units
- *ENTA_generate_manual_tables.do*: a documentation utility, not part of the calculation. It reads *ENTA_aggregates.dta* and writes the per-capita values of the aggregate overview tables of Chapter 2 as LaTeX table rows, so that those tables are taken from the data rather than transcribed by hand
- *ENTA_aggregates.dta*: the output file containing all aggregate values in million Euro, nominal
- *ENTA_units.dta*: the output file containing the unit conversion factors

Unit conversion (real, PPS, percent of income, percent of GDP) is not done in this folder — every value here is a nominal total. The ENTA dataexplorer calculates each unit on demand from these totals and the conversion factors in *ENTA_units.dta*; see Appendix A (Units) for the formulas.

3.1 ENTA_aggregates_compilation.do

ENTA_aggregates_compilation.do is the aggregate backbone of ENTA. It organises the non-financial sector accounts as a condensed sequence of accounts adapted to ENTA:

Income account

1. Primary income

Transfer accounts (redistribution accounts)

2. Government transfer contributions (taxes)

3. Government transfer benefits in cash (social benefits)
4. Contributions and benefits in funded pension systems
5. Transfers between private sectors
6. Government transfers in kind / public consumption
7. Disposable income and transfer indicators

Use of disposable income accounts

8. Private consumption
9. Saving

The required input data from *ENTA_input_data/ENTA_input_datasets* are:

- *esa_aggregates.dta* (ESA sector accounts)
- *taxstructure_wide.dta* (tax revenues by function)
- *inctax_nonempl_decomp.dta* (decomposition of income taxes paid by the non-employed, based on EUROMOD output data)
- *gov_expenditure.dta* (government expenditure by function)
- *consumption_hh_coicop.dta* (household consumption by COICOP)

After combining and saving the nominal aggregates (*ENTA_aggregates.dta*, in *ENTA_aggregates/*), the file continues by cleaning a small number of known-missing values (early years of some COFOG-derived benefit categories in a few countries) and saving the result for its two other consumers: *ENTA_output_data/ENTA_aggregates.csv* — the single canonical CSV, used by both the ENTA dataexplorer and external publication (e.g. Zenodo) — and *ENTA_output_data/ENTA_aggregates.dta* as the general long-format output (same data as the CSV, different format; distinct from *ENTA_aggregates/ENTA_aggreg* above despite the similar name).

3.2 ENTA_units.do

ENTA_units.do generates the conversion factors required to calculate ENTA data in different units for comparisons across countries and years — it does not calculate the units themselves. The conversion factors include:

- Exchange rates, year-specific and fixed at 2020 (from *ppp_exchange_rates.dta*)
- HICP price index with base year 2020 (from *hicp.dta*)
- Purchasing power parities, year-specific and fixed at 2020 (from *ppp_exchange_rates.dta*)

- Primary income (total economy and household sector) and GDP, as nominal aggregate totals, not per capita (from *ENTA_aggregates.dta*)
- Total population, all ages (POPtot)
- NTA norm: average labour income at ages 30–49 (from age profiles)
- ENTA norm: average male employment income at ages 30–49 (from age profiles)

The output is saved as *ENTA_units.dta* in the *ENTA_aggregates* folder, and also exported to *ENTA_output_data* (as *ENTA_units.dta* and *ENTA_units.csv*) for the ENTA dataexplorer. See Appendix A (Units) for a full codebook of *ENTA_units.dta* and the exact formula for every unit.

4 Calculation of ENTA Age Profiles (ENTA_ageprofiles)

The folder *ENTA_ageprofiles* contains the do-files that generate age- and sex-specific profiles for all ENTA components. Each do-file processes one country at a time and is called in a loop over all countries. The output files are saved in the subfolder *ageprofiles*.

4.1 Income (ENTA_income.do)

The file *ENTA_income.do* uses the prepared EU-SILC data (*silc_[CNT2].dta*) to generate sex-specific age profiles for employment income and asset income (YE, YM, YEM, YP, YO), as well as for equivalised household income (EHI). The output is saved in *ENTA_ageprofiles/ageprofiles/inc_[CNT2].dta*.

4.2 Taxes (ENTA_taxes.do)

ENTA_taxes.do uses EU-SILC microdata and combines them with the age-specific tax rates (file *taxr.dta*, generated in *Micro_EUROMOD.do*) to estimate taxes paid on income from employment (TYEM), pensions and other benefits combined, public and funded (TBSPPEN), and non-pension benefits (TBSN). The age profiles are saved in *ENTA_ageprofiles/ageprofiles/tax_[CNT2].dta*.

4.3 Benefits (ENTA_benefits.do)

ENTA_benefits.do uses EU-SILC microdata to calculate age- and sex-specific averages of total social benefits (BSP), pension benefits incl. mandatory funded pensions (BSPPEN), family benefits (BSFAM), unemployment benefits (BSU), and other benefits (BSX). The age profiles are saved as *ben_[CNT2].dta* in *ENTA_ageprofiles/ageprofiles*.

4.4 Consumption and intra-household transfers (ENTA_consumption_intrahousehold_transfers.do)

This file calculates the age profiles for private consumption and intra-household transfers. It uses several data sources:

- HBS data for age-specific consumption expenditure on education services (CHE) and health (CHH)
- EU-SILC data for imputed rent (CHR), based on the EU-SILC variable HY030
- EU-SILC with imputed consumption values (from *CHX1015_[CNT2].dta*) for other consumption (CHX)

Intra-household transfers are calculated as the difference between each household member's consumption and their individual income, net of saving and inter-household transfers. Household members whose consumption exceeds their income receive intra-household transfers (BPW); those whose income exceeds their consumption pay them (TPW).

The output is saved as *ctpriv_[CNT2].dta* in *ENTA_ageprofiles/ageprofiles* and contains the age profiles of education consumption (CHE), health consumption (CHH), imputed rent (CHR), other consumption (CHX), intra-household transfers received (BPW), and intra-household transfers paid (TPW).

4.5 Public consumption

Unlike the other age profiles described in this section, public consumption (education CGE, health CGH, and other CGX) is not built by a separate do-file in *ENTA_ageprofiles* — it is calculated directly within the Government sector section of *ENTA_compilation1_esa.do*, using the following supplementary data:

Education consumption profiles are calculated by combining enrolment rates by age, sex, and ISCED level (from *enrollment.dta*) with per-student expenditure by ISCED level (from *gov_expenditure.dta*).

Health consumption profiles are based on the age-specific public health data (from *CGH_AWG.dta*, built by *Combine_public_health.do*, see above), which is merged into the compilation and scaled to the government health aggregate. All countries use the profile from the 2018 Ageing Report.

Other public consumption (CGX) is allocated equally per capita across all ages.

The resulting CGE, CGH, and CGX variables become part of *ENTA_priv_wide.dta*, alongside the other ENTA indicators, rather than being saved as a separate per-country file.

4.6 Employment from LFS (ENTA_employment.do)

ENTA_employment.do uses the Labour Force Survey microdata to generate age- and sex-specific profiles of employment rates and working hours. These profiles serve as supplementary information for interpreting the ENTA results and are not part of the ENTA accounting framework. A person's full-time equivalent is the usual weekly working hours divided by 35, capped at one. The profiles are not smoothed. The output is saved as *lfs_[CNT2].dta* in *ENTA_ageprofiles/ageprofiles*.

Part III

Information needed for the interpretation of the results

E. Country-specific patterns and events

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Some ENTA series contain abrupt level shifts, sign changes or single-year spikes that look like data errors but are not. They reflect national policy reforms, Europe-wide macroeconomic events, or the way a particular country reports to Eurostat. This chapter documents the cases identified so far, so that users encountering them can distinguish a real feature of the data from a defect.

A note on reading them. A common source of confusion is that a series changes level permanently and then *every* year of one era looks anomalous against the average of the whole period. Several entries below are of that kind. Before treating an unusual value as an error, check whether the series has a structural break — and if so, compare the year against its own era rather than against the full time span.

The entries are grouped by cause rather than by country, because most of them affect several countries at once. Sections 5 and 6 cover changes that happened in the economy or in national law: the data move because the world moved. Section 7 covers the other kind, breaks produced by the way the data are built or reported; these are documented in full in Chapter 5 and appear here only as signposts, so that a user who has noticed something odd can find the explanation from the symptom.

Each entry separates what is observed in the data from what is inferred about it. Where an explanation has not been checked against a documented source it is labelled as an interpretation, and the

entry says so explicitly. Nothing here should be read as a verified account of a national measure unless it is stated as one.

5 Changes caused by the economy

5.1 The 2008–09 financial crisis, EU-wide

Pattern. Cash benefits and household saving move together across nearly every country ENTA covers. Between 2008 and 2009, social benefits (`BS`) rise as a share of household primary income in all 26 countries (median +7%), pensions in all 26 (+5%), and unemployment benefits in all 26, by far the largest move (+31%). Household disposable income (`YDb`) falls as a share of household primary income in all 26 countries, and interest paid on mortgages (`YPp`) falls in 23 of 26 (median −39%), consistent with the interest-rate cuts described below.

Explanation. The 2008–09 global financial crisis reduced employment and income across Europe and triggered the automatic and discretionary expansion of unemployment and other social benefits that is standard in a recession. This is well-established economic history and is not treated further here.

Consequence for analysis. 2009 is a genuine cyclical trough, not a data break, and should be read as such in any series that shows it. The three country-specific entries below — the male-intensive employment collapse, Irish and Romanian property income, and German corporate income — are particular symptoms of this same crisis.

5.2 The 2009 male-intensive employment collapse

Pattern. The male share of household primary income fell in 23 of the 26 countries in 2009. The falls were largest in Lithuania (−6.70 points) and Latvia (−3.50), ahead of Ireland (−2.86), Bulgaria (−2.44) and Estonia (−1.93). In Lithuania the move sits almost entirely in employment income (the male share of `Y` falls by 6.70 points against 6.54 for `YE`), and it has a mirror image on the benefit side, where the male share of benefits *rises* as men move onto benefits (Lithuania +17.1 points, Latvia +9.7). It unwinds gradually rather than snapping back: Lithuania's male income share runs 49.49, 49.65, 50.91, 51.95, 52.22, 53.11 over 2009–2014.

What is established. The fall in the male income share, its concentration in employment income, its mirror image in benefit receipt, and the multi-year gradual unwind are all in the data, decomposed exactly for Lithuania.

Interpretation. This fits the international pattern in which recessions hit male-dominated sectors such as construction and manufacturing harder, sometimes called the "mancession."[1] The sectoral

[1] H. Hoynes, D. L. Miller and J. Schaller, "Who Suffers during Recessions?," *Journal of Economic Perspectives* 26(3), 2012, pp. 27–48. The mechanism is documented there for the United States; construction employment is over 90% male in every EU member state (Eurostat, Employment by sex, age and economic activity (NACE Rev. 2), dataset `lfsa_egan2`), consistent with a sectoral rather than a general channel in these countries too.

composition of the Lithuanian and Latvian labour markets in 2009 specifically has not been checked directly here.

Consequence for analysis. Sex-specific profiles for 2009–2013 in the countries above should not be read as a permanent shift in the gender distribution of income; several years of gradual convergence are part of the same recession, not a new equilibrium.

5.3 Interest rate collapse, 2008–2011

Pattern. Household asset income falls in several countries in the years policy rates were cut to near zero after the financial crisis — the mirror image of the 2022–2023 rate normalisation below. Property income received falls in Cyprus (2009), Denmark (2010), Spain (2009), Finland (2009), Lithuania (2009–2010) and Luxembourg (2009); taxes on asset income and assets fall with it in Cyprus, Hungary (2009–2011), Ireland (2009), Italy (2009), Romania (2009) and Slovenia (2009).

What is established. The falls are in the source data, appear in several countries in the years immediately following the financial crisis, and mirror the 2022–2023 rises documented below, applied in reverse.

Explanation. The ECB's main refinancing rate was cut from 4.25% to 1.00% between October 2008 and May 2009, and reduced further to 0.00% by March 2016; non-eurozone central banks among the ENTA countries followed broadly similar paths over the same years.[2] A fall in household interest-bearing asset income in exactly this window is the expected consequence of a policy-rate move of that size.

Consequence for analysis. Asset income and the taxes on it are not comparable across this boundary either, for the same reason as the 2022–2023 shift below: the change in level reflects interest rates, not a change in the underlying wealth or its distribution by age.

5.4 Ireland and Romania: negative net property income, 2008

Pattern. In 2008 households in both countries paid more in property income — essentially mortgage interest — than they received. Ireland recorded 8,599 received against 9,036 paid; Romania 2,648 against 3,051. Ireland 2007 was already close to the boundary, at 8,180 against 7,908.

What is established. The negative net values are in the source data for both countries, and 2008 sits at the peak of both countries' property markets.

Explanation. The two countries reached the same outcome by different routes. Irish household mortgage debt roughly tripled between 2002 and 2008, and the household debt-to-disposable-income ratio was among the highest in the eurozone, peaking at around 200%; at the 2008 peak, mortgage interest payments plausibly exceeded the household sector's total property income received.[3] Romanian household debt was much lower in relative terms, but the National Bank of Romania's policy

[2] European Central Bank, "Key ECB interest rates," `https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html`.

[3] P. Honohan, Central Bank of Ireland, "The Irish Banking Crisis: Regulatory and Financial Stability Policy 2003–2008," report to the Minister for Finance, 31 May 2010.

rate stood at 10.25% in mid-2008, among the highest in the EU, with household lending rates higher still;[4] the combination of rapid recent credit growth and a high spread between lending and deposit rates is a plausible account of a similar outcome at a much lower debt level, though this has not been checked against a specific Romanian household-debt figure for 2008.

Consequence for analysis. These are the only country-years in which net property income is negative for the household sector, and they require care in any calculation that uses it as a proportion or a denominator.

5.5 Germany: negative corporate primary income, 2009

Pattern. The primary income of the German corporate sector is negative in 2009, at −137 million euro.

Explanation. This is the same 2008–09 recession as the entries above, recorded on the corporate side: German corporations' operating surplus and property income together turned negative in the trough year. The value is genuine, and it produces a corporate age profile that is negative at every age, which is the correct representation.

5.6 COVID-19, 2020–2021

Pattern. Two distinct effects. First, household saving rises steeply in 2020 in most countries. Second, in some countries taxes less subsidies on production (`TPN`) turn strongly negative for the household sector. Poland is the clearest case: household `TPN` falls to −8,919 in 2020, against −301 in 2019 and −2,955 in 2021.

The saving effect is close to universal: household saving (`SH`) rises as a share of primary income in 25 of the 26 countries in 2020 (median +5.75 points), and social benefits (`BS`) rise in all 26 (median +2.32 points), driven by pensions and non-pension benefits together (+2.40 and +0.84 points respectively).

Explanation. The Polish figure can be traced through the source data. Other subsidies on production (D.39) received by the household sector rise from 4,057 in 2019 to 12,573 in 2020 and fall back to 6,446 in 2021, while other taxes on production (D.29) paid by that sector stay flat at about 1,550 throughout. The change in `TPN` between 2019 and 2020 (−8,618) is accounted for almost entirely by the change in D.29 paid less D.39 received (−8,511). On the paying side, government D.39 payments rise from 2,801 to 18,729 in the same year.

Two features of the ESA framework put that flow into the household sector. D.39 is defined to cover subsidies on payroll or workforce, and short-time work schemes and compensation for the self-employed were recorded as other subsidies on production during the pandemic. Unincorporated businesses belong to sector S.14, so support paid to them arrives in the household sector rather than in the corporate sector. Which individual Polish programmes make up the 2020 figure has not been traced here.

[4] National Bank of Romania, monetary policy rate history, `https://www.bnr.ro`.

Consequence for analysis. ENTA treats taxes less subsidies on production as part of primary income, adding them to mixed and employment income and then removing them again as a transfer to government. The subsidy therefore does not appear under taxes on assets, where it has no place, but inside taxes on employment income. For Poland this matters for the reading of 2020: taxes on employment income fall from 95,451 to 87,943 between 2019 and 2020, a drop of 7.9 per cent, but net of the production subsidy the same figures are 95,752 and 96,861, a small rise. Most of the apparent collapse in labour taxation that year is the subsidy inflow, not a change in what employees and the self-employed paid.

A related effect appears in "other" private consumption, which falls in 2020 and recovers above its pre-pandemic level by 2023 as travel, restaurants and recreation resumed.

5.7 Energy price and cost-of-living relief, 2022

Pattern. Miscellaneous current transfers (D.75) received by households spike in 2022 and fall back partly in 2023. The effect is largest in Austria (+311% against the 2019–2021 average), Germany (+197%), Malta (+105%) and Czechia (+84%). Where the spike is large it turns net other current transfers (`TX`) sharply negative.

What is established. The spike is in the aggregate source data, it is confined to miscellaneous current transfers from government to households, it appears in many countries in the same year, and it partly reverses in 2023.

Interpretation. The timing and the confinement to this one transfer category point to the cost-of-living and energy relief payments made in 2022. The individual national measures behind the figures have not been identified here, so the attribution should be read as the natural reading of the timing rather than as a verified decomposition. A further caveat applies to comparisons across countries specifically: governments did not all book one-off relief under the same ESA transaction. Where a country recorded its relief as a social benefit (D.62) rather than as a miscellaneous current transfer (D.75), it would not appear in the D.75 pattern described here at all, so the size differences between countries in this entry may partly reflect a bookkeeping choice rather than the size of the relief itself; this has not been checked country by country.

Slovakia moves the other way. Household `TXH` fell in Austria, Germany, the Netherlands and Sweden in 2022 (by 0.31–1.34 points of household primary income), consistent with relief being paid. Slovakia rose instead, by 1.57 points. A Slovak government policy brief on 2022 inflation documents three measures pointing the same way: free school meals in primary schools were cancelled from September 2021; tobacco excise rose in February 2021 and again in March 2022; and pensions, material-need benefits, parental allowance and child allowance are indexed to the *previous* year's inflation, so the 2022 price spike only reached those transfers in 2023, temporarily reducing their real value. The brief records no offsetting Slovak household relief package as of June 2022.[5] This

[5] V. Plutzer, J. Šilan and D. Hojdan, Ministry of Finance of the Slovak Republic, Institute for Financial Policy, Policy brief 2022/7, "How did high inflation come about?", 16 June 2022, `https://www.mfsr.sk/files/archiv/47/Policy_brief_Inflation_2022.pdf`.

establishes the direction of the Slovak move but not the exact mechanism: the brief describes benefits and excise duties, while `TXH` is built from the broader D.7 other-current-transfers item.

Consequence for analysis. ENTA allocates both components of other current transfers (`TXH` and `TXNPISH`) with a receipt profile, labour income plus pensions, which is zero below age 15. A one-off payment of this kind is therefore booked entirely on adults, and it reaches children through the intrahousehold transfers rather than directly. Where the underlying programme paid amounts tied to the number of children, the age pattern in 2022 is therefore approximate. Slovakia should not be read as part of the EU-wide relief pattern above; its 2022 move has the opposite cause.

5.8 Interest rate normalisation, 2022–2023

Pattern. Asset income rises sharply in 2022 and again in 2023, in every country. Taxes on asset income, interest paid on mortgages, and household saving rise with it. In the age profiles the increase is concentrated at older ages, where asset holdings are largest.

Household property income received in 2023, relative to the 2019–2021 average, rose in every country covered by ENTA. The largest increases were Ireland (+154%), Czechia (+150%), Slovenia (+138%), Hungary and Poland (+127%), Spain (+94%), France (+89%) and Belgium (+86%); the smallest were still above +60%.

Explanation. The ECB's main refinancing rate rose from 0.00% to 4.50% between July 2022 and September 2023, after more than six years at zero; non-eurozone central banks among the ENTA countries followed broadly similar paths.[6] Household interest income roughly doubled, and interest paid on mortgages rose in parallel. This is a genuine economic change, not a break in the data, and it reverses the interest-rate collapse of 2008–2011 described above.

Consequence for analysis. Asset income and everything derived from it — taxes on asset income, disposable income, saving — are not comparable across the 2022 boundary without acknowledging the rate shift. Czechia is the clearest case: property income received runs 7,163 in 2021, 12,446 in 2022 and 15,780 in 2023, with taxes on property income rising from 469 to 2,063 over the same period.

6 National policy and law

6.1 Hungary: nationalisation of the funded pension pillar, 2011

Pattern. All funded-pension variables — contributions, benefits and the associated taxes — fall by roughly half between 2010 and 2012 and remain at the lower level thereafter. Contributions run 2,117, 2,331, 2,810, 2,405 and 2,271 for 2006–2010, then 1,102 in 2011, 766 in 2012 and approximately 690 in later years.

Explanation. Hungary's mandatory private pension pillar was nationalised in 2011 and the great majority of members were transferred back to the public system.

[6] European Central Bank, "Key ECB interest rates," `https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html`.

A wider pattern. Hungary is the clearest case of a broader phenomenon: funded-pension contributions and benefits step to a new level, in one year, in most countries that have a funded pillar or experienced a pension-related fiscal event. Verified against *ENTA_aggregates* on 2026-09-23, 30 country-years carry a step of this kind, and 22 of the 30 are large relative to the series' own year-on-year variation ($|z| \geq 2$). In six of them, contributions and net funded-pension benefits move in near-exact mirror image (for example Poland 2011: -0.82 and $+0.83$ points of household primary income; Slovakia 2013: -1.57 and $+1.56$), the signature of a change in contribution rates rather than a data fault.

Six of these steps have been traced to a documented reform or fiscal event: Hungary's 2011 pillar nationalisation (above); Poland's 2011 cut of the contribution rate directed to the Open Pension Funds (OFE) from 7.3% to 2.3% of gross salary, and its 2014 transfer of 51.5% of OFE assets to the public pension institution ZUS, with 2015 the first full year of the resulting lower contribution flow rather than a separate reform;[7] Lithuania's 2009 cut of the second-pillar contribution rate from 5.5% to 2%;[8] Slovakia's 2012–2013 cut from 9% to 4%, alongside a window allowing participants to switch back to the public-only system;[9] and Portugal's 2011 transfer of major banks' pension fund assets, worth around 3.5% of GDP, to the state Social Security system as a one-off fiscal measure.[10]

Three further cases have a documented change but not one that is clearly a funded-pillar contribution or benefit reform: Czechia introduced a small, low-enrolment voluntary second pillar in 2013, abolished from 2016;[11] Greece consolidated its auxiliary pension funds into a single fund (ETEA) and cut their benefits in 2012–2013 as part of the adjustment programme; and Spain reformed the legal framework for employer-sponsored occupational pension plans in 2022. In these three the size of the ENTA step has not been checked against the size of the underlying reform.

The remaining country-years — Czechia 2023, Slovenia 2011–2014, Germany 2009, Romania 2023 and Cyprus 2013 — have not been matched to a named reform. Germany has no mandatory funded pillar, so its 2009 step more plausibly reflects the recession than a policy change; Cyprus's 2013 step coincides with that year's banking crisis and bail-in rather than any pension reform found. These five should be treated as unexplained until checked further.

Consequence for analysis. The Hungarian funded-pension series should be treated as two separate regimes. Years before 2011 are not comparable with years after it, and a year such as 2008 is only a peak within its own era, not an anomaly. The same caution applies to the other countries listed above, whether or not their step has been traced to a named reform.

6.2 Estonia: voluntary second-pillar pension withdrawal, 2021

Pattern. Estonian taxes on property income (`TYP`) more than triple in 2021 and roughly halve again in 2022 — the underlying DG TAXUD capital-income series runs 51.9, 95.9, 321.8, 167.2 and 150.1

[7] OECD, *Pensions at a Glance 2013* (for the 2011 reform) and *Pensions at a Glance 2015* (for the 2014 reform), OECD Publishing, Paris.
[8] OECD, *Pensions at a Glance 2011*, OECD Publishing, Paris. doi: `10.1787/pension_glance-2011-en`.
[9] OECD, *Pensions at a Glance 2013*, OECD Publishing, Paris. doi: `10.1787/pension_glance-2013-en`.
[10] OECD, *OECD Economic Surveys: Portugal 2012*, OECD Publishing, Paris.
[11] OECD, *Pensions at a Glance 2013* and *Pensions at a Glance 2015*, OECD Publishing, Paris.

million euro over 2019–2023 — while the DG TAXUD series for labour taxes on the non-employed, where ordinary pension taxation would appear, stays flat over the same years.

Explanation. Estonia's mandatory funded second pension pillar became voluntary in 2021, and a one-off window let members withdraw their accumulated savings before retirement age. Withdrawals taken before retirement age are subject to a 20% income tax.[12] About 20% of participants used the window, withdrawing roughly 1.3 billion euro in total (about 4.5% of GDP) in the final months of 2021.[13] A 20% tax on 1.3 billion euro is about 260 million euro, close to the roughly 225 million euro excess over baseline that the capital-income series shows in 2021, plus a further 71 million in 2022 as the window continued.

Consequence for analysis. The 2021–2022 level of Estonian property income and the taxes on it reflects a one-off pension withdrawal, not a change in ordinary capital income, and should not be extrapolated. It is also the opposite kind of reform from the Hungarian entry above: Estonia made its funded pillar voluntary rather than abolishing it.

6.3 Poland: the Rodzina 500+ family benefit, 2016

Pattern. Polish family benefits (`BSFAM`) rise sharply as a share of household primary income in 2016, from 1.84% to 3.45%.

Explanation. Poland's "Family 500+" (Rodzina 500+) programme was adopted by law in February 2016 and began paying out in April 2016: a universal monthly child-raising benefit of PLN 500 for every second and subsequent child, and for the first child in lower-income families, roughly twice the average family benefit paid per beneficiary the year before.[14] The size and timing of the jump in `BSFAM` match a scheme of this kind precisely: the benefit is paid from April, so 2016 carries nine months of it, consistent with a rise smaller than a full doubling.

Consequence for analysis. Polish family-benefit levels before and after 2016 are not comparable, and cross-country comparisons of family benefits should treat Poland's 2016 jump as a policy change rather than a break in survey coverage.

6.4 Bulgaria: no taxes on social benefits

Pattern. Taxes on social benefits are zero for Bulgaria in every year.

Explanation. The zero comes from the source and is a statement rather than a gap. ENTA takes the aggregate from the DG TAXUD tax revenue by economic function data, category LNON, taxes on labour income of the non-employed. Bulgaria is reported there as exactly zero in every year from 1995 to 2024. It is the only country in the dataset for which this holds: every other country reports a

[12] Estonian Ministry of Finance, "Restructuring the second pillar of the funded pension," `https://www.fin.ee/en/restructuring-second-pillar-funded-pension`.

[13] G. Moss, "A cautionary tale from Estonia," *IPE*, 12 May 2026, `https://www.ipe.com/analysis/a-cautionary-tale-from-estonia/10136677.article`.

[14] Act of 11 February 2016 on State Assistance in Raising Children; A. Sowa, "'Family 500+': A new family income-supporting benefit in Poland," European Social Policy Network (ESPN) Flash Report 2016/45, European Commission, June 2016.

positive figure in every year, including the smallest ones (Cyprus 24, Croatia 97, Latvia 122 in 2023). Bulgaria reports ordinary positive values for every other economic function in the same dataset — taxes on labour of employees, of employers, on capital income of households and of the self-employed — so this is not a country missing from the collection. The reported decomposition is also internally consistent: total labour taxes equal the employee plus employer components exactly, leaving nothing unattributed. Where a country genuinely fails to report, the data look different: Iceland shows a total of zero while its components sum to 937.

This is consistent with Bulgarian tax law. Article 13(1) of the Income Taxes on Natural Persons Act exempts income from compulsory and voluntary insurance, including pensions, together with unemployment benefits and compensation, and aid and maintenance allowances paid under separate legislation such as the Persons with Disabilities Act and the Family Allowances for Children Act.[15]

Consequence for analysis. A zero age profile is the correct representation for Bulgaria and should not be treated as a gap to be filled. Bulgaria should nevertheless be left out of cross-country comparisons of taxes on benefits, since its zero reflects tax law rather than a measured amount.

6.5 Denmark: early-retirement pay (*efterløn*)

Pattern. Danish unemployment benefits (`BSU`) are high at ages 60–65 — 7,000 to 12,000 euro per person at ages 63–65 in 2008–2017 — and fall to almost zero at 66, where pensions rise by a similar amount. The age at which the high values start moves up over the period: in the survey data `BSU` rises gradually from age 60 in 2008–2012, but in steps at 62–63 in 2014 and 2016 and at 62–64 in 2018.

Interpretation (not checked against a documented source here). At these ages `PY090` records mainly *efterløn*, the Danish voluntary early-retirement pay, which is paid through the unemployment insurance funds and ends when the public pension starts. The upward shift of the starting age is consistent with the retirement reform that raised the earliest age for *efterløn* from 60 in steps from 2014 onwards.

Consequence for analysis. Danish unemployment benefits at ages 60–65 are not comparable with unemployment benefits elsewhere: they measure early retirement rather than unemployment. Comparisons across countries are better made on total benefits (`BS`), which do not depend on how the two schemes are classified. How the smoothing rules treat this pattern is described in Section 5.5.4.

7 Breaks caused by the data

The entries above are things that happened. The breaks below are the other kind: a series shifts because the way it is measured, reported or built changed, not because the economy did. Each one is documented in full in Chapter 5; this list exists so that a user who has noticed the symptom can find the explanation from it.

[15] Bulgaria, Income Taxes on Natural Persons Act (*Zakon za danatsite varhu dohodite na fizicheskite litsa*), Art. 13(1); summary at `https://www.bulgaria-tax-law.bg/taxes-other-income.html`.

- **Germany, 2019: the sex split of pensions and asset income shifts.** The male share of pensions is 56–58% in 2014–2018 and 52.3% in 2019, and stays at the lower level. Two changes in the German EU-SILC coincide in that income year: the oldest age group moves from "73 and older" to single years, and the survey method changed. Profiles before and after 2019 should not be compared by sex. See Section 5.2.
- **Germany, from 2022: profiles become step functions.** EU-SILC delivers five-year age groups above 20, with 16–19 collapsed, so German profiles are smoothed only up to 2021. See Section 5.2.
- **Netherlands, Czechia, Ireland and Slovakia: employment profiles are step functions.** The public-use LFS files for these four carry only five-year age groups, and everything derived from employment inherits the step. See Section 5.3.
- **Bulgaria, 2010: the age and sex pattern of some income components is lost.** The survey's income components do not sum to the published household total; the level is unaffected because profiles are scaled to the national accounts. See Section 5.2.
- **Bulgaria ends in 2022.** The 2023 ESA sector accounts were not published when the aggregates were built. See Section 5.2.
- **Croatia starts in 2009, and 2009–2010 tax profiles are imputed.** Croatian EU-SILC begins with the 2009 income reference year; EUROMOD begins in 2011, so the tax rates and the pension share of taxes on benefits for 2009 and 2010 are carried back from 2011. Only the age pattern is imputed — the level comes from each year's own aggregate. See Section 5.4.
- **Denmark, before 2019: property income is built differently.** Danish EU-SILC reports interest and dividend income net of interest paid, so property income received is built from rental income with imputed rent added instead. See Section 5.3.
- **Hungary 2018, France 2020, Germany 2007–2008: mortgage interest is missing.** Property income paid would be zero at every age; the affected profiles are replaced by the average of the neighbouring years. See Section 5.3.
- **Ireland, 2018: private pensions jump.** EU-SILC reports private pension income for 609 persons against 124–171 in the neighbouring years, shifting the age and sex pattern of property income received towards older men. The 2018 profile is replaced by the average of 2017 and 2019. See Section 5.3.
- **Ireland, 2011: the split of benefits into pensions and other benefits breaks.** Public pensions fall from 16,313 to 12,017 million euro while total benefits are almost unchanged, following near-offsetting moves in the Irish COFOG data. Total benefits are unaffected; the split is not comparable across 2010 and 2011. See Section 5.5.
- **Denmark, 2016–2018: benefits dip at 62 and jump at 63.** The rise in early-retirement pay has become a step at 62, which the smoothing flattens by one age: `BS` at 62 is about 7–8%

below the mean of its neighbours. Read ages 61–63 together. See Section 5.5.4.

- **Greece, 2014: taxes on assets halve and taxes on production rise.** About 2.7 billion euro of household taxes move to a different transaction in the national accounts from 2014. `TA`, `TYA` and `TPN` are not comparable across 2013 and 2014; total taxes are hardly affected. See Section 5.4.
- **Czechia and Slovakia: taxes on benefits are high relative to the microsimulation.** The rate implied by the national accounts is about 10% against 0.2% for Czechia and 19% against 0.5% for Slovakia. The level should be read as the national accounts concept, not an out-of-pocket household payment. See Section 5.4.
- **The pension and non-pension split of taxes on benefits is absent in early years.** The split comes from EUROMOD, which starts between 2005 and 2007 depending on the country, and 2009 for Croatia. Before a country's first EUROMOD year the total is published on its own. See Section 5.4.
- **Germany and Ireland: the split of other current transfers begins late.** `TXH` and `TXNPISH` are separated only from 2012 for Germany and 2010 for Ireland; before that the whole of `TX` sits in `TXH`. German `TXH` moves from −14,374 million euro in 2011 to +6,762 million in 2012 for this reason alone. Use `TX`. See Section 5.7.
- **Sweden: private education consumption is a structural zero.** The HBS provides no education consumption for Sweden, so `CHE` is reported as zero and the amount is carried in other consumption. Sweden must be excluded from cross-country comparisons of private education spending. See Section 5.9.

8 How to read a country-specific result

Three questions cover most cases where a single country looks unusual.

Is the level or the shape unusual? Levels come from the national accounts. A level that looks odd is a property of the aggregate source, not of the ENTA construction, and the entries earlier in this chapter are the place to look. A shape that looks odd is more likely to come from one of the treatments in Chapter 5.

Is the country on a coarser age grid? Germany after 2021, and the Netherlands, Czechia, Ireland and Slovakia for employment, are step functions rather than single-year profiles. Comparisons of profile curvature across countries are affected; comparisons of totals are not.

Is the series a structural zero? Swedish private education consumption, Bulgarian taxes on social benefits, and the benefit-tax components before each country's first EUROMOD year are the cases where a variable is reported as zero or missing by construction rather than measured as such.

References

Abio, G., Patxot, C., Souto, G., & Istenič, T. (2021). The role of gender, education and family in the welfare organization: Disaggregating national transfer accounts. *The Journal of the Economics of Ageing*, *20*, 100348.

Dramani, L., & Oga, I. J.-B. (2017). Understanding demographic dividends in africa: The nta approach. *Journal of Demographic Economics*, *83*(1), 85–101.

d'Albis, H., & Moosa, D. (2015). Generational economics and the national transfer accounts. *Journal of Demographic Economics*, *81*(4), 409–441.

European Commission. (2018). *The 2018 ageing report: Economic and budgetary projections for the EU member states (2016–2070)* (European Economy No. 8/2018). Brussels: European Commission, Directorate-General for Economic and Financial Affairs. Retrieved from `https://economy-finance.ec.europa.eu/publications/2018-ageing-report-economic-and-budgetary-projections-eu-member-states-2016-2070_en`

European Commission. (2026). *Data on Taxation.* `https://webgate.ec.europa.eu/taxation_customs/redisstat/databrowser/view/tax_ec_func/default/table`. Taxation and Customs Union.

EUROSTAT. (2013). *European system of accounts: ESA 2010.* Publications Office of the European Union. `https://ec.europa.eu/eurostat/documents/3859598/5925693/KS-02-13-269-EN.PDF`.

Hammer, B., Christl, M., & De Poli, S. (2023). Public redistribution in europe: Between generations or income groups? *The Journal of the Economics of Ageing*, *24*, 100426.

Hammer, B., Spitzer, S., Vargha, L., & Istenič, T. (2020). The gender dimension of intergenerational transfers in europe. *The Journal of the Economics of Ageing*, *15*, 100234.

Istenič, T., Hammer, B., Šeme, A., Dolinar, A. L., & Sambt, J. (2019). *European national transfer accounts.* Agenta. Retrieved from `https://www.jstor.org/stable/pdf/26928909.pdf`

Ladusingh, L., & Narayana, M. (2012). Demographic dividends for india: evidence and implications based on national transfer accounts. In *Aging, economic growth, and old-age security in asia* (pp. 203–230). Edward Elgar Publishing.

Lee, R. D., & Mason, A. (2011). *Population aging and the generational economy: A global perspective.* Edward Elgar Publishing. `https://idl-bnc-idrc.dspacedirect.org/server/api/core/bitstreams/168f97d0-da44-48d4-aa73-fb3eb4606b88/content`.

Loichinger, E., Hammer, B., Prskawetz, A., Freiberger, M., & Sambt, J. (2017). Quantifying economic dependency. *European Journal of Population*, *33*(3), 351–380.

Mason, A., Lee, R., & members of the NTA network. (2022). Six ways population change will affect the global economy. *Population and development review*, *48*(1), 51–73.

Mérette, M., & Navaux, J. (2025). Did seniors benefit from consumption and public transfers at the expense of youth? national transfer accounts in canada's provinces (1997–2017). *The Journal of the Economics of Ageing*, 100609.

Nelisse, R. (2021). *Non-Financial Corporations Split Into Subsectors.* https://www.cbs.nl/-/media/_pdf/2022/38/report-subsectoring-nfcs.pdf.

Oosthuizen, M. (2019). *Inequality and the generational economy: race-disaggregated national transfer accounts for south africa, 2015* (No. 2019/24). Wider working paper.

Shraberman, K., & Weinreb, A. A. (2024). The fiscal consequences of changing demographic composition: Aging and differential growth across israel's three major subpopulations. *The Journal of the Economics of Ageing*, *27*, 100500.

United Nations. (2011). *System of National Accounts 2008.* https://unstats.un.org/unsd/nationalaccount/docs/sna2008.pdf.

United Nations. (2013). *National Transfer Accounts Manual: Measuring and Analysing the Generational Economy.* http://www.un.org/en/development/desa/population/publications/development/NTA_Manual.shtml.

Vanhuysse, P., Medgyesi, M., & Gál, R. I. (2023). Taxing reproduction: the full transfer cost of rearing children in europe. *Royal Society Open Science*, *10*(10), 230759.